

वार्षिक प्रतिवेदन
Annual Report
2009-2010



राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी
National Academy of Agricultural Research Management

www.naarm.ernet.in

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Preface

The agriculture sector is facing numerous challenges so is the National Agricultural Research System. Issues such as climate change, diversion of food crops for biofuel, trans-boundary diseases and entry of multi-nationals, besides growing demand-supply gap of key food commodities, are of serious concern. The greatest challenge is accelerating innovations for higher, sustainable and inclusive agricultural growth. This calls for a paradigm shift in prioritizing research portfolio, improving efficiency of research system and promoting innovative institutions for effectively linking research organizations with all stakeholders, including farmers, agri-industry and remunerative domestic and global markets. The Academy is responding to the emerging challenges by undertaking need-based research, reorienting capacity building programmes, imparting education for next-generation professionals, providing consultancies, and promoting public-private partnership. To meet the challenge and strengthen National Agricultural Research System, the Academy has a well trained and strong faculty and best infrastructure facilities.

The Academy, during the year, took a great leap in transforming its role from capacity-building of different stakeholders in the National Agricultural Research System to an institution of global repute through its effective and responsive key roles in capacity-building, research, academic programmes and public-private partnership. As many as 53 capacity-building programmes were organized for 1581 participants of different levels of hierarchy in the National Agricultural Research System. Research was undertaken on several key areas of concern to agricultural research and education policy, planning and management. Important ones include learning and capacity building, manpower planning in agriculture, assessment of frontier technologies for agricultural research, geospatial data and knowledge management and leadership in change management. Scientists of the Academy were successful in leaving a mark in the scientific arena with their contributions through publications in national and international journals.

The year was a landmark for the Academy on education front. Two post-graduate diploma programmes on 'Information Technology Management in Agriculture' and 'Intellectual Property and Technology Management in Agriculture' were successfully completed and all the students were placed in reputed national and multinational organizations. This year, the Academy also launched a two-year post-graduate programme in Management (Agriculture). To promote public-private partnership and link all the stakeholders in agri-supply chain, the Academy conceptualized and put into operation an Agribusiness Knowledge Centre (AKC). It is envisioned that AKC would improve the efficiency of agri-supply chain and accelerate innovations by effectively linking relevant stakeholders. Besides these programmes, the Academy provided support to National Agricultural Innovation Project through helpdesk, and extended consultancy to several institutions for intellectual property protection and commercialization. The Academy also successfully conducted the examination of senior research fellow for post-graduate studies of Indian Council of Agricultural Research.

In accomplishing its targets and launching new programmes, the Academy received overwhelming support from the Council. We are especially grateful to Dr S. Ayyappan,



Director-General, Indian Council of Agricultural Research and Secretary, Department of Agricultural Research and Education, Government of India, for extending unexceptional support, and extending continuous encouragement and guidance to make the Academy more vibrant and responsive to meet new challenges. The Academy owe to Dr Mangala Rai, Former Director-General and Secretary, Department of Agricultural Research and Education for his constant support in expanding the programmes of the Academy. Dr Arvind Kumar, Deputy Director-General (Education) provided full support in strengthening the Academy's programmes. Prior to him, Dr S. P. Tiwari, former Deputy Director-General (Education) was always helpful in promoting the programmes of the Academy. We offer our sincere thanks to the Secretary, Indian Council of Agricultural Research, and personnel and finance divisions. Our special thanks are to Dr S. M. Ilyas, former Director of the Academy, who conceptualized and launched number of programmes to transform the Academy.

My colleagues, Dr G. P. Reddy and Mrs G. Aneeta took the responsibility of compiling and editing the report; Mr P. Namdev brought the report to the present form and Mr M. Ravi provided the photographs. Dr N. H. Rao, Dr P. Manikandan, Dr D. Rama Rao and Dr R. Kalpana Sastry supervised and provided valuable suggestions in finalizing the report. I sincerely acknowledge their incredible contribution in brining out this report. I also acknowledge the outstanding contributions and team efforts of all the staff of NAARM for their overwhelming support in fulfilling the mandate of the Academy.

I am sure that the contribution of the Academy will be useful for our partners and stakeholders. We always look forward for their continuous support and valuable suggestions in accomplishing our mission.

30 June 2010
Hyderabad



(PKJoshi)
Director





कार्यकारी सारांश

कार्यकारी सारांश

नार्म की दीर्घावधि परिकल्पना तथा XI पंचवर्षीय योजना के अनुमोदित रणनीति के अनुरूप वर्ष 2009-2010 के लिए नार्म की गतिविधियों का महत्व, मुख्यतः क्षेत्रों पर रहा है:

- i रा.कृ.अनु.प्र से प्रभावी एवं प्रतिक्रियात्मक राष्ट्रीय कृषि नवोन्मेषी प्रणाली के संगठनात्मक नवीनीकरण सुविधा हेतु परामर्श एवं क्षमता निर्माण, अनुसंधान नीतिगत सहयोग एवं परामर्श।
- ii अकादमी का कृषि प्रबंधन के उत्कृष्ट संघटित संस्थान के रूप में रूपांतरण जिससे समतुल्य विश्वविद्यालय के स्तर से स्नातकोत्तर एवं शोध स्तर के पाठ्यक्रम चलाए जाए।

1. क्षमता निर्माण :

वर्ष 2009-2010 में 1581 सहभागियों के लिए 53 कार्यक्रम आयोजित किए गये। इनमें 4 बुनियादी पाठ्यक्रम (279 कृ.अनु.से प्रशिक्षुओं के लिए), 24 वरिष्ठ स्तर के प्रशिक्षण कार्यक्रम (654 सहभागी), 11 प्रबंधन विकास कार्यक्रम/कार्यशालाएँ (363 सहभागी), एक कार्यकारी विकास कार्यक्रम (11 सहभागी), 11 गैर परिसरीय प्रशिक्षण कार्यक्रम (265 सहभागी) एवं 2 अन्तर्राष्ट्रीय प्रशिक्षण कार्यक्रम सम्मिलित रहें। इनमें से, 12 कार्यक्रम शिक्षा एवं क्षमता निर्माण, एन.ए.आई.पी. के (एल एवं सी बी) परियोजना के तहत संपन्न हुए, जिसमें नार्म कनसोरशियम अभिनेता तथा आई.आई.एम, लखनऊ, मैनेज हैदराबाद एवं रा.ग्रा.वि.सं, हैदराबाद साझेदार संस्थान रहें। एल एवं सीबी परियोजना के कनसोरशियम लीडर के रूप में नार्म ने वर्ष 2009-2010 के दौरान 3 साझेदार संस्थानों द्वारा आयोजित 15 प्रबंधन विकास कार्यक्रमों में सुविधक की भूमिका निभाई। प्रबंधन विकास/प्रशिक्षण कार्यक्रमों में निम्नांकित विषय क्षेत्र सम्मिलित थे। कृषि अभिनवता को उन्नत करने में नेतृत्व विकास, सफल अनुसंधान प्रस्तावों का विकास, अनुसंधान आधारित कनसोरशिया के प्रबोधन एवं मूल्यांकन का प्राथमिकता निर्धारण, भू-अंतरिक्षीय ज्ञान प्रबंधन, मल्टीमीडिया विषय सार विकास, ई-शिक्षा माड्यूलों का विकास, सार्वजनिक निजी साझेदारी



प्रबंधन, कृषि उद्यमीकरण, एकस्वाधिकार प्रारूपण, कृषि व्यापार शिक्षा, उच्च कृषि शिक्षा के पुनः रूपांतरण, प्रौद्योगिकी प्रबंधन एवं वाणिज्यिकरण एवं अन्य। इसके अतिरिक्त नार्म द्वारा एल एवं सीबी परियोजना के द्वारा प्रमुख क्षेत्रों में रा.कृ.अनु.प्र के लगभग 100 वैज्ञानिकों को अन्तर्राष्ट्रीय प्रशिक्षण भी प्रदान किया गया।

2. अनुसंधान :

अकादमी के अनुसंधान के प्रमुख क्षेत्र कृषि अनुसंधान एवं शिक्षा नीति, योजना एवं प्रबंधन पर केंद्रीकृत हैं। इनमें शिक्षा एवं क्षमता निर्माण, कृषि में मानवशक्ति योजना, कृषि अनुसंधान के लिए उभरती प्रौद्योगिकी में विकास का मूल्यांकन, भूअंतरिक्षीय डेटा एवं ज्ञान संगठन प्रबंधन, ज्ञान का सांख्यिकीय विस्तारण, रा.कृ.अनु.प्र के सांख्यिकीय संगणन की सुदृढ़ता, कृषि शिक्षा का कार्यनिष्पादन मूल्यांकन एवं अन्य क्षेत्र सम्मिलित है। अनुसंधान, 9 परियोजनाओं के तहत किया गया, जिसमें 6 रा.कृ.न.प. द्वारा तथा एक भा.कृ.अनु.प. द्वारा निधि पोषित रहें। एन.ए.आई.पी द्वारा वित्त पोषित 6 परियोजनाओं में, दो परियोजनाओं में नार्म कनसोरशिया प्रतिनिधि तथा शेष 4 में सहभागी रहा है। वर्ष 2009-2010 की प्रमुख उपलब्धियाँ इस प्रकार हैं :

- ◆ कृषि महाविद्यालयों के गुणवत्ता निर्धारण के लिए सिस्टम डॉयनामिक्स सिमुलेशन मॉडल
- ◆ कृषि विश्वविद्यालयों के कार्यनिष्पादन मूल्यांकन के लिए प्रणाली एवं संकेतक
- ◆ कृषि एवं नॉनो जैव प्रौद्योगिकी, खाद्यान्न सुरक्षा के लिए नॉनो प्रौद्योगिकी के मूल्यांकन संबंधी नॉनो प्रौद्योगिकी एकस्वाधिकार एवं संदर्भिका स्रोत का डेटाबेस
- ◆ बीटी कपास एकस्वाधिकार का आई.पी. भू-दृश्य निर्माण
- ◆ कृषि अनुसंधान, शिक्षा एवं प्रौद्योगिकी हस्तांतरण (आंध्रप्रदेश राज्य के लिए) के विशेष संदर्भ में अंतरिक्षीय एवं विशेष डेटा प्रदान करने हेतु नार्म भू-अंतरिक्षीय पुस्तकालय (एन जी एस एल) ग्रामीण सामुदाय द्वारा किए



गये निर्णय के सहयोग एवं ग्रामीण ज्ञान केंद्रों में अनुप्रयोग के लिए भू-अंतरिक्षीय ग्रामीण ज्ञान प्रबंधन पद्धति।

- ◆ जीविका विश्लेषण के लिए अंतरिक्षीय संतुलित प्रतिचयन के लिए प्रक्रियाविधि।
- ◆ मानवशक्ति योजना के लिए प्रणाली एवं प्रणालीगत अनुरूपक मॉडल
- ◆ विकासशील देशों (सी ए एस - आई पी के साथ सहयोगिता) में आई पी रेजैम पर सार संग्रह

केस अध्ययन (8)

- ◆ जी आई उत्पाद (नवारा धान, पोक्कली धान एवं अलाहाबाद सफेदा) के लिए तीन केस अध्ययन
- ◆ एशियाई पेरिफेरिक क्षेत्र (नार्म-मोनसांटो के साथ सहयोगिता में) पी वी पी विधि पर केस अध्ययन
- ◆ कृषि मूल्य कड़ी में दो केस अध्ययन : आं.प्र. में फुटकर खाद्य प्रबंधन, कुक्कुटपालन में करारगत कृषि (नार्म एवं सगुणा खाद्य के साथ सहयोग)
- ◆ जीविका असुरक्षितता का भू-अंतरिक्षीय मूल्यांकन
- ◆ वेबसाइट विकास (2) आईपूर ग्राम (धान फाउंडेशन एवं एस ए आई आर डी के साथ सहयोग से) में एस एच जी के लिए ग्राम ज्ञान केंद्र एवं वेब पोर्टल।
- ◆ ई-शिक्षा माड्यूलों का विकास (60) : मूडल एल एम एस, दुपाल सी एन एस में वास्तविक शिक्षा के लिए शिक्षा माड्यूल (एम एस एक्सेल प्रयोग, ओपन स्रोत सांख्यिकीय साफ्टवेयर (वास्तविक शिक्षा केंद्रों से नार्म वेबसाइट की सुगमता) के प्रयोग से कृषि प्रबंधन, कृषि में सूचना प्रबंधन



तथा कृषि में बौद्धिक संपदा प्रबंधन में स्नातकोत्तर कार्यक्रमों के 59 पाठ्यक्रम माड्युल

- ◆ प्रयोक्ता मान्युल एवं शिक्षा संसाधन : मूडल (ओ.एस.एस) द्वारा ई-शिक्षा पद्धति वेब एवं मल्टीमीडिया विषय सार विकास, जी.आई.एस. डेटा मैनिंग एवं अन्य क्षेत्र।
- ◆ डेटाबेस का विकास : आं.प्र. के नलगोंड़ा के जिला स्तर के ग्रामीण जीविका असुरक्षितता मूल्यांकन के लिए जी.आई.एस. उत्पाद एवं असुरक्षितता का मानचित्र
- ◆ प्रश्नावली/ डेटा संचयन उपकरण : संगठनात्मक परिवर्तन, नेतृत्व, कार्यनिष्पादन मूल्यांकन, संगठनात्मक नागरिकता व्यवहार, अभिनवता गुणवत्ता से संबंधित उपकरण

3. शिक्षा :

वर्ष 2009-2010 अकादमी के लिए ऐतिहासिक वर्ष रहा है, चूंकि शिक्षा प्रबंधन में अकादमी ने अपनी प्रारंभिक पहल में ही विश्वस्त एवं संकेतिक योगदान की छाप छोड़ी है। निम्नांकित दो एक वर्षीय स्नातकोत्तर कार्यक्रम सफलातपूर्वक संपन्न हुए।

- ◆ 24 विद्यार्थियों से युक्त कृषि में सूचना प्रौद्योगिकी प्रबंधन (पीजीडी-आईटीएमए) में स्नातकोत्तर डिप्लोमा।
- ◆ 16 विद्यार्थियों से युक्त कृषि में बौद्धिक संपदा एवं प्रौद्योगिकी प्रबंधन में (पीजीडी-आईपीटीएमए) का स्नातकोत्तर डिप्लोमा।

सभी स्नातक विद्यार्थी, प्रख्यात राष्ट्रीय एवं बहु राष्ट्रीय संगठनों में सफलतापूर्वक स्थानापन्न हुए।

यह वर्ष इसका साक्षी है कि ए.आई.सी.टी.ई के अनुमोदन से ही प्रबंधन (कृषि) के दो वर्षीय स्नातकोत्तर डिप्लोमा का शुभारंभ हुआ। जुलाई 2010 में विद्यार्थियों के



द्वितीय बैच के लिए अखिल भारतीय प्रवेश परीक्षा (नार्माट - 2010) भी सफलतापूर्वक आयोजित की गई।

4. परामर्श एवं नीतिगत सहयोग :

नार्म के सहायता डेस्क द्वारा कई एन.ए.आई.पी. परियोजनाओं को मैत्रीपूर्ण सहयोग प्रदान किया गया तथा बौद्धिक संपदा एवं वाणिज्यिककरण के लिए कई संस्थानों को भी सहयोग दिया गया।

5. कृषि व्यवसाय ज्ञान केंद्र (ए के सी) :

नार्म ने एक कृषि व्यवसाय ज्ञान केंद्र (ए के सी) को संकल्पनात्मक रूप प्रदान करने तथा उसके प्रचलन हेतु ज्ञानटेक इनफार्मेशन सिस्टम्स प्रा.लिमिटेड (जी.आई.एस.पी.एल) के साथ सार्वजनिक निजी सहभागिता (पीपीवी) का करार किया। कृषि आपूर्ति कड़ी के पणदाताओं में मूल्य हास, व्यवसाय विकास एवं वृद्धि की सुविधा हेतु एक पर्यावरणात्मक अनुकूल मंच प्रदान करने की आवश्यकता से ही कृषि व्यवसाय ज्ञान केंद्र का अविर्भाव हुआ है। इसके दो घटक हैं।

1. निजी क्षेत्र, गैर सरकारी संगठनों एवं कृषि संगठनों के कुछ पणदाताओं को नार्म में भौतिक सुविधाएँ जिसके द्वारा प्रशिक्षण तथा अन्य डेटा उत्पाद की सहायता।
2. वास्तविक घटक, जो कृषि व्यवसाय ज्ञान विनिमय (ए.के.एक्स) के रूप में प्रचलित होगा, पणदाताओं को ऑनलाइन द्वारा ज्ञान उपलब्ध कर सकें या प्रतिक्रियात्मक सहयोग श्रुतिबद्ध कर सकें।

एड्ड कलात्मक आई.सी.टी. आधारित कृषि व्यवसाय ज्ञान विनिमय (आई.के.एक्स) मंच की स्थापना की गई जिसका उद्देश्य विभिन्न पंजीकृत पणदाताओं के ज्ञान विनियम हेतु कृषि व्यवसाय ज्ञान केंद्र में पंजीकरण, विचार-विमर्श, ज्ञानोपार्जन तदोपरांत अनुभव का आदान प्रदान एवं समयानुरूप वेब आधारित प्रबोधन सहित विभिन्न पणदाताओं के बीच वाणिज्यिक कार्य व्यवहार की शुरुवात तथा उसके प्रबोधन की सुविधा प्रदान करता है। कृषि व्यवसाय ज्ञान केंद्र (ए.के.सी) एवं कृषि व्यवसाय ज्ञान विनिमय (ए.के.एक्स) द्वारा नार्म भा.कृ.अनु.प



एवं कृषि व्यवसाय कंपनियों के लिए अनुसंधान एवं शिक्षा कार्यक्रमों की अग्रवर्ती एवं पाश्चगामी कड़ी स्थापित करने हेतु एक मंच प्रदान किया गया।

वर्ष 2009-2010 के लिए 12 केंद्रों में अखिल भारतीय प्रतियोगी भा.कृ.अनु.प.-एस.आर.एफ (पी जी एस) परीक्षा संपन्न हुई, जिसमें पी.एचडी उपाधि हेतु फेलोशिप के लिए 174 अभ्यर्थी तथा बिना फेलोशिप के पी.एचडी कार्यक्रम में प्रवेश हेतु 374 अभ्यर्थियों ने योग्यता प्राप्त की।

नार्म वैज्ञानिकों ने राष्ट्रीय एवं अन्तर्राष्ट्रीय पत्रिकाओं में प्रकाशित लेखों के अपने योगदान से वैज्ञानिक क्षेत्र में एक सफल छाप छोड़ी है। 10 पत्रिकाओं में लेख 4 पुस्तक अध्याय, पाँच क्रियात्मक लेख, आठ अनुसंधान प्रतिवेदन, सम्मेलन, कार्यशाला एवं संगोष्ठियों में प्रस्तुत 17 लेख।

अकादमी, को दिसंबर 12 से 15, 2009 के दौरान राष्ट्रीय डेयरी अनुसंधान संस्थान, करनाल में भा.कृ.अनु.प. अंतर-क्षेत्रीय खेलकूद प्रतियोगिताओं एवं फरवरी 23-27, 2010 के दौरान केंद्रीय मास्यिकीय प्रौद्योगिकी संस्थान (सीआईएफटी) द्वारा कोचीन में आयोजित भा.कृ.अनु.प. अंतर - संस्थानीय खेलकूद प्रतियोगिताओं में कई पुरस्कार प्राप्त हुए। नार्म को उत्तम संस्थानीय उद्यान (बेस्ट इंस्टीट्यूशनल गॉर्डन), हाईब्रिड टी (एच.टी) रोजेस का उत्तम संचयन एवं फ्लोरिबाण्ड रोजेस के रोलिंग ट्राफिस के साथ सिकंदराबाद बागवानी संघ द्वारा आयोजित ज़्झ्ज् वार्षिक रोज शो में 'क्वीन ऑफ द शो' पुरस्कार प्राप्त हुआ।

रा.कृ.अनु.प्रणाली की क्षमता निर्माण में अकादमी ने अपनी भूमिका में महत्वपूर्ण परिवर्तन किया है जिससे की अकादमी द्वारा क्षमता निर्माण, शैक्षणिक कार्यक्रम एवं अनुसंधान में प्रभावी एवं उत्तरदायी योगदान से वैश्विक ख्याति के संस्थानों का विकास संभव हो सके।





Executive Summary

Executive Summary

The thrust of the Academy's activities in 2009-10 is on two broad fronts in accordance with the long term vision for NAARM, and the approved strategy for the XI Five Year Plan:

- (i) Capacity building, research, policy support and consultancy to facilitate the organizational renewal of NARS to a more efficient and responsive National Agricultural Innovation System,
- (ii) Transforming the Academy into a fully integrated Institution of excellence in Agricultural Management, offering courses at both post-graduate as well as doctoral level, with a Deemed to be University status.

1. Capacity building: 53 programmes were organized in 2009-10 for 1581 participants. These include 4 Foundation Courses (for 279 ARS probationers), 24 Senior Level Training Programmes (654 participants), 11 Management Development Programmes/ Workshops (363 participants), one Executive Development Programme (11 participants), 11 off-campus Training Programmes (265 participants) and 2 International Training Programmes. Of these, 12 programmes were organized under the Learning and Capacity Building (L&CB) project of NAIP, of which NAARM is the Consortium Leader and IIM Lucknow, MANAGE Hyderabad and NIRD Hyderabad are partner institutions. As the consortium leader of the L&CB project, NAARM provided oversight and facilitated the organization of 15 Management Development Programmes by the 3 partner institutions during 2009-10. The Management Development/Training programmes covered the following themes: Leadership Development for promoting agricultural innovation; Developing winning research proposals; Priority setting and Monitoring and Evaluation of consortia based research; Geospatial Knowledge Management; Multimedia content development; Developing e-learning modules; Managing Public-Private Partnerships; Agri-preneurship; Patent drafting, agribusiness education, revitalizing higher agricultural education, Technology management and commercialization, and others. In addition, NAARM also facilitated the international training of nearly 100 scientists of NARS in frontier areas through the L&CB project.

2. Research: The research themes of the Academy address several areas of key concern to agricultural research and education policy, planning, and management. These include: learning and capacity building, manpower planning in agriculture, assessing developments in emerging technologies for agricultural research, geospatial data and knowledge organization and management, digital dissemination of knowledge, strengthening statistical computing of NARS, performance assessment of agricultural education, and others. The research is organized under 9 projects of which six are funded by NAIP, and one by ICAR. Of the six projects funded by NAIP, NAARM is the consortium leader in two and partner in 4. Key outputs for 2009-10 are:

- ♦ System dynamics simulation model for quality assessment of agricultural college
- ♦ Methodology and indicators for performance assessment of agricultural universities



- ♦ Databases of nanotechnology patents and bibliographic sources of relevance to agriculture and nanobiotechnology; assessments of nanotechnology implications for food security
- ♦ IP landscaping of Bt cotton patents
- ♦ NAARM Geospatial Library (NGSL) for delivering spatial and attribute data of relevance to agricultural research, education and technology transfer (for the State of Andhra Pradesh)
- ♦ Geospatial Village Knowledge Management System for application in village knowledge centres to support decision making by rural communities
- ♦ A procedure for spatially balanced sampling for livelihoods analysis
- ♦ Methodology and systems simulation model for manpower planning
- ♦ A compendium on IP regimes in developing countries (in association with CAS-IP)
- ♦ Case studies (8)
 - ♦ three case studies for GI products [Navara rice, Pokkali rice and Allahabad safeda]
 - ♦ case study on PVP laws in Asia-Pacific region: (NAARM in collaboration with Monsanto)
 - ♦ two case studies in agricultural value chains: retail food management in AP, contract farming in poultry (NAARM in collaboration with Suguna foods)
 - ♦ geospatial assessment of livelihoods vulnerability
- ♦ Web sites developed (2): Village Knowledge Centre and web portal for SHGs in Aipur village (in collaboration with Dhan Foundation and SAIRD); Web portal: for digital media resources centre in Plone CMS
- ♦ e-learning modules developed (60): 59 course modules of PG programmes in agricultural management, information management in agriculture and intellectual property management in agriculture using MOODLE LMS; Learning modules for virtual learning in Dhrupal CMS (MS Excel Applications; open source statistical software (accessible from Virtual Learning Centre on NAARM web site)
- ♦ User manuals and learning resources: for e-learning systems through Moodle (OSS), Web and Multimedia content development, GIS, Data mining, and other areas
- ♦ Databases developed: databases, GIS products and vulnerability maps for assessing vulnerability of rural livelihoods at block level for Nalgonda district of AP and at district level for AP State
- ♦ Questionnaires/data collection instruments: related to organizational change -



leadership, performance assessment, organizational citizenship behaviour; innovation quality

3. Education: The year 2009-10 is a landmark year for the Academy as it embarked on and consolidated its role in management education. The following two one-year Post Graduate Programmes were successfully completed:

- ♦ Post Graduate Diploma in Information Technology Management in Agriculture (PGD-ITMA) with 24 students, and
- ♦ Post Graduate Diploma in Intellectual Property and Technology Management in Agriculture (PGD-IPTMA) with 16 students.

All the graduating students were successfully placed in reputed national and multinational organizations.

The year also saw the launch of the *two-year Post Graduate Diploma in Management (Agriculture) after approval by AICTE*. The first batch of students (24) was admitted in July 2009. The All India Admission Test (NAARMAT2010) was successfully conducted for admission to the second batch of students in July 2010.

4. Consultancy and Policy Support: Handholding support was provided to several NAIP projects through the helpdesk being operated from NAARM. Similar support has also been provided to several institutions for intellectual property protection and commercialization.

5. Agribusiness Knowledge Centre (AKC): NAARM conceptualized and put into operation, an Agribusiness Knowledge Centre (AKC) in Public-Private Partnership (PPP) mode. The AKC is the outcome of the recognition of the need to create a platform that can provide a conducive ecosystem that will nurture and facilitate value addition, business development and growth among the stakeholders across the agricultural supply chain. It consists of two components:

- (i) a physical component to house some of the stakeholders from the private sector, NGOs and farmer organizations at NAARM and provide knowledge services like training, data products, and others
- (ii) A virtual component, called the Agribusiness Knowledge Exchange (AKX) to orchestrate the collaboration interactively and on-line to provide value adding knowledge to the stakeholders.

A state-of-art ICT based Agribusiness Knowledge Exchange (AKX) platform is being built with the aim to facilitate registration into the Agribusiness Knowledge Center, Exchange of knowledge between different registered stake holders, start discussion blogs, post knowledge sharing experiences, and initiate/monitor commercial transactions between different stake holders, including web based monitoring, on almost real time basis. The AKC and AKX will provide a platform for NAARM/ICAR and agribusiness companies to establish backward and forward linkages to the research and education programmes.



All-India competitive ICAR-SRF (PGS) Examination was conducted successfully in 12 centres for the year 2009-10 where 174 candidates were selected for fellowship for pursuing their Ph. D. degree programme and 374 candidates were qualified for admission to Ph. D. programme without fellowship.

NAARM scientists were successful in leaving a mark in the scientific arena with their contributions through published papers in national and international journals. Ten journal articles; four book chapters; five working papers; eight research reports; 17 Papers presented in Conference, Workshop, Seminar, etc.

The Academy won several prizes in ICAR Inter-Zonal Sports Tournament held at NDRI, Karnal from December 12 to 15, 2009, and also at ICAR Inter-Institutional Sports Meet organized by Central Institute of Fisheries Technology (CIFT) at Cochin from February 23 to 27, 2010. NAARM won Queen of the Show rolling trophy in the XXIV Annual Rose Show conducted by Secunderabad Horticultural Society along with rolling trophies for the best Institutional garden, best collection of Hybrid Tea (HT) Roses, and Floriband roses.

The Academy has witnessed a great leap in transforming its role from capacity building of NARS to institution of global repute through its effective and responsive key roles in capacity-building, academic programmes and research.





NAARM - an Overview

NAARM – An Overview

Indian National Agricultural Research System (NARS) is one of the largest agricultural systems in the world employing over 25,000 scientists. Effective and efficient management of this vast pool of manpower poses a formidable challenge to the NARS system. Keeping in view this objective, the Government of India has established the National Academy of Agricultural Research Management (NAARM), Hyderabad, in 1976. Over the years, the Academy has expanded its horizon of activities. Through its innovative need-based Human Resources Development programmes, and demand-driven research initiatives in the specialized areas of Agricultural Research and Education Management, it has helped the systems in evolving appropriate policies to improve the efficiency and effectiveness of research and education in particular and agriculture at large.

The scope of the Academy has been further widened to encompass tailor-made training programmes, especially for the international participants from the developing countries in Asia, Africa and Latin America. Agricultural research management has formed another major activity of the Academy. This has provided an input for improvising training and the policy support. The Academy extends a facilitative role to the Indian Council of Agricultural Research (ICAR) in developing policy guidelines on the various issues related to personnel, organizational and management reforms, human resource development and assessment system, aimed at improving the overall efficiency and effectiveness of the NARS. NAARM has been developing itself as a centre of excellence in management studies. The Academy also undertakes and offers specialized consultancy services in the mandated activities on the strength of the experienced and expert faculty.

The Academy strives to generate a sense of fraternity and inculcate a scientific work culture amongst the agricultural scientists and teachers across the country. Being a unique institution of its kind, the Academy is well poised to augment its usefulness and aims to emerge as an institution *par excellence* to facilitate and support a culture of dynamic management in agricultural research and education.

Vision

“By the year 2025, the Academy will emerge as India's premier institute of excellence in the field of agricultural management, known and sought nationally and globally for its expertise in management of agricultural education, research, training, consultancy, public policy programmes and agri-business, and will contribute significantly to promote the sustainable growth and development of agriculture”.

Mission

“Committed to promote professionalism in the management of agricultural education, research and extension programmes and agri-business, and enhance the efficiency, responsiveness, and performance of the National Agricultural Research System (NARS)”.



Mandate

The Academy is mandated to enhance the efficiency and effectiveness of NARS through:

- ◆ To be an integrated institution of agricultural management focusing on creation, dissemination and application of knowledge through its education, training, research, consultancy and policy support programmes
- ◆ To serve as an apex resource center for collection, compilation, documentation and dissemination of innovative learning resources and practices in agricultural management followed in India and other countries
- ◆ To work as a catalyst for building and enhancing the competence of individual scientists and the capability of institutions of NAS for addressing contemporary issues in agricultural management
- ◆ To facilitate the organizational renewal of the NAS and management of change
- ◆ To serve as a Think Tank for the NAS and provide research-based inputs and advice to agricultural policy makers, planners, administrators, and others
- ◆ To emerge as global thought and knowledge leader in agricultural management
- ◆ To establish and foster functional partnerships and effective networking with leading management institutes of the world in order to emerge as global thought and knowledge leader

Objectives

Commensurate with the mandate, the following objectives are set for the Academy:

- ◆ To impart agricultural management education
- ◆ To enhance the teaching-learning effectiveness through proper management of agricultural education
- ◆ To plan and organize need-based, multi-tier, stakeholder-driven and customized on-campus and off-campus training programs
- ◆ To facilitate knowledge and technology dissemination management through innovative use of Information and Communication Technologies (ICTs)
- ◆ To undertake research on agricultural and technology management, and address emerging concerns in agriculture
- ◆ To offer consultancy and manage dialogues to backstop training and to provide policy support to NARS
- ◆ To develop suitable management tools, practices and processes for facilitating organizational effectiveness
- ◆ To assemble quality resource material and function as a resource center of information and knowledge



- ◆ To promote facilitative work culture for fostering creativity and innovativeness
- ◆ To enhance administrative and financial management in the system
- ◆ To forge and strengthen partnerships, linkages and networking at regional, national and global levels
- ◆ To take up other related activities for fulfilling the mandate

Organization and Management

The Director leads the Academy supported by Joint Director. The faculty comprises well qualified and competent scientists who are supported by a number of technical and administrative personnel. Administration, auxiliary and supporting personnel help in attaining Academy's mandated objectives of training, research, education, consultancy, and related programmes and activities. Following organogram of NAARM, details the organizational structure of the Academy:



The Academy receives guidance for its effective functioning from the Institute Management Committee (IMC) and Research Advisory Committee (RAC), comprising eminent scientists, management experts, developmental agencies, and administrative personnel from within and outside the Academy. There are also Academic and Research Councils to guide the academic training, research and education related activities of the Academy.



Linkages

NAARM has a rewarding and rich experience of having partnership with many Indian and international institutions. Strong linkages have been developed with:

National

- ◆ Management Institutions like Administrative Staff College of India (ASCI), Indian School of Business (ISB), Indian Institutes of Management (IIMs), National Institute of Agricultural Extension Management (MANAGE), National Institute of Rural Development (NIRD), Institute of Public Enterprise (IPE)
- ◆ Private sector and NGOs
- ◆ Department of Science and Technology (DST)
- ◆ The ICAR system
- ◆ Agricultural Universities in India and foreign countries

International

- ◆ The CGIAR institutions including International Crop Research Institute for the Semi-Arid Tropics (ICRISAT) and International Food Policy Research Institute (IFPRI)
- ◆ The World Bank, and The Food and Agriculture Organization of the United Nations (FAO)
- ◆ South Asian Association for Regional Cooperation (SAARC)
- ◆ Department of International Development (DFID, UK)
- ◆ NARS of developing countries like Sri Lanka, Nigeria, Yemen, Tanzania, Nepal, etc.

Infrastructure and Logistics

The facilities available on the campus include state-of-art lecture halls with modern audio-visual aids, conference halls and auditorium; wi-fi enabled broad band internet service with access both in class rooms and hostels, indoor and outdoor games, excellent hostel facilities to students, in-campus hospital and a well equipped library. NAARM has also a digital library which has many publications in digital format. As a member of the consortium for e-resources in agriculture, NAARM has direct access to over 10,000 journals from EBSCO, CSIRO, Springer, Cera and Open J-Gate.

Excellent laboratory facilities and other facilities are available for:

- ◆ Video production
- ◆ Geographical Information Systems
- ◆ Digital multimedia
- ◆ Audio-visual aids
- ◆ Photography
- ◆ Web design and development
- ◆ Offset printing
- ◆ Information search and retrieval
- ◆ Organizational behaviour exercises



Human and Financial Resources

Human Resources (as on 31.03.2010)

Sl.No.	Category Posts	Sanctioned Strength	Filled Positions	Vacant Positions
1.	Research Management	02	02	-
2.	Scientific	40	17	23
3.	Technical	51	42	09
4.	Administrative	43	40	03
5.	Supporting	39	39	--
	Total	175	140	35

Budget Allocation, Expenditure during the Year 2009-2010

Expenditure Statement - Non-Plan (as on 31.03.2010)

(Rs. in lakhs)

Sl.No.	Head of Account	Budget Allocation	Expenditure
1	(a) Establishment charges (b) Wages	888.65 81.48	888.65 81.42
2	(b) OTA	0.43	0.43
3	Traveling allowance	5.30	5.25
4	Other charges including equipment	240.64	241.55
5	Maintenance of residential / non-residential buildings Pension & Retirement benefits Loans & Advances Total	16.14 35.18 233.70 0.32 1501.84	15.64 34.82 233.63 0.24 1501.63

Expenditure Statement - Plan (as on 31.03.2010)

(Rs. in lakhs)

Sl. No.	Head of Account	Budget Allocation	Expenditure
1	Establishment charges	453.44	453.44
2	Traveling Allowance	38.56	38.56
3	Other charges including equipments	49.14	49.14
4	Furniture & fixtures	3.60	3.60
5	Library books & scientific journals	38.26	38.26
6	Works	115.00	115.00
7	Other Items including HRD	2.00	2.00
	Total	700.00	700.00



Resource Generation 2009-10

i) Off-campus and Sponsored Programmes

Sl.No.	Programme Title	Agency	Amount (Rs)
1	Off-campus Specialized Short Programme on "Improving Efficiency of Scientific Personnel"	IGFRI, Jhansi	1,36,742
2	Off-campus Specialized Short Programme on "Improving Efficiency of Technical Personnel"	IGFRI, Jhansi	1,09,776
3	Programme on "e-Content Creation in Standard Formats"	Fisheries College & Research Institute, Thoothukudi	1,76,600
4	Programme on "e-Content Management"	College of Fisheries, Mangalore	1,76,600
5	Programme on "Multimedia Content Development"	Fisheries College & Research Institute, Thoothukudi	3,82,100
6	"HRD training for Programme Officers of North Eastern Region"	Zonal Coordinating Unit (Zone-III, ICAR)	2,82,000
7	Programme on "Multimedia Content Development"	College of Fisheries, Mangalore	3,82,100
8	Programme on "Multimedia Content Creation"	Home Science Faculty, GBPUAT, ANGRAU, UAS, Dharwad	2,13,322
9	Programme on "e-Content Management"	Home Science Faculty, GBPUAT, ANGRAU, UAS, Dharwad	1,56,516
10	Programme on "e-Content Management"	College of Horticulture, Mudigere, Solan and TNAU	1,85,304
11	Programme on "Multimedia Content Development"	College of Horticulture, Mudigere, Solan and TNAU	3,34,209
12	Programme on "E-learning"	Dairy Science Faculty, NDRI, Karnal; SVVU, Tirupati and AAU, Anand	3,30,000
13	Off-campus Specialized Short-term Training for "Improving Efficiency of Technical Officers"	IARI, New Delhi	1,54,398
	Total amount generated	30,19,667	



ii) International Programme

Sl.No.	Programme Title	Dates	Amount (Rs)
1	International training on agricultural research systems and its management	20-4-09 to 02-5-09	2,34,135

iii) Education programmes (as Fees)

S.No.	Name of the Programme	Amount (Rs Lakhs)
1	PGD(ITMA)	48.0
2	PGD(IPTMA)	16.0
3	PGDM(A)	60.06

iv) Resource Generation through Sponsored Research Projects

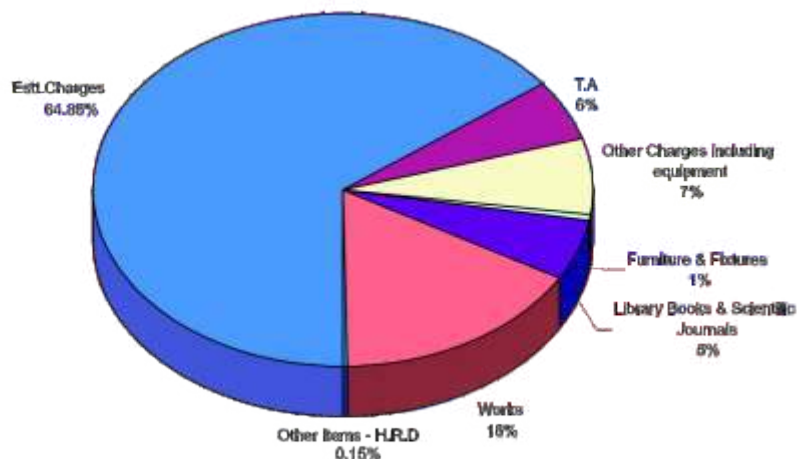
Sl.No.	Project	Amount released in 2009-10 (Rs Lakhs)
1	Learning and Capacity Building project	422.44
2	Re-designing the farmer-extension-agricultural research / education continuum in India with ICT-mediated knowledge management	24.6
3	Assessments of Development in Nanotechnology for Agricultural R & D	6.29
4	Agroweb – Digital Dissemination System for Indian Agricultural Research	11.89
5	Assessment of Future Human Capital Requirements in Agriculture	12.25
6.	Strengthening Statistical Computing for NARS	20.41
7.	ICAR XI Plan Scheme “Intellectual Property Management and Transfer / Commercialization of Agricultural Technology Scheme”	5.5
	TOTAL	503.38

v) Resource Generation from other activities

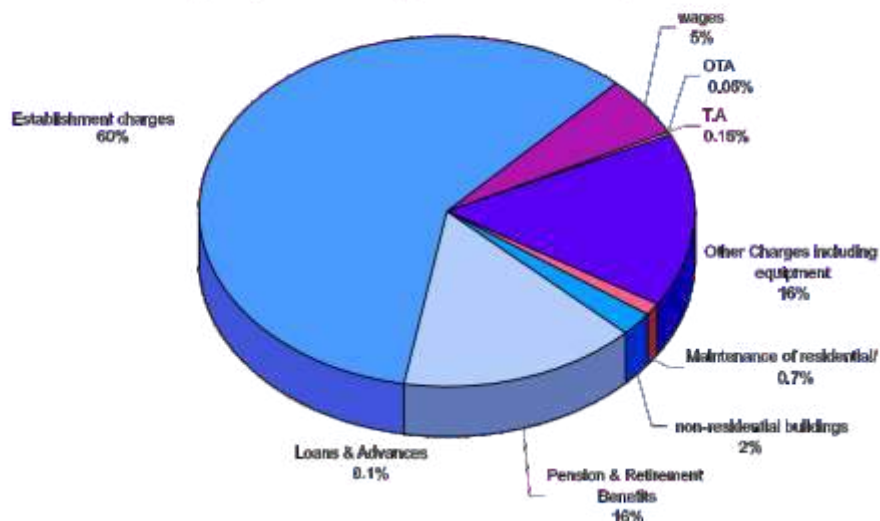
Sl.No.	Programme	Amount (Rs)
1	ICAR-SRF Examination	12,75,500



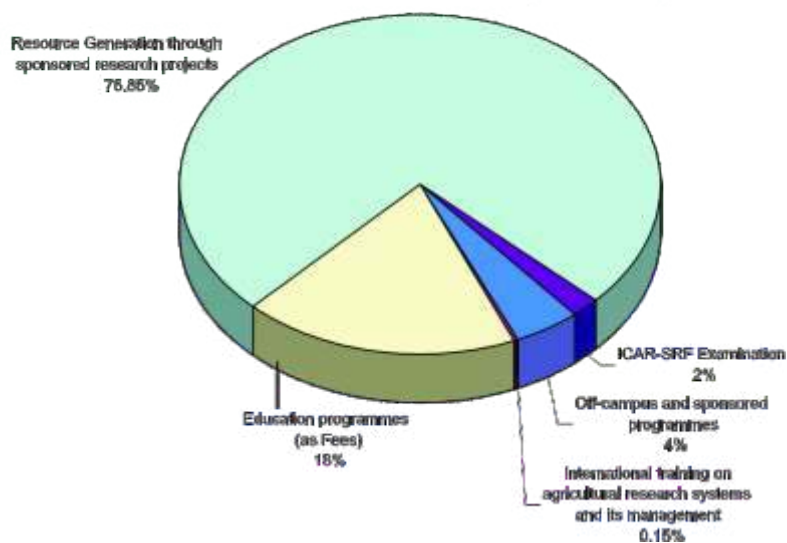
Budget (2009 – 10): Plan Expenditure



Budget (2009 – 10): Non Plan Expenditure



Resource Generation (2009-10)





Capacity Building

Capacity Building

One of the key mandates of the Academy is to organize capacity building programmes for the benefit of research managers, scientists, teachers, technical, administrative and finance personnel in the National Agricultural Research System. The glimpse of various programmes organized at the Academy during the reporting period is given below:

Sl.No.	Type of Programme	No. of Programmes	No. of Participants
1	<i>Foundation Course for Agricultural Research Service</i>	4	279
2	Senior-level Programmes	24	654
3	Workshops	11	363
4	Executive Development Programme (EDP)	1	16
5	International Programmes	2	4
6	Off-campus Programmes	11	265
	Total	53	1581

A. Foundation Course for Agricultural Research Service (FOCARS)

FOCARS, the flagship programme of the Academy, is designed for the newly recruited entry-level scientists in Agricultural Research Service of the ICAR.

Sl.No.	Title	Period	Number of Participants	Programme Directors
1	86 th FOCARS	April 21 to August 18, 2009	65	J. Challa M.N. Reddy
2	87 th FOCARS	June 23 to October 20, 2009	39	S.K. Nanda R.V.S. Rao
3	88 th FOCARS	November 4, 2009 to March 3, 2010	48	G.R.K. Murthy K.M. Reddy
4	89 th FOCARS	December 15, 2009 to April 13, 2010	127	S.K. Soam A. Dhandapani

The four-month programme emphasizes on important and related areas that are crucial for executing research and other functions more effectively in the organization. The training also provides an exposure on the concepts and principles of project management with special emphasis on project formulation and implementation. The programme is intended to emphasize on areas such as agricultural research systems policies and management, human resource management, and information and communication management.





The first phase of the programme stresses on aspects like Indian and global agriculture, agricultural research system, issues of WTO and IP management, research project management, communication management, computer-based information technology, organizational and behavioral issues, administrative and financial rules, and participatory rural appraisal techniques.

During the second phase of the programme, the scientists undergo Field Experience Training (FET) in various centres of ICAR Institutes/Agricultural Universities/KVKs, wherein they interact with farmers, extension agencies, input supplying agencies, scientists, and others to identify the problems and develop interdisciplinary research projects. In the third phase of the programme at the Academy, the trainees focus on sharing of their experiences, and visit various institutions of repute located in-and-around Hyderabad. After the successful completion of the programme, they are posted in different constituent research institutes of Indian Council of Agricultural Research (ICAR).

B. Senior Programmes

During the reporting period, Academy organized a number of programmes for senior-level positions in the NARS.

Sl.No.	Name of the Programme	Duration	No. of participants	Programme Director(s)
1	Developing Winning Research Proposals	June 18-23, 2009 October 22-27, 2009	18 18	J. Challa D. Rama Rao
2	MDP on Managing Public-Private-People Partnerships in Agricultural Research	July 1-7, 2009	22	K.H. Rao G.P. Reddy
3	Geo-spatial Knowledge Management for Sustainable Livelihoods Security (Knowledge Management)	July 1-10, 2009	21	M.N. Reddy N.H. Rao
4	Web-based E-Learning and Content Management Under NAIP (Agroweb)	July 14-24, 2009 November 3-13, 2009	19 18	G.R.K. Murthy K.M. Reddy D. Rama Rao G.R.K. Murthy
5	HRD Training Programme for Programme Coordinators of KVKs in the North East Region	July 25- August 1, 2009	35	D. Rama Rao N. Sandhya Shenoy
6	Multimedia Content Management for the Faculty of Fisheries	August 3-14, 2009 August 20-29, 2009 December 2-11, 2009	26 24 26	K.M. Reddy V.K.J. Rao



Sl.No.	Name of the Programme	Duration	No. of participants	Programme Director(s)
7	Special Module on Agripreneurship: Managing Agricultural Innovation and Enterprise Development	August 24-26, 2009	30	R. Kalpana Sastry
8	Web Design Methodologies and Protocols	September 8-18, 2009	19	N. Sandhya Shenoy M.N. Reddy
9	E-Content Management for Home Science Faculty	September 22-26, 2009	21	G.R.K. Murthy K.M. Reddy
10	Technical and Administrative Support for Consortia-based Research in Agriculture	October 26-31, 2009 February 22-27, 2010	20 24	R.V.S. Rao S.K. Soam and R. Kalpana Sastry
11	Vigilance Administration	October 29 to 31, 2009	35	Ashish Roy
12	IT-based Decision Support Systems (DSS) for Multimedia Content Development	November 17-27, 2009 January 20 -30, 2010 March 17-27, 2010	37	K.M. Reddy V.K.J. Rao
13	MDP in Agricultural Research	December 17-23, 2009	19	J. Challa K.H. Rao
14	Decision Support Systems for Geo-spatial Knowledge Management for Sustainable Livelihoods Security (Knowledge Management)	February 16-26, 2010	21	M.N. Reddy N.H. Rao
15	E-Learning for Dairy Science Faculty	March 2-12, 2010	22	D. Rama Rao

Developing Winning Research Proposals

It is vital for the scientists of agricultural research organizations to develop research projects that can attract external funding for specific research efforts in the light of increasing competition. To develop skills through hands-on experience in writing winning research proposals that can win funds from donors focusing on the needs of the stakeholders and to give practice in writing various components of a research proposal, two training programmes on



Developing Winning Research Proposals were organized under L&CB Component of NAIP. These programmes also laid emphasis on the use of log frame and Project Evaluation and Review Technique (PERT) in research programme planning and enlightened the participants on how to develop a good value chain design and budget estimate that is rationally accepted. Scientists from academic and research institutes, in both public and private sectors of NARS, participated in these training programmes.



MDP on Public-Private Partnerships (PPP) for Innovation in Agriculture

Public-Private Partnerships are increasingly being emphasized as appropriate interventions for strengthening research and innovation in agriculture sector in developing countries. Partnerships between public research institutions, farmer groups, private agribusiness firms and non-government organizations offer an opportunity to tap the strength and core competencies of diverse partners. To discuss how the role of public-private partnerships in food and agriculture can be expanded and sustained to improve the reach, effectiveness, and efficiency of research and innovation for agricultural growth and development, a Management Development Programme on Public-Private Partnerships (PPP) for Innovation in Agriculture was organized. The programme aimed to deliberate on various critical issues for effectively managing PPP with a win-win mindset, by broadly addressing: (i) role and relevance of PPPs in agricultural research and development; (ii) develop criteria for identifying PPP opportunities in agricultural research; and (iii) prioritize key issues in effectively negotiating and managing PPP projects. The programme included imparting knowledge on various economic, mathematical, statistical and econometric techniques. Senior-level scientists, involved in PME activities of approved National Agricultural Innovation Project (NAIP) consortia projects and Nodal officers of PME cells of ICAR institutes and SAUs' attended the programme.

Geospatial Knowledge Management for Sustainable Livelihoods Security



Geographical Information Systems (GIS) provide an effective framework to integrate diverse spatial data sets into decision support systems to define and resolve the complex issues related to sustainable management of agricultural systems. The Sustainable Rural Livelihoods (SRL) framework provides the base to assess rural livelihood systems. To understand the concepts of spatial

database design and management and their applications in agriculture, as well as to integrate the sustainable livelihoods framework and participatory rural GIS for knowledge management and community decision-support, a training programme on Geo-spatial Knowledge Management in Sustainable Rural Livelihoods Security was organized. The programme introduced case studies of GIS-based decision support systems for sustainable livelihoods security, and training modules along with hands-on experience of GIS software and hardware. The programme was organized in four modules: 1) Database and GIS concepts and practice; 2) Remote sensing and image processing concepts and practice; 3) Sustainable livelihoods framework concepts and participatory GIS; and 4) Case studies of geo-spatial knowledge management for sustainable livelihoods security. Scientists and Technical Officers of ICAR, Agricultural Universities, Governmental and Non-governmental organizations attended the training programme.

Web-based E-Learning and Content Management

The processes of learning and knowledge management have acquired new dimensions with the advent of fast developing trends in Information and Communication Technologies



(ICTs). The technological advances in internet and mobile communication can make the benefits of e-Learning multifold. Further, it provides the access to the knowledge to the remotely placed stakeholders of the agriculture through e-Learning in distance mode. To familiarize with web design methodologies using open source software for content development and management in the context of e-learning and to familiarize with content



management software in the web environment, two training programmes on Web based E-Learning and Content Management were organized under NAIP Component 1: Agroweb Digital Dissemination System for Indian Agricultural Research (ADDSIAR) (AGROWEB). These programmes widely covered an introduction to content creation tools, web standards for content and e-Learning; basics of open source software in content management including software installation, course creation and administration, assignments, evaluation, grading and web hosting along with software on on-line learning evaluation and monitoring. Consortia partners of NAIP, scientists, teaching faculty, technical officers and network administrators from ICAR Institutes took part in these training programmes.

HRD Training Programme for Coordinators of KVKs

A HRD training programme for Coordinators of Krishi Vignan Kendras (KVKs) of North East region of India was organized. Dr Managala Rai, DG, ICAR inaugurated the programme. Dr K.D. Kokate, DDG (Ag. Extension), ICAR was one of the resource persons. The programme included theoretical and practical sessions on the topics like ICTs for agriculture, web designing, content creation, etc. The



participants were provided with resource material and a CD containing presentations of the speakers and Open Source Software for reference and use. Thirty five Programme Coordinators of the KVKs from the North East region of India were benefited through this programme.

Multimedia Content Management

Two training programmes on Multimedia Content Management were conducted for the faculty of College of Fisheries, Mangalore, Karnataka, and for the Home Science faculty of the Acharya N.G. Ranga Agricultural University, Andhra Pradesh; University of Agricultural Sciences, Dharwad, Karnataka; and G.B. Pant University of Agriculture and Technology, Pantnagar, Uttarakhand. The faculty members were familiarized with the content management software in the web environment, and were facilitated to develop case studies on content management in project mode with relevance to agriculture education.



Agripreneurship: Managing Agricultural Innovation and Enterprise Development

A training module on 'Agripreneurship: Managing Agricultural Innovation and Enterprise Development' was conducted as a part of the course curriculum for the benefit of students and faculty of PG Diploma in Intellectual Property and Technology Management in Agriculture (PGD-IPTMA). Dr Richard Cahoon, Professor, Cornell University, was the moderator. Scientists working in Information Technology Management Units of the ICAR Institutes located in Hyderabad also got benefited through this programme.

Web Design Methodologies and Protocols

The ICAR institutes' websites need to be developed to have features such as uniformity in content management, format and updating mechanism, facility for content search, retrieval and processing tools for database management for creating a common gateway - a 'one-stop window' for getting access to the information on National Agricultural Research and Education System in India. In order to provide exposure on new generation web technologies and to build capacity in design, development and management of websites, a training programme on Web Design Methodologies and Protocols was organized. The course content included an introduction to new generation web technologies, commercial/open source/ freeware/ shareware web management; systems and Content Management Systems and Learning Management Systems (CMS&LMS); scripting languages and comparison, management of on-line and searchable databases, demo and hands-on practice for development of websites and hosting web based contents. Nineteen participants associated with web development, system administration and management were benefited through this programme.

E-content Management

A training programme on E-content Management was organized with a view to provide an understanding of learning standards, teaching and learning methods; to familiarize with open source software for e-content development and management; and to familiarize the trainees with online evaluation. Faculty members of Home Science College of Acharya N.G. Ranga Agricultural University were trained on e-learning aspects relevant to their domain knowledge. The concepts of content management systems with reference to e-learning, innovative e-content creation tools and learning management systems using open source systems were covered during the programme.

Technical and Administrative Support for Consortia-based Research in Agriculture



The quality of research depends on the quality of technical, administrative and financial support available to the scientists of NARS. Therefore, it is important to sensitize the technical and administrative staff on their changing role in the pluralistic approach. To meet this goal, two training programmes on Technical and Administrative Support for Consortia-based Research in Agriculture



were organized. The broad objectives of these programmes were to sensitize and enhance the understanding of the technical and administrative staff of NARS about the consortia mode of research under NAIP. The programme also aimed to develop personality skills and competency in the participants that would help in enhancing their effectiveness in supporting the activities under NAIP. Administrative Officers, Finance & Accounts Officers and Technical Officers from ICAR institutes attended programme.

Programme on Vigilance Administration

A special training programme on Vigilance Administration was organized for the vigilance officers of the ICAR institutes. The main objective of the programme was to provide an overview of vigilance administration and management, and to update and acquaint the vigilance officers with the latest Central Vigilance Commission guidelines. The programme also aimed at improving the efficiency of vigilance administration and taking appropriate action on anonymous and pseudonymous complaints, and matters related to procurement processes from vigilance angle. Thirty five vigilance officers of ICAR institutes attended this training programme.

IT-based Decision Support Systems for Multimedia Content Development

The advent of worldwide packet networks, optical disks and other mass storage devices, interactive digital video, imaging, digitizing, scanning, and other computer graphic technologies have paved the way for development in multimedia. To develop an understanding and appreciation of use of multimedia in e-learning and knowledge management, three training programmes on IT-based Decision



Support Systems for Multimedia Content Development was organized at the Academy. The programme also exposed the participants on digitizing, synthesizing, animating and packing images, sound, graphics, videos and text for both web and CD based presentations, and familiarized them with multimedia production tools in various categories and enabled them to develop their own modules for ICDS (Information Communication and Dissemination Systems) for participatory knowledge management. The programme consisted of interactive lectures, discussions, demonstrations and extensive hands on exercises leading to the development of a multimedia module for web / CD based presentations. Scientists, teaching faculty, research managers, and technical officers from NAIP Consortia, ICAR institutes, Agricultural Universities participated in these training programmes.

Management Development Programme in Agricultural Research

The 13th Management Development Programme in Agricultural Research was organized to develop middle level management competencies in the organizational set-up of research institutions; to apprise on issues of administration and financial management and also to



sensitize on the behavioural issues of people at work. The programme was structured around four thematic areas: (i) research management; (ii) human resource management; (iii) administration and financial management; and (iv) information and communication technology. Heads of Departments and Programme/Zonal Coordinators of ICAR, Heads of Departments and Associate Directors of Research of SAUs, numbering 19, took part in this programme.

Decision Support Systems for Geospatial Knowledge Management for Sustainable Livelihoods Security



Geographical Information Systems (GIS) is a tool to integrate diverse spatial data sets into decision support systems to define and resolve the complex issues related to sustainable management of agricultural systems. The Sustainable Rural Livelihoods (SRL) Framework provides the base to assess rural livelihood systems. This training programme on Geospatial Knowledge Management for Sustainable Livelihoods Security was designed with the

twin goals of understanding concepts of GIS and other spatial technologies as well as the SRL framework, and the integration of these concepts into decision support systems for sustainable rural livelihoods security. This training programme was organized under Learning and Capacity Building project of NAIP. The course was organized in 4 modules: (i) database and GIS concepts and practices; (ii) remote sensing and image processing concepts and practices; (iii) sustainable livelihoods framework concepts and participatory GIS; and (iv) case studies of geospatial knowledge management for sustainable livelihoods security. The hands-on practice was given on GIS and image processing software tools like ArcGIS, ArcView, IDRISI, etc. Scientists, faculty of ICAR, SAUs, governmental and non-governmental organization partners of NAIP consortia partners attended this training programme.

E-learning

To bring awareness in E-learning and to build the capacity in content creation and management in teaching, research, and technology transfer activities, a training programme on E-learning was organized for dairy science faculty members. The programme introduced content creation tools, web standards for content and e-learning; and also basics of open source software in content management - software installation, course creation and administration, assignments, evaluation, grading and web hosting, and also software on on-line learning evaluation and monitoring. The methodology of the training followed was a blend of lecture sessions, demonstrations and hands-on practical exercises. Twenty two faculty members belonging to National Dairy Research Institute, Karnal, Haryana; Sri Venkateswara Veterinary University, Tirupati, Andhra Pradesh; and Assam Agricultural University, Jorhat, Assam attended this training programme.



C. Workshops

Sl.No.	Name of the Workshop	Duration	No. of Participants	Coordinator(s)
1	Launch workshops on Assessment of Future Human Capital Requirements in Agriculture	May 8, 2009 and November 11, 2009	57	D Rama rao SK Nanda
2	Brainstorming workshop on Post Graduate Diploma in Management (Agriculture)	May 25, 2009	20	GP Reddy
3	Workshop on Patent Drafting and Technical Skills	June 26-27, 2009	60	R Kalpana Sastry
4	Policy and Prioritization, Monitoring and Evaluation (PME) Support to consortia-based Research Projects in Agriculture	August 31- September 05, 2009 October 5-9, 2009	15 19	SK Nanda SK Soam BS Sontakki GP Reddy
5	Advanced Workshop on Patent Drafting and Technical Skills (NAARM & Confederation of Indian Industry (CII)- AP Technology Development & Promotion Center	October 5-6, 2009	58	R Kalpana Sastry Subodh Kumar (APTDC-IPR)
6	Brainstorming Workshop on Agribusiness Education	October 22, 2009	37	GP Reddy KH Rao
7	Workshop on Management of Stress-related Disorders	November 5-6, 2009	7	A Debnath
8	Workshop on Leadership for Transition to NAIS	November 17-21, 2009	24	P Manikandan RVS Rao
9	Sensitization Workshop on Networking and Web Hosting	February 2-3, 2010	45	GRK Murthy
10	National Workshop on Revitalizing Higher Agricultural Education in India – Issues and Challenges	February 24-25, 2010	40	J Challa
11	Interactive Dialogue-cum-Workshop on Technology Management and Commercialization in Agriculture	March 4, 2010	26	SK Soam R Kalpana Sastry



Workshops on Assessment of Future Human Capital Requirements in Agriculture



NAARM, in association with Institute of Applied Manpower Research (IAMR), New Delhi, is carrying out the project “Assessment of future human capital requirements in agriculture” under component-1 of NAIP. The project was launched in May, 2009. Two workshops were organized to identify the broad issues and parameters for technology forecasting and to fine-tune the manpower planning methodology. Accordingly, a concept

paper was developed giving a brief review on the manpower forecasting methodology and also a framework for use in the current study. Five questionnaires were developed for data collection from individuals, self-employers, universities, institutions, tracer studies and other stakeholders. A brochure was also prepared and circulated to all stakeholders. Project information is made available online through project website. Initiated collaboration with State Agricultural Universities (SAUs), established association with 32 universities and received data from 10 SAUs. A survey plan was developed in dialogue with stakeholders from concerned organizations. A nation-wide survey is in progress in 27 states to assess supply of trained graduates, current and changing employment scenarios and manpower structure and changing economic environment (including sector growth). Issues for focus group discussions were finalized in consultation with experts. NAARM has organized 25 Focus Group Discussions in various states so far.

Brainstorming Workshop on Post Graduate Diploma in Management (Agriculture)

A brainstorming session on Post Graduate Diploma in Management (Agriculture) was organized at the Academy to discuss and finalize the curriculum for the same. Twenty eminent personalities representing academia and industries participated in the workshop to develop curriculum. This programme was scheduled to be launched in the Academic year 2009-10, in the third year of the XI Plan. The main outcomes of the workshop were: (i) remodel the course structure keeping in view with clients requirements and workload to the students, with maximum credit hours restricted to 100 hours; (ii) restructure the course curriculum restricting number of credit hours to maximum 18 credits per trimester; and (iii) restricted number of courses maximum to seven per trimester.



NAARM-APTDC IP Capacity Workshop on Patent Drafting and Technical Skills

IP Capacity Workshop on Patent Drafting and Technical Skills was organized with Joseph Earnest Root, Patent Expert, USA as main resource person. The workshop objectives were: (i) Improving Patent Drafting Skills; (ii) understanding what



drives the framing of critical drafting; (iii) acquiring theoretical & practical skills related to patent application procedures and developing claim drafting skills; and (iv) core skills on scope of patent, application procedures, different types of claims, drafting, in compliance with procedural requirements. The themes deliberated upon included writing for patents; specification drafting; claim interpretation; patent reform; navigating the minefield, etc.

MDP Workshop on Policy and Prioritization, Monitoring and Evaluation Support to Consortia-based Research in Agriculture

Prioritization, Monitoring and Evaluation (PME) efforts are critical issues in management of agricultural research and development. In times of fiscal constraint, more emphasis is placed on accountability and programme effectiveness. PME is an integrated management approach that can be used to validate programme existence and adjust research agenda for enhanced effectiveness especially in case of public agricultural research system. Two MDP



Workshops on Policy and Prioritization, Monitoring and Evaluation (PME) Support to Consortia-based Research in Agriculture were organized at the Academy from August 31 to September 05, 2009 and from October 5 to 9, 2009. The main objectives of these workshops were to (i) sensitize the participants to the O&M reforms for consortia-based research policy, priority setting, monitoring and evaluation as envisaged in NAIP; (ii) orient the participants to the concepts and description of PME of agricultural R&D; (iii) equip them with major techniques of PME in different areas of agricultural research and development; (iv) discuss empirical studies and to synthesize experiences in application of PME techniques; and (v) bring together the viewpoints of researcher, administrator and donor to make PME effective and rewarding. These workshops provided an excellent opportunity to know theoretical logic and empirical application of various PME techniques and their analysis along with discussion on emerging issues to scientists involved in PME activities of approved NAIP Consortia projects and In-charges of PME cells of ICAR Institutes and SAUs who took part in the programmes.

Advanced Workshop on Patent Drafting & Technical Skills

A workshop on Advanced Patent Drafting & Technical Skills was organized with an objective to enhance drafting of patent applications for USPTO. The workshop also dealt with prior art search, patent search, case studies, etc. The workshop was jointly organized by Confederation of Indian Industry (CII) and AP Technology Development & Promotion Center (APTDC) with NAARM. Scientists, researchers, leading lawyers, technology managers, inventors, IP attorneys, SME's, private and corporate IP practitioners and students of IPTMA courses of NAARM participated in this workshop.



Brainstorming Workshop on Agribusiness Education

A Brainstorming Workshop on Agribusiness Education was organized in association with Ernst & Young Private Limited to obtain vital inputs for designing an effective post-graduate programme and curriculum in agricultural management. The participants included invitees from agribusiness companies, academic institutions and others. Senior Officials from the following companies/institutions participated in the Workshop:

- ◆ Adayana
- ◆ Barwale Foundation
- ◆ Basix Microfinance
- ◆ Competent Agri Solutions Pvt Limited
- ◆ Creamline Dairy
- ◆ Heritage Foods India Limited
- ◆ Indian farmers Fertilizer Cooperative Limited
- ◆ Indian School of Business, Hyderabad
- ◆ International institute of Information Technology
- ◆ ITC-Agribusiness division
- ◆ JK Agrigenetics
- ◆ Nandan Biometrics
- ◆ National Dairy Development Board
- ◆ Reliance Retail
- ◆ Tilda Riceland
- ◆ YES bank



The main purpose for the brainstorming workshop was on (i) highlight key emerging issues and opportunities in agribusiness which are of relevance to agribusiness education; (ii) talent gaps in the agricultural value chain; (iii) list existing and potential needs of the industry which a post graduate programme can address. The workshop also addressed the niche areas for NAARM in view of its strength. The workshop also deliberated on (i) ideal

balance of the course structure; (ii) admission criteria and process; and (iii) possible internships and career opportunities for agribusiness graduates.

Recommendations:

- (I) Increasing growth and changing composition within the industry and the economy, new legislations, and changing customer demands point to increased need for talented manpower – agribusiness managers with holistic perspective and problem solving abilities.
- (ii) In changing circumstances, there is a need to develop a strategic, interdisciplinary and problem solving mindset among agribusiness graduates.
- (iii) Curriculum and pedagogy must be contextualized to meet the competency requirements. The curriculum must balance general management and agribusiness specialization courses.



- (iv) Increase industry exposure of faculty through increased interactions, consultancies and sponsored research
- (v) To become a leading agricultural management institute, NAARM must ensure a stream of high quality student entrants and faculty. Further it must strive to ensure that agribusiness management is branded as career of choice. Publicity through brochures and stakeholder meets must be done in the right manner and at regular intervals both to prospective students as well as to future recruiters.
- (vi) Placements should be faculty run, with minimum interference by students.
- (vii) NAARM needs to partner with the industry through greater faculty interactions, and by extending invitations to organizations to participate in both student and institutional activities

Workshop on Management of Stress-related Disorders

Stress is the root cause for many diseases like hypertension, diabetes, cardiac, musculo-skeletal and other psycho-somatic diseases. A workshop on Management of Stress-related Disorders was organized to develop in-depth understanding of the forces causing stress and its effects – physical, social and organizational; to understand the various stress related disorders, and to develop constructive means for coping up with stress and achieve productive and enriched life-style. Stress-releasing exercises were demonstrated and were practiced during the workshop. Senior managers like Directors, Joint Directors, Heads of Departments, Zonal Coordinators and Project Coordinators, CAOs and FAOs, and Principal Scientists of ICAR institutes; Heads of Departments and Associate Directors of Research, Registrars and Comptrollers of SAUs got benefited in this workshop.

Workshop on Leadership for Transition to National Agricultural Innovation System

National Agricultural Innovation Project (NAIP) has introduced “consortia-based research” approach, which demands involvement of all the stack holders in the food supply chain. It is important to provide opportunities for all consortia partners of NAIP to explore the leadership issues and the needed leadership skills to facilitate the success of transition of NARS to National Agricultural Innovation Systems (NAIS). Hence, a workshop on “Leadership for Transition to NAIS” was organized. The workshop aimed at exploring the concept of leadership and personal effectiveness with a view to adapt ideas of leadership for successful management of change and innovation. The workshop consisted of a blend of lectures, self-exploration exercises, case analysis, experiential learning, and group discussion. The workshop had benefited 24 senior-level scientists and teachers from ICAR institutes, Agricultural Universities and NGOs.



Workshop on Sensitization on Networking and Web Hosting



A workshop on “Sensitization on Networking and Web Hosting” was organized to (i) address the issues pertaining to websites of ICAR institutes; (ii) understand the mechanism of portal and website integration; and (iii) identify the reusable and replicable resources developed as a part of the project so that they can be shared by participating institutes. It is worthy to note that websites of most of the institutes were

developed by adhering to uniformity guidelines developed earlier and by using CMS technologies. Through this workshop, participants were sensitized on various web-related issues, CMS and Portal Integration mechanism. Inputs from industry and various government agencies were taken to acquaint the participants about the web-related issues. Forty-five participants attended this workshop.

Workshop on Revitalizing Higher Agricultural Education in India – Issues & Challenges

A significant change is likely to take place in the field of higher agricultural education in India. To deliberate and discuss the present trends and future challenges in agricultural education, a National Workshop on Revitalizing Higher Agricultural Education in India – Issues and Challenges was organized. Dr Arvind Kumar, DDG (Education), ICAR, was the Chief Guest during the inaugural ceremony. The workshop deliberated on four theme areas:

- Constraints and limitations of the present agricultural education system in India
- Successful models of agricultural education system in India vis-à-vis US pattern and other developed countries
- Center-State relationship and Organization and Management issues faced by the Agricultural universities
- Networking and public-private partnership in education

The delegates deliberated on the four theme areas and recommended number of recommendations. Forty renowned educationists and academicians, Vice Chancellors, Deans and Senior Professors of agricultural universities, policy advisors attended the workshop.

Workshop on Technology Management and Commercialization in Agriculture

An interactive Dialogue-cum-Workshop on Technology Management and Commercialization in Agriculture was organized at the Academy. Dr Keith Jones, Executive Director, Washington State University Research Foundation, Pullman, USA, was the Chief Resource Person, who addressed the audience on technology management and commercialization in Washington State University. The objectives of the workshop were:



(I) share the knowledge about technology commercialization in USA; (ii) discuss the implications of agricultural technology management in Indian NARS with special focus on IP strategies in public-private partnership; and (iii) document the areas of studies to be taken up by the NAARM for further in depth analysis. The major recommendations emerged from the workshop were:



- (i) Valuation of technologies and models for benefit sharing to be taken-up as one of the priority areas. NAARM would conduct a workshop on this topic with some relevant partners;
- (ii) Appropriate methods are to be evolved to share breeding lines with private sector. The ICRISAT consortia model needs to be studied with respect to its feasibility for wider application in NARS;
- (iii) Evolve mechanisms for sharing genetic stocks and registration of breeds in animal sector;
- (iv) Develop a product franchise with proper costing and complete transparency and need to document successful business models which are functioning in several organizations.

D. Executive Development Programme

Sl.No.	Name of the Programme	Duration	No. of Participants	Course Director
1	Agricultural Research Management	August 21-25, 2009	16	J Challa

Executive Development Programme on Agricultural Research Management

As a part of the Academy's endeavour towards promoting learning and capacity building of the NARS, 11th Executive Development Programme in Agricultural Research Management was organized for the benefit of senior personnel in Research Management Position such as Directors and Joint Directors of ICAR Institutes and Deans and Directors of Agricultural Universities. The Programme was designed for conceptual awareness and competency building in the theme areas of scientific endeavours and institute management, administration and finance, institution building, and legislative acts. The programme aided the delegates to develop a visionary approach towards perspective planning for their research institutes. The programme was structured into interactive sessions by in-house and guest faculty, case analysis, and presentations. Sixteen Directors of ICAR institutes, Assistant Directors General of ICAR, Deans and Directors of agricultural universities took part in the programme.



E. International Training programmes

Sl.No.	Name of the Programme	Duration	Participants	Course Director
1	International Training Programme for Nigerian Scientists	April 13-25, 2009	2	T Balaguru
2	Observation Visit of officials from NARC, Nepal	September 8-10, 2009	2	BS Sontakki

International Training Programme on Research Project and Agricultural Research System and its Management



An International Training Programme on Research Project and Agricultural Research System and its Management was organized for senior research managers of the Nigerian Institute of Oil Palm Research, Nigeria. The participants were sensitized to important issues like agricultural research project management, technological forecasting, public-private partnership in agricultural research, research prioritization, monitoring and

evaluation, impact assessment of agricultural research, and inter-disciplinary team building in research organizations. The participants also visited the national and international agricultural research institutions such as Directorate of Rice Research, Directorate of Oilseeds Research, and Project Directorate on Sorghum, and International Crop Research Institute for the Semi-Arid Tropics (ICRISAT).

Observation Visit of Officials from Nepal Agricultural Research Council, Nepal

Mr Parashuram Lal Karna, Director, Financial Administration and Mr. Narendra Lakhe, Administrative Officer from the Nepal Agricultural Research Council (NARC), Nepal, visited the Academy, and had detailed interaction and discussion on issues related to administrative and financial management in ICAR. The participants also attended the workshop on administrative and financial management.



F. Off-campus Training Programmes

Sl.No.	Name of the Programme	Duration	No. of Participants	Course Director(s)
1	Specialized Short Programme on Improving Efficiency of Scientific Personnel at IGFR, Jhansi	April 20-23, 2009	25	P. Manikandan R.V.S. Rao
2	Specialized Short Programme on Improving Efficiency of Technical Personnel at IGFR, Jhansi	April 20-23, 2009	25	P. Manikandan R.V.S. Rao
3	Training Programme for Scientific and Technical Manpower of CIFRI Barrackpore, Kolkata	April 29 to May 1, 2009	37	B.S. Sontakki M. Suresh Kumar
4	Training Programme on E-Content management at Fisheries College & Research Institute, Thoothukudi; TN College of Fisheries, Mangalore, Karnataka; College of Horticulture, Mudigere, Karnataka	April 29- May 6, 2009	26	G.R.K. Murthy K.M. Reddy
		May 30–June 4, 2009	28	
		October 23-28, 2009	30	
5	Training Programme for Administrative and Accounts Personnel of CIFRI, Barrackpore, Kolkata	May 2-5, 2009	39	M. Suresh Kumar
6	Special Training Programme on Vigilance Administration and Management for Vigilance Officers of ICAR Institutes Organized at Bangalore	May 25-27, 2009	27	Ajay Kumar, Director DARE M. Suresh Kumar
7	Training Programme on Improving Efficiency of Administrative Personnel of DRR & PDP, Hyderabad Organized at DRR, Hyderabad.	June 10-15, 2009	31	M. Suresh Kumar K.H. Rao
8	Training Programme for Administrative and Accounts Personnel of ICAR Research Complex, Goa	June 23-25, 2009	26	M. Suresh Kumar Bilgrami, SFAO, IIHR Bangalore
9	Training Programme on Multimedia Content Development (Sponsored by College of Fisheries & Research Institute, Thoothukudi, TN)	June 26-July 7, 2009	28	K.M. Reddy V.K.J. Rao
10	Training Programme on Improving efficiency of Technical Officers of IARI, New Delhi	March 25 – 27, 2010	25	P. Manikandan K.H. Rao



Specialized Short Programme on Improving Efficiency of Scientific Personnel

An off-campus specialized short programme on improving efficiency of scientific personnel at IGFRI was organized. The programme was specifically organized to orient the scientific personnel to the various issues related to project and personnel with a view to enhance their efficiency and effectiveness.

Specialized Short Programme on Improving Efficiency of Technical Personnel

A programme for technical personnel at IGFRI, Jhansi, was organized. The major purpose of this programme was to orient the technical personnel to diverse issues that would help enhance their human relations at work and their efficiency. The programme provided an opportunity for the participants to understand the various dimensions of human relations at work through structured experiences and experiential learning.

Training Programme for Scientific and Technical Manpower

One of the most critical problems faced by the research organizations is the development of general management capability for dealing with the demands of rapidly changing external and internal working environment. To improve skills and efficiency of the scientists, technical personnel in discharging their functions and providing constructive research and administrative support to their superiors in meeting the organizational priorities and needs, an off-campus training programme for scientific and technical staff was organized at the Central Inland Fisheries Research Institute (CIFRI), Barrackpore. The course aimed to strengthen and familiarize with latest developments, procedures and Right-to-Information Act 2005. Along with NAARM faculty, eminent speakers from renowned institutions were drawn as resource persons to this programme.

Training Programme on E-content Management

Three off-campus programmes on E-content Management were organized at Fisheries College and Research Institute (TNVASU), Thoothukudi, and College of Fisheries, Managalore, and College of Horticulture, Mudigere, Karnataka. These three programmes related to e-learning and content creation were conducted to (i) understand learning standards, teaching and learning methods; (ii) familiarize with open source software for content development and management for e-learning; and (iii) familiarize the trainees with online evaluation. Faculty members of these colleges were trained on e-learning aspects pertaining to their domain knowledge. The concepts of content management systems with reference to e-learning, innovative e-content creation tools and learning management systems using open source systems were covered during the training programme.

Training Programme for Administrative and Accounts Personnel

An off-campus training programme for administrative personnel was conducted at the Central Inland Fisheries Research Institute (CIFRI), Barrackpore. The main objective of this course was to strengthen and familiarize with latest developments and procedures. This programme aimed at improving skills and efficiency of the administrative personnel in discharging their functions and providing constructive research and administrative support to their superiors in meeting the organizational priorities and needs. The programme was exclusively designed for CIFRI, in which thirty six Administrative, Accounts and Finance Assistants, and other administrative personnel participated.



Special Training Programme on Vigilance Administration and Management

Vigilance is not simply a police function but a key element of management of human conduct. To familiarize the Vigilance Officers of research institutes with vigilance administration and disciplinary matters of Government of India *mutatis-mutandis* applicable to ICAR, an Off-Campus Special Training Programme on Vigilance Administration and Management was organized at National Institute of Animal Nutrition and Physiology, Aduodi, Bangalore. The main objective of the programme was to provide an overview of vigilance administration and management and to update and acquaint the vigilance officers of the Institutes with the latest Central Vigilance Commission guidelines. This programme also aimed at improving the efficiency of vigilance administration and taking appropriate action on anonymous and pseudonymous complaints and matters related to procurement processes from vigilance angle. Vigilance Officers and Presenting Officers of ICAR Institutes located in the states of West Bengal, Orissa, North Eastern regions, Andaman & Nicobar Group of Islands, Goa and Southern States got benefited through this training programme.

Training Programme on Improving Efficiency of Administrative Personnel

An off-campus training programme on Improving Efficiency of Administrative Personnel was organized at the Directorate of Rice Research to orient the participants to different management skills and work simplification in discharging their functions and to provide constructive services in meeting the organizational priorities and needs and also to strengthen the existing administrative and financial management processes in ICAR institutes. The programme broadly covered stress management, motivation and interpersonal relationships, systematic approach to do work, financial management and budgetary analysis, office automation, pay fixation, deputation, foreign service matters, conduct rules and disciplinary matters, Right-to-Information Act 2005. Thirty one administrative and accounts personnel of Directorate of Rice Research and Project Directorate on Poultry underwent the training.



Training Programme for Administrative and Accounts Personnel of ICAR Research Complex, Goa

A training programme for scientists, administrative and accounts personnel of ICAR Research Complex, Goa, was organized to orient them to different management skills and work simplification in discharging their functions to provide constructive services to their superiors in meeting the organizational priorities and needs and also to strengthen the existing administrative and financial management processes in ICAR institutes through continuous training programmes on modern management techniques. Thirty staff members of the institute got benefited by this programme.

Training Programme on Multimedia Content Development

A training programme on Multimedia Content Development was organized at College of Fisheries & Research Institute, Thoothukudi, Tamil Nadu. This programme related to





multimedia content creation was conducted to understand learning standards, teaching and learning methods; to familiarize with open source software for content development and management for e-learning, and to familiarize the trainees with online evaluation. A total of 54 faculty members of fisheries from both the campuses were trained on e-learning aspects pertaining and relevant to their domain knowledge. The concepts of content management systems

with reference to e-learning, innovative e-content creation tools and learning management systems using open source systems were covered during the training programme.

Training Programme on Improving Efficiency of Technical Officers

Keeping in view the specific requirements of Technical officers of the Indian Agricultural Research Institute, an off-campus programme was organized specifically for them. The programme was organized with the specific purpose of providing an opportunity for the technical officers to undergo a capacity building programme that would help them enhance their efficiency and effectiveness. The programme oriented the technical officers to the importance of healthy human relationships at work place, with a focus on team work, positive personality development and a healthy work orientation. Various issues related to human relations management were presented through structured experiences to make the learning relevant and meaningful for the participants. The programme benefited 25 senior-level Technical Officers of Indian Agricultural Research Institute.





Research

Research

Research Projects

Research in the Academy are conducted in the following three Divisions:

- (I) **Agricultural Systems Management and Policies:** The component research themes include: Agricultural scenario and policy analysis; Agribusiness management; Research project management; Research prioritization, monitoring, evaluation and impact assessment; Identifying new policy initiatives for productivity enhancement of NARS; analysis of agri-food supply chains; commodity trading, policy studies on agricultural production-consumption systems and sustainable rural livelihoods systems; Technology forecasting and management; Intellectual property management; Agrobiodiversity management.
- (ii) **Information and Communication Management:** The component research themes include: ICTs in agri-supply chains; ICT policy for NARS; Information Technology based decision support systems; digital multimedia resources for agricultural development; Geographical Information Systems in agricultural management; ICT applications for village knowledge centres, Development of digital multimedia resources for agriculture; geospatial knowledge discovery and management; Participatory technology development and transfer; designing e-learning systems and Distance training.
- (iii) **Human Resource Management:** The component research themes include: HRD Strategies for NARS;. Impact assessment of NAARM training; Performance appraisal and accountability in agricultural research and education; Organizational culture; Evolving systems for HRD; and Educational Technologies for enhancing learning, manpower planning in agriculture.

Besides the funds available from the XI Plan allocations for research, the faculty of the Academy have obtained significant funds as extra mural competitive grants. The extramural funding for the period 2008-12 presently stands at above Rs 55 crores. The research programmes are broadly divided into following five areas:

- Agricultural science and technology policy
- Accelerating agricultural innovations through ICTs and institutional change
- Organization and management for strengthening agricultural research
- Agri-marketing and value chain management
- Governance and institutional arrangements in areas of topical issues

The detailed list of all the research projects in operation during 2009-10 along with the funding agencies is given in the Annexure -1.



NAARM

Agricultural Science and Technology Policy

Assessment of Developments in Nanotechnology for Agricultural R&D

N.H. Rao and R. Kalpana Sastry

Objectives:

- ❖ To develop a framework and process based methodology for nanotechnology assessment to facilitate its integration into agricultural research domain
- ❖ To design and develop bibliographic and patents databases of nanotechnology research, to map nanoresearch areas to the agricultural supply chain
- ❖ To engage, share, and facilitate stakeholders of NARS in developing a new nanoscience and technology research agenda to enhance, complement and integrate with current agricultural research
- ❖ To assess implications for institutional policies, society and environment, to facilitate technology transfer

Progress:

Case studies on nanotechnology and food security, food delivery systems and nano-biotechnology were developed. Specially designed database model was used to organize information from R&D indicators in nanotechnology. Knowledge mapping concepts were applied and three case studies were developed: (i) for food security; (ii) for disease diagnosis and (iii) disease treatment. This has resulted in R&D projects being initiated in the system. It is also realized that studies on ethical, environmental and risk assessment of nanotechnology in agriculture needs to be encouraged in NARS.

Knowledge Discovery and Knowledge Management Tools to Characterize Livelihoods Systems

M.N. Reddy, N.H. Rao, G.P. Reddy, K.H. Rao and K.V. Kumar

Objective:

- ❖ To develop knowledge discovery and knowledge management tools to characterize livelihoods systems

Specific Objectives:

A study was undertaken to develop an objective method for characterizing vulnerability of livelihoods systems based on application of data mining techniques and the SRL framework for characterizing livelihoods systems. The specific objectives of the study were:

- ❖ To develop an objective method of analysis to segment the study region into homogeneous clusters with respect to the five types of livelihoods assets of the SRL framework: natural, social, human, physical and financial
- ❖ To develop an index to characterize the vulnerability of the resultant clusters



Progress:

A case study based approach was adopted to illustrate the development and application of the framework and analytical procedure. The data of 59 blocks of Nalgonda district was used for the case study. The framework is scalable and can be adapted at village level or larger district and state levels.

Five major livelihoods assets considered in this study were natural capital, human capital, financial capital, physical capital, and social capital. Each of these assets is associated with many attributes. Since each attribute is measured on a different scale and they are of different units, it is necessary to standardize each attribute into an index. The following commonly used normalization method in data mining is adopted for such normalization:

$$V_{ij} = \frac{x_{ij} - \text{Min}(x_{ij})}{\text{Max}(x_{ij}) - \text{Min}(x_{ij})}$$

$$i = 1, 2, \dots, n$$

$$j = 1, 2, \dots, m$$

Where, x_{ij} represents the value of the j^{th} attribute in the i^{th} administrative block

V_{ij} is normalized attribute and $0 \leq V_{ij} \leq 1$ for all i and j

$\text{Max}(x_{ij})$ is the maximum value of the j^{th} attribute in the i^{th} block.

$\text{Min}(x_{ij})$ is the minimum value of the j^{th} attribute in the i^{th} block

The EM algorithm

The EM (Expectation Maximization) algorithm, a popular model based clustering method extends the K-means paradigm in a different way. The K-means algorithm assigns each object to cluster based on distance criteria, whereas in EM each object is assigned to each cluster according to a weight representing its probability membership. In other words there are no strict boundaries between clusters. The EM algorithm is a popular iterative refinement algorithm that can be used for finding the parameter estimates. It can be viewed as an extension to K-means algorithm. EM starts with an initial estimate or “guess” of the parameters of the mixture model. It iteratively rescores the objects against mixture density produced by the parameter vector. The recorded objects are then used to update the parameter estimates. Each object is assigned a probability that would possess a certain set of attributes values given that it was a member of a given cluster.

Vulnerability Index

Assessments of Vulnerability have been carried out in a variety of contexts, such as vulnerability of livelihoods with respect to climate change and variability], the world food programme's vulnerability analysis and mapping for targeting food aid (World food programme, http://en.wikipedia.org/wiki/World_food_Programme). The common procedure in these analysis is to quantify multidimensional issues into some kind of indicators and then combine into a composite index allowing these diverse variables to be integrated. For example the Human Development Index, incorporates life expectancy, health, education and standard of living indicators for an overall picture of national well being (UNDP,

http://en.wikipedia.org/wiki/Human_Development_Index)



Let n_k be the number of administrative blocks belonging to k^{th} cluster of say s^{th} livelihoods asset (s = social, natural, physical, human and financial in this case).

$$\sum n_k = n \quad \text{for a given livelihoods asset say, } s$$

Where, n is the total number of administrative blocks.

Then the vulnerability index for the k^{th} cluster for a given asset is the average of all the normalized values, V_{ij} associated with m_s attributes of s^{th} asset, and can be represented as follows:

$$V_i(s) = \frac{1}{n_k} \frac{1}{m_s} \sum \sum V_{ij} \quad i=1, 2, \dots, n_k; \text{ and } j=1, 2, \dots, m_s$$

Where, $V_i(s)$ is the vulnerability index for the k^{th} cluster of s^{th} livelihoods asset,

n_k is the number of blocks associated with k^{th} cluster of s^{th} asset, and

m_s is the number of administrative blocks associated with s^{th} asset.

Based on the magnitude of values of the index VI , clusters of each livelihoods asset can be ordered according to their vulnerability. If the direction of the attributes are positively associated with increasing degree of vulnerability, then the greater the value of $VI_k(s)$, the greater is the degree of vulnerability of that cluster.

Case Study

Nalgonda district of Andhra Pradesh is selected for this case study. This district is one of the 100 districts identified by the Government of India as backward and vulnerable. The district comprises 59 administrative blocks (mandals). The source of the data for this case study is "Hand book of Statistics" Nalgonda district of Andhra Pradesh, India (Compiled and published by the Chief Planning Officer, Nalgonda, 2005-2006).

The number of clusters for the five types of assets varied from two to four. The minimum number of clusters (2) is associated with the physical and financial assets. Maximum numbers of clusters (4) are associated with the social assets; the range in variability of vulnerability index (Table 1) among the four clusters associated with social asset is large (0.501) compared to other assets. The range is between 0.22 to 0.27 for the natural, financial and human assets and least for the physical assets. Apart from ordering of clusters according to vulnerability based on the vulnerability index, the value of the vulnerability index and the range of vulnerability index among the clusters for each asset are important in decision-making. For example, in the case of natural assets, soil quality and livestock density are relatively more vulnerable compared to the other four attributes. There are nine blocks, which are relatively more vulnerable with respect to natural assets and among the associated attributes, soil quality and livestock density is critical.

Table 1: Number of clusters, value of likelihood function, vulnerability index and number of blocks for each cluster



S.No	Livelihoods Asset	Number of Clusters	Value of likelihood function	Vulnerability index	Number of blocks (%)
1	Human	3	1.3964	0.456	47
				0.342	31
				0.186	22
2	Natural	3	2.6651	0.744	15
				0.590	71
				0.523	14
3	Financial	2	0.3616	0.414	68
				0.184	32
4	Physical	2	3.1611	0.628	42
				0.516	58
5	Social	4	2.1622	0.706	14
				0.364	25
				0.349	34
				0.205	27

With respect to social assets, eight blocks are characterized as highly vulnerable and all the four attributes associated with this asset, population density, average land holding, agricultural labor and workers are equally critical (Fig.1). About 35 blocks are relatively more vulnerable compared to the remaining 24 blocks in the case of physical assets. In this case, the attributes related to communication, education and health are relatively more critical compared other attributes. In this study there are only two attributes associated with each of the financial and human assets. Both the financial attributes, average income and loans are equally important and female literacy is more critical attribute for human assets.

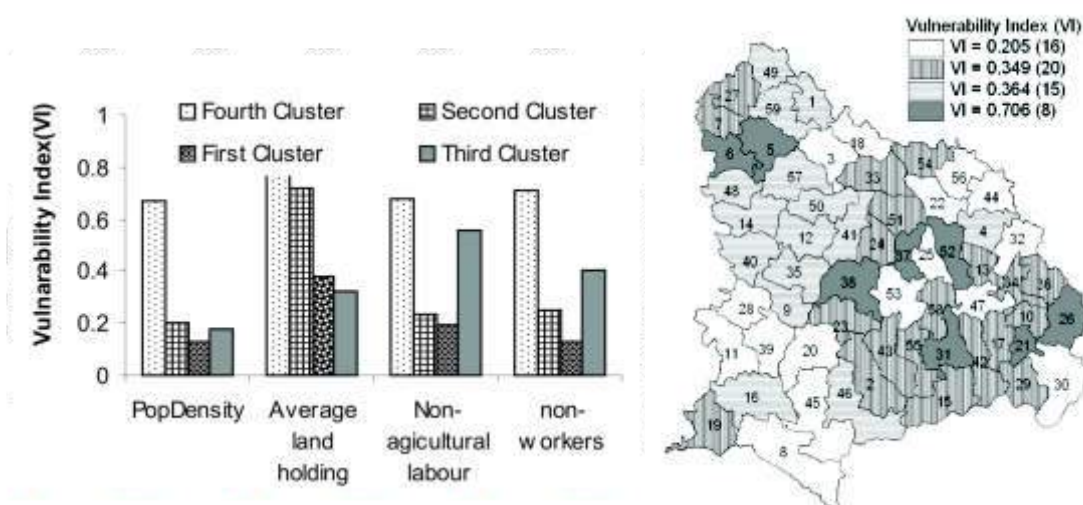


Fig.1 Characterization of vulnerability of livelihoods with respect to social asset of administrative blocks (mandal) in Nalgonda district, Andhra Pradesh in 2006

The method of analysis proposed is an objective method to prioritize developmental plans based on the vulnerable clusters. Further, this analysis provides information on relative importance of various attributes associated with vulnerability of each livelihoods asset and can lead to more need based decisions by scientists and policy makers. While the study identified clustered in which the administrative blocks are geographically contiguous, there may be spatial dependencies within a cluster. Hence extending the cluster methods by incorporating spatial dependencies in the covariance structure may result in statistically better results. One of the outputs of this study is the quantitative index for assessing the degree of vulnerability. The index lies between zero and unity. Zero indicates relatively no vulnerability and the value of one indicates maximum vulnerability.

Spatial Framework for Assessing Vulnerability and Adaptive Capacity of Livelihoods Systems to Climate and Market Stressors

N.H. Rao, M.N. Reddy, G.P. Reddy, K.H. Rao and K.V. Kumar

Objective:

- ❖ To develop a spatial framework for assessing vulnerability and adaptive capacity of livelihoods systems to climate and market stressors

Progress:

In recent years, assessments of vulnerability and adaptive capacity of biophysical and social systems have become the key components of sustainability assessments. A sufficiently general, analytical, and knowledge based decision-support framework that can be applied across different scales, for assessing the sustainability of rural livelihoods systems is developed.

The sustainable rural livelihoods (SRL) framework of DFID is integrated with a GIS platform to allow for flexible and interactive design and mapping of indicators of vulnerability and adaptive capacity of systems in multiple dimensions and at different scales. The SRL framework allows characterization and analysis of rural livelihoods by integrating the vulnerability and adaptive capacity concepts. Integration with GIS provides for the creation of an effective database of biophysical, and socioeconomic indicators as well as an interactive interface for assessing and mapping the vulnerability and adaptive capacity of livelihoods systems. The framework is applied to data from the administrative Blocks of Nalgonda district in Andhra Pradesh.

The SRL framework is used to identify the variables that characterize vulnerability and adaptive capacity of livelihoods systems in the blocks of the Nalgonda district (Fig 2). For each type livelihoods asset (natural, human, social, physical, financial capital), a set of variables or attributes were identified. Each variable/attribute may be characterized either by a single indicator or by a set of indicators. All the indicators were normalized to be in the range of 1-100, and classified on a scale of 1-5 to allow comparisons. Next, the classified indicators for each variable and for all the variables representing an asset were summarized into an index. For example, for the livelihoods asset natural capital, the variables were defined by soils, rainfall, and groundwater. For each variable, specific



indicators were identified; e.g. for soils the indicators are soil depth, soil type, available water holding capacity, slope, and erosion status. These were integrated into a Soil Quality Index. All the indicators were normalized to have values only in the range of 0 to 100. The normalized indicators for each attribute were then classified into a vulnerability class. Finally the indices for each type of asset were summarized into a corresponding asset index. A similar process was adapted for variables characterizing the adaptation capacities. The indicators were mapped to assess relative vulnerabilities of the blocks of the Nalgonda district (Fig 3).

The results revealed that an aggregated index of vulnerability may not be an adequate measure of the relative vulnerability of different blocks in the district. It would be necessary to examine the indicators at a more disaggregated level to differentiate between the vulnerabilities of different blocks. This is because regions that may relatively more vulnerable with respect to one type of livelihoods assets (for example natural capital), may be less so with respect to other assets (physical, financial, human or social capital). On the other hand, the aggregate measure of adaptive capacity captures the differences between blocks to enable differentiation and prioritization. The GIS based framework allows mapping the relative variations of different indicators which in turn allows prioritizing and targeting development of specific assets by region to reduce the vulnerability of livelihoods and increase the adaptive capacity of the local populations.



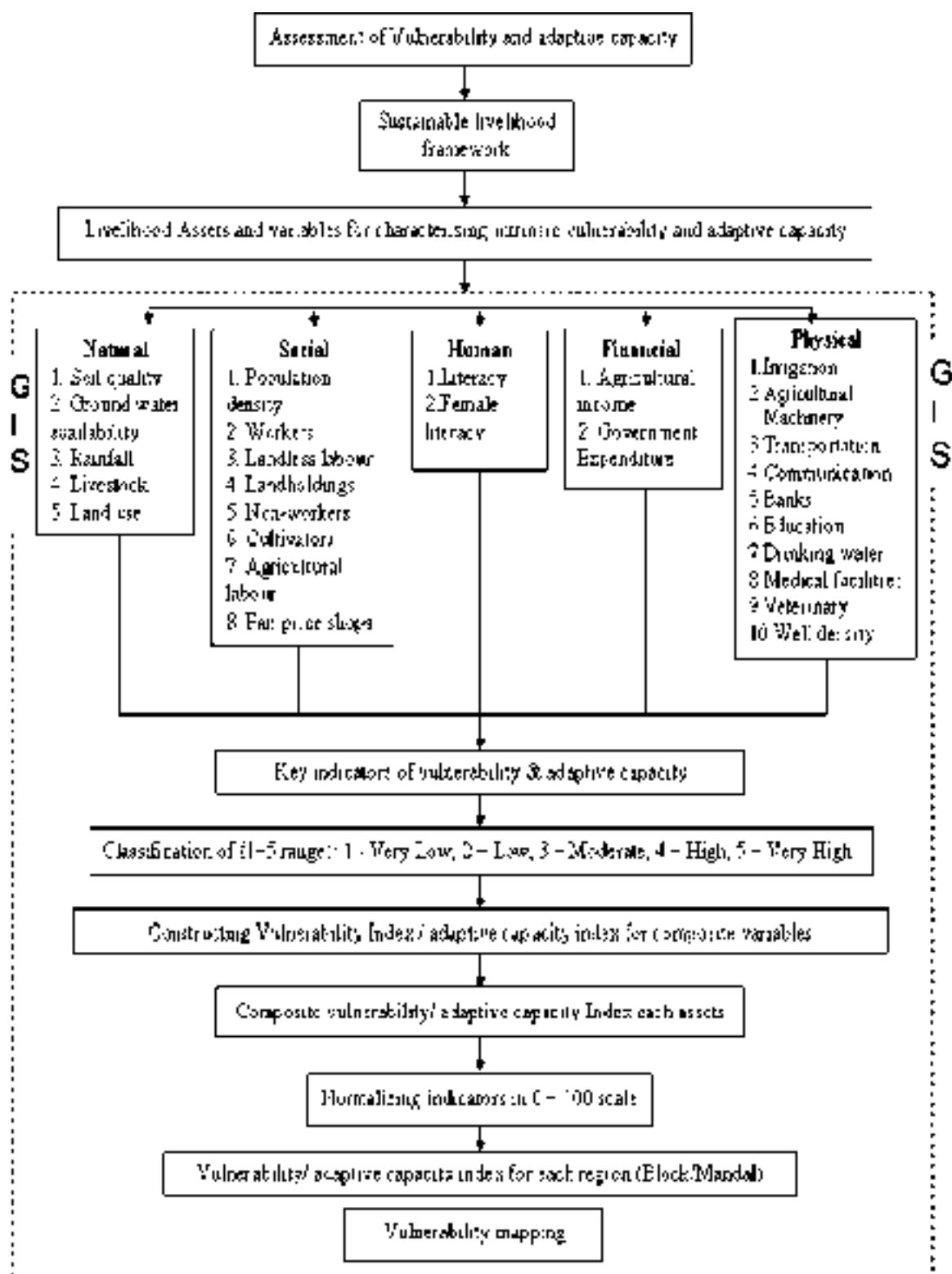


Fig 2 - Framework for assessment of vulnerability of livelihoods systems

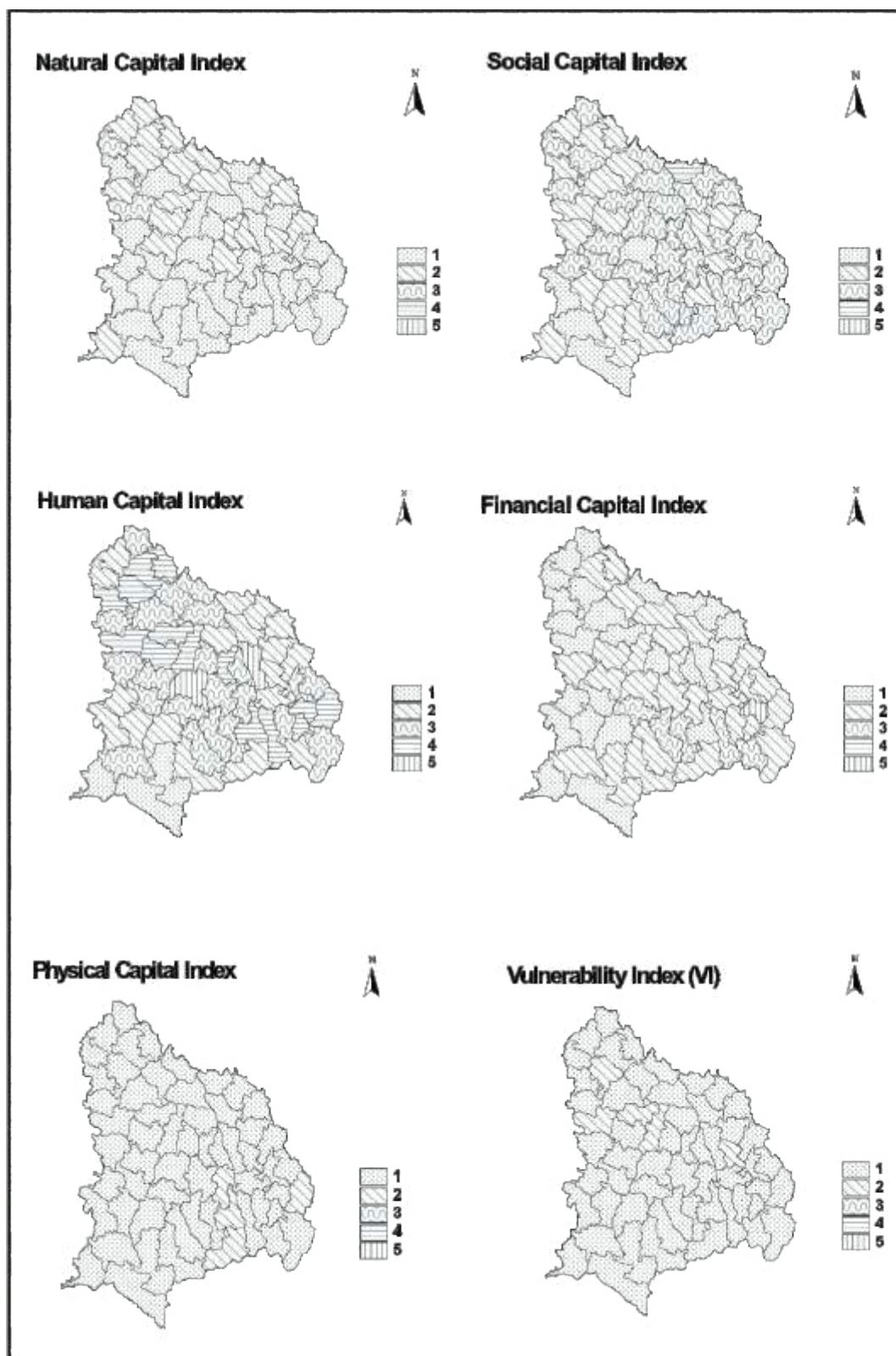


Fig 3 Distribution of indices of vulnerability of the five types of livelihoods assets and the vulnerability index among the blocks of Nalgonda district



Development of Procedure for Spatially Balanced Sampling for Livelihoods Analysis

N.H. Rao, M.N. Reddy, K.V. Kumar, V.K.J. Rao, K.H. Rao, G.P. Reddy and S.K. Soam

Objective:

- ❖ To develop procedure for spatially balanced sampling for livelihoods analysis

Progress:

Sample surveys for livelihoods assessment and analysis are usually based on random stratified random sampling from a list of farmers in the village. This results in a skewed spatial distribution of the selected farmers. A spatially balanced procedure is developed that uses the Reversed Randomized Quadrant-Recursive Raster (RRQRR) algorithm in ArcGIS to generate a spatially balanced sample of farmers based on survey numbers. This design enables selection of sample farmers for questionnaire surveys from the spatial distribution of land use parcels as identified by survey numbers rather than the list of farmers. The increasing accuracy with sample size and the relative improvement in sampling effectiveness spatially balanced sampling (SBS) over stratified random sampling (SRS) and for two sample sizes (30 and 60) are illustrated in Fig 4.

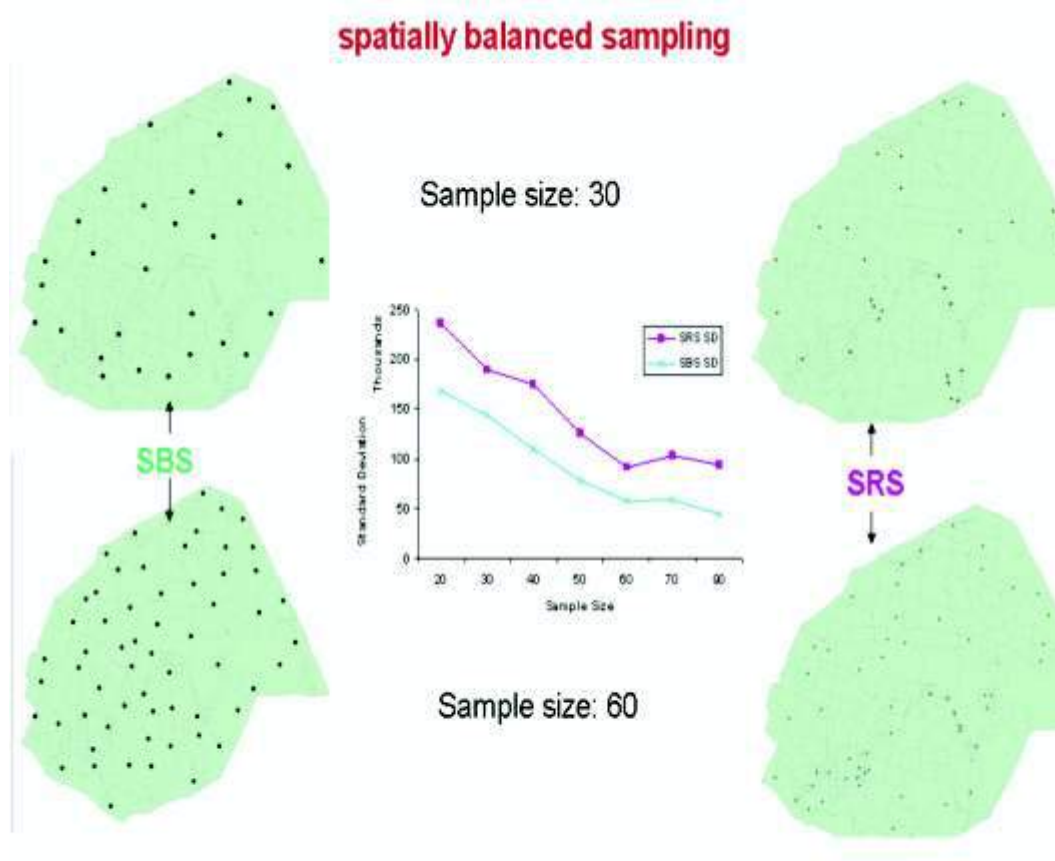


Fig 4: Improvement in sampling accuracy with spatially balanced sampling

Accelerating Agricultural Innovations through ICTs and Institutional Change

Agoweb Digital Dissemination System for Indian Agricultural Research

G.R.K Murthy, D. Rama Rao, N. Sandhya Shenoy and P. Mohan Singh, NAARM
R.C. Agrawal, NBPGR, New Delhi

Objectives:

- ❖ To identify standards, develop uniform guidelines, content management strategies and a model template for websites of ICAR institutes
- ❖ To develop model website of all consortium partners to meet requirements of stakeholders
- ❖ To design and develop ICAR 'Portal' and integrating the websites of consortium partners
- ❖ To build capacity of personnel in ICAR institutes in design, development and management of websites

Progress:

Online modules were developed like online databases for retrieval of ICAR-SRF Hall ticket generation and download module in consortia mode; Online database to publish the profiles of scientific professionals of ICAR; Online PDF search to provide quicker and easy access to scanned resources of Academy's training activities; Online Examination module to facilitate online evaluation; Online training management module has been developed and integrated into the website; Guidelines for uniformity of ICAR websites was developed in consortia mode; Model templates for homepages of websites of ICAR institutes were developed and circulated to the consortia partners. New NAARM website was developed using Joomla CMS.

Geospatial Data and Knowledge Organization Systems for Agricultural Research, Education and Technology Transfer (Geospatial Library and PGIS)

Re-designing the farmer-extension-agricultural research /education continuum in India with ICT-mediated knowledge management

N.H. Rao, M.N. Reddy, K.V. Kumar, V.K.J. Rao, K.H. Rao, G.P. Reddy and S.K. Soam

Partners: ICRISAT, Patancheru; IIT, Kanpur; IIT, Bombay; IIIT, Kozhikode; UAS, Dharwar; GBPUA&T, Pantnagar

Objectives:

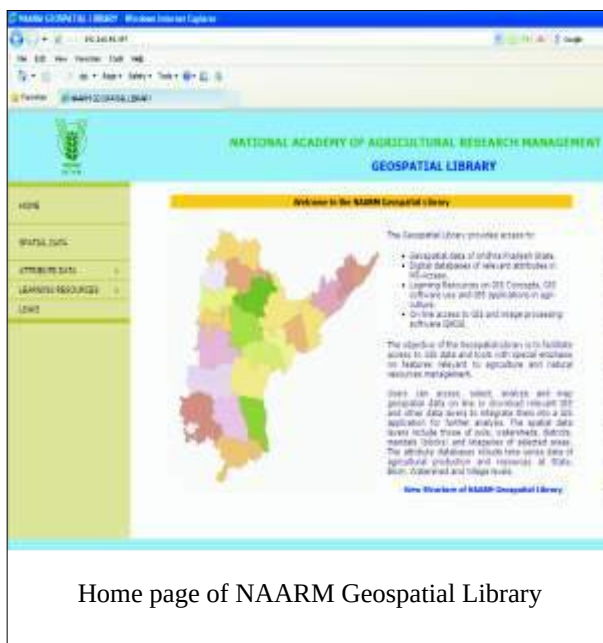
- ❖ To build a geospatial library and provide web-based access to geospatial data, services, and learning and knowledge resources to support agricultural research, education and technology transfer in a region covered by an SAU
- ❖ To build a participatory rural GIS using PRA processes for village knowledge centers to support agricultural decision-making by local communities
- ❖ To build the capacity of NARS personnel in digital content development and knowledge management for effective technology generation and transfer



Progress:

A framework and structure was designed for the study. The geospatial library was developed as web application accessible on NAARM campus LAN. Geospatial data and services are available at four levels for the State of Andhra Pradesh: district, block, watersheds and village for the features such as soils, administrative boundaries, mini and microwatersheds, markets. A prototype Geospatial Village Knowledge management System for the Villages in Penpahad mandal of Nalgonda district was developed and tested in association with an NGO namely SAIRD.

(i) NAARM Geospatial Library (NGSL)



Home page of NAARM Geospatial Library

A prototype geospatial library was designed to provide web-based access to geospatial data, services, and learning and knowledge resources to support agricultural research, education and technology transfer. The prototype is based on data of a region covered by a State Agricultural University (SAU)/ Government Department/ Institution. The State of Andhra Pradesh, (which is the region covered by Acharya NG Ranga Agricultural University) is chosen to develop the spatial and attribute data resources for the Geospatial Library. The Library is accessed via a Website designed for delivery of spatial data, attribute data and learning resources.

The NAARM Geospatial Library organizes spatial and attribute data, and knowledge resources in three main categories

- ❖ Spatial data (of soils, watersheds – sub, mini, micro watersheds, agricultural markets, agro-eco-subregions at State, District, Block/Mandal and village level)
- ❖ Attribute data (for state, district, block, village data of crops, socioeconomic data)
- ❖ Learning resources (eg. training manuals for GIS and image processing software, applications in agriculture)

The structure of NAARM Geospatial Library (NGSL) is given in Fig 5.



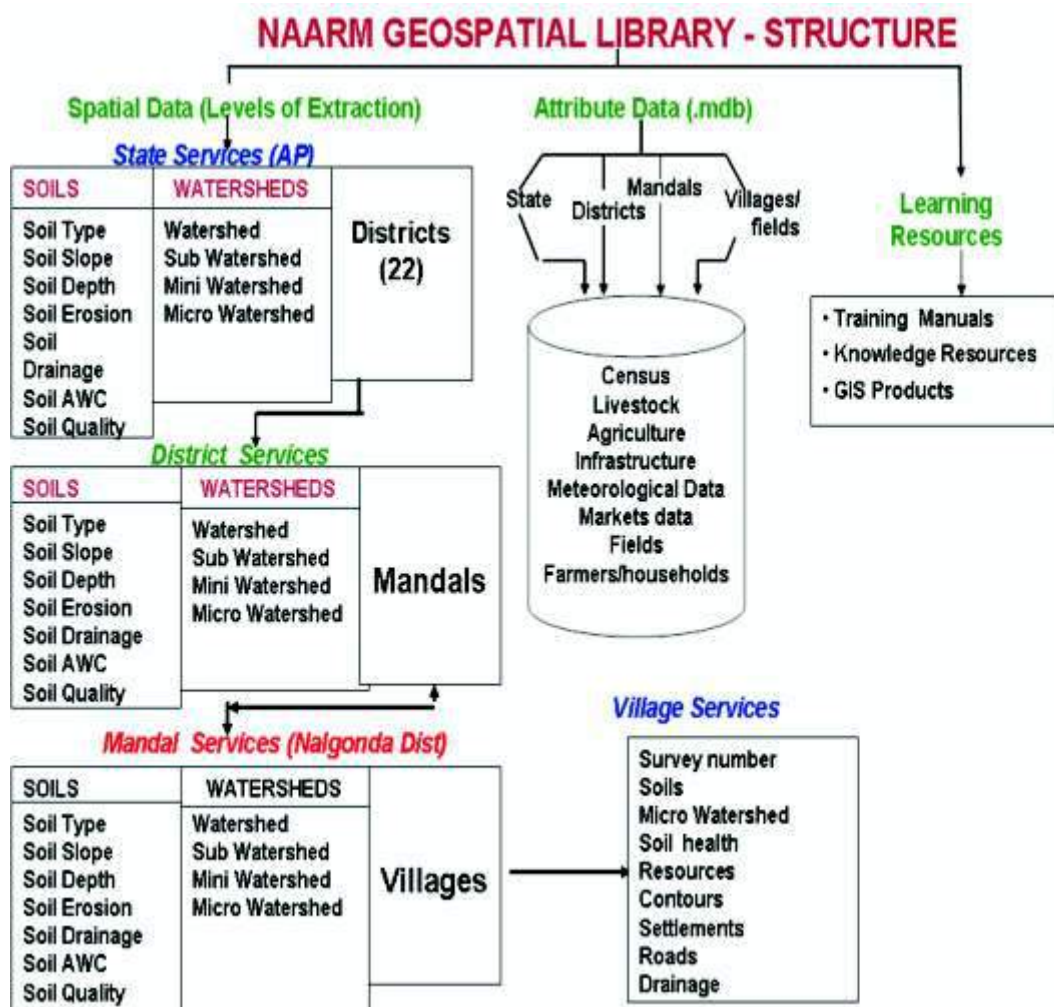


Fig 5: Structure of NAARM Geospatial Library

The spatial data of different themes was available at state, district, block village and field levels (Fig 6). Attribute data were available on various aspects of agricultural production. The spatial data was delivered to users on client systems in a convenient format (shape files) via an interface provided by ArcIMS. The downloaded data can be used with any GIS software to enhance agricultural research, management and education by integrating them with other spatial and attribute data available with users to address specific problems. Simple thematic mapping and geospatial analysis can also be carried out on line by the users after downloading the java viewer from the server. The NGSL has been tested and used in training programmes of NAARM, by scientists in NARS and ICRISAT, and by students of NAARM and other institutions.



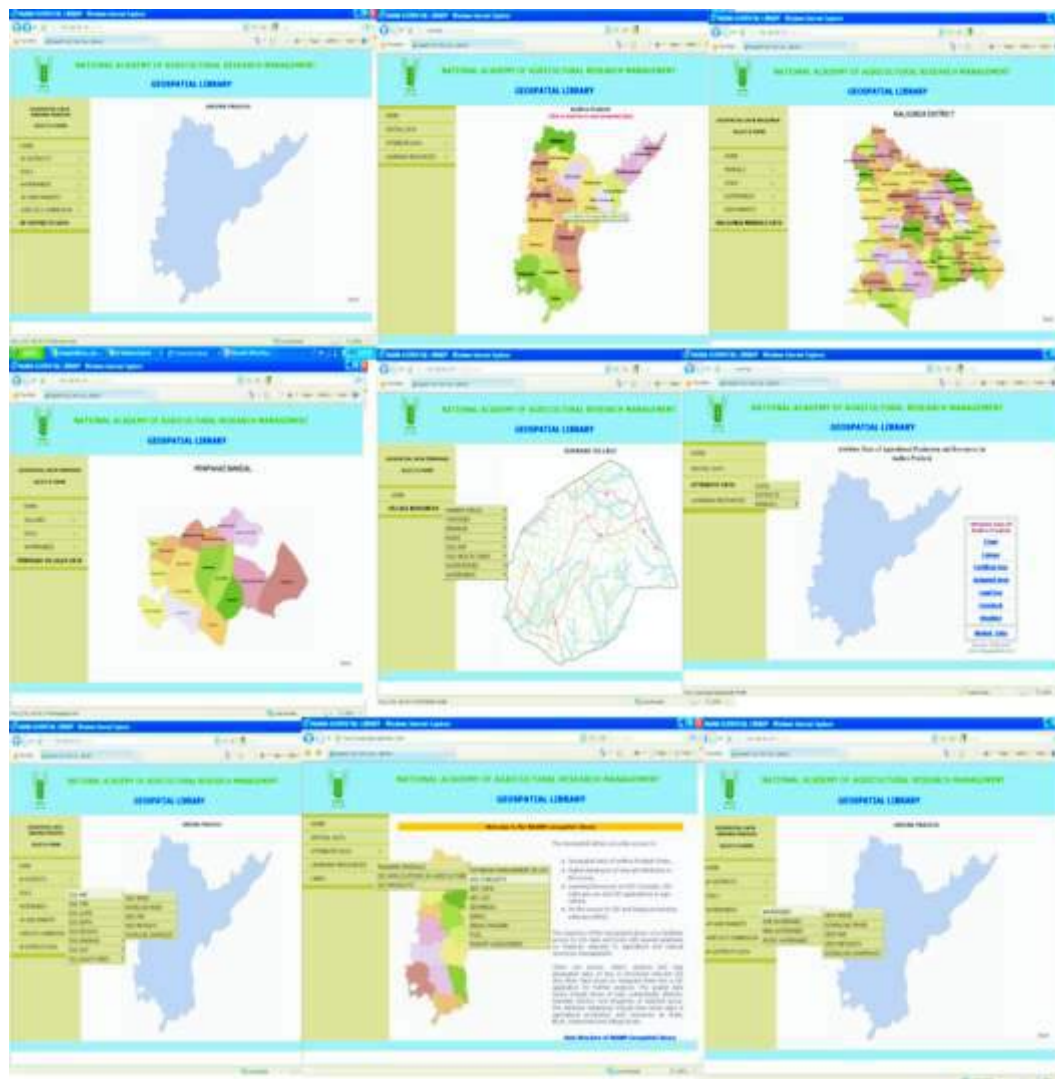


Fig 6: User interfaces of the NGSL for spatial and attribute data

(ii) Geospatial Village Knowledge Management System (GVKMS)

While the Geospatial library primarily addressed the supply side concerns of agricultural knowledge management, the demand side spatial data and knowledge management at the village level and farm level is equally important for local level planning supporting rural livelihoods to raise rural incomes. Enabling access by local village communities to spatial data of people and resources in the VKCs has the potential to transform the way by which farms can be integrated into the supply chains of agribusinesses, rural communities can make decisions on agricultural and natural resources management and by which development agencies and institutions can target technologies and interventions more effectively to raise rural incomes and improve the quality of life in rural areas.

A prototype Geospatial Village Knowledge Management System (GVKMS) was developed to support agricultural decision-making by rural communities. The GVKMS is a deployable windows application developed using MapObjects and Visual Basic in addition to standard GIS software. The GIS software was used to create the spatial data layers relevant to the decision processes. These were then integrated along with other attribute data and



learning resources (including indigenous knowledge resources) into the deployable application in the Village Knowledge Centre. At the VKC itself, there is no need to install complex GIS software. All that is required is a computer operating on windows operating system.

The basic structure of the database provided in the Geospatial Village knowledge Management System consists of three main categories: spatial data, attribute data, and knowledge resources (Fig 7). A prototype GVKMS is developed for the Duppahad village in Penpahad mandal of Nalgonda district. The digitized spatial data layers included in the prototype comprise the village boundary, drainage, contours, roads, settlements, soils including information from soil health cards, micro watersheds and cadastral maps. Several external databases are included to connect to these layers (Fig 8). These include village statistics on population, livestock, land use statistics, and survey number wise land use information. In addition knowledge resources on technologies available, indigenous knowledge from local communities obtained by participatory processes are also included in the GVKMS. The entire set of geospatial data, attribute data and knowledge resources are packaged into a deployable application that can be easily installed in a village knowledge centre running on MS Windows or ported easily to a laptop.

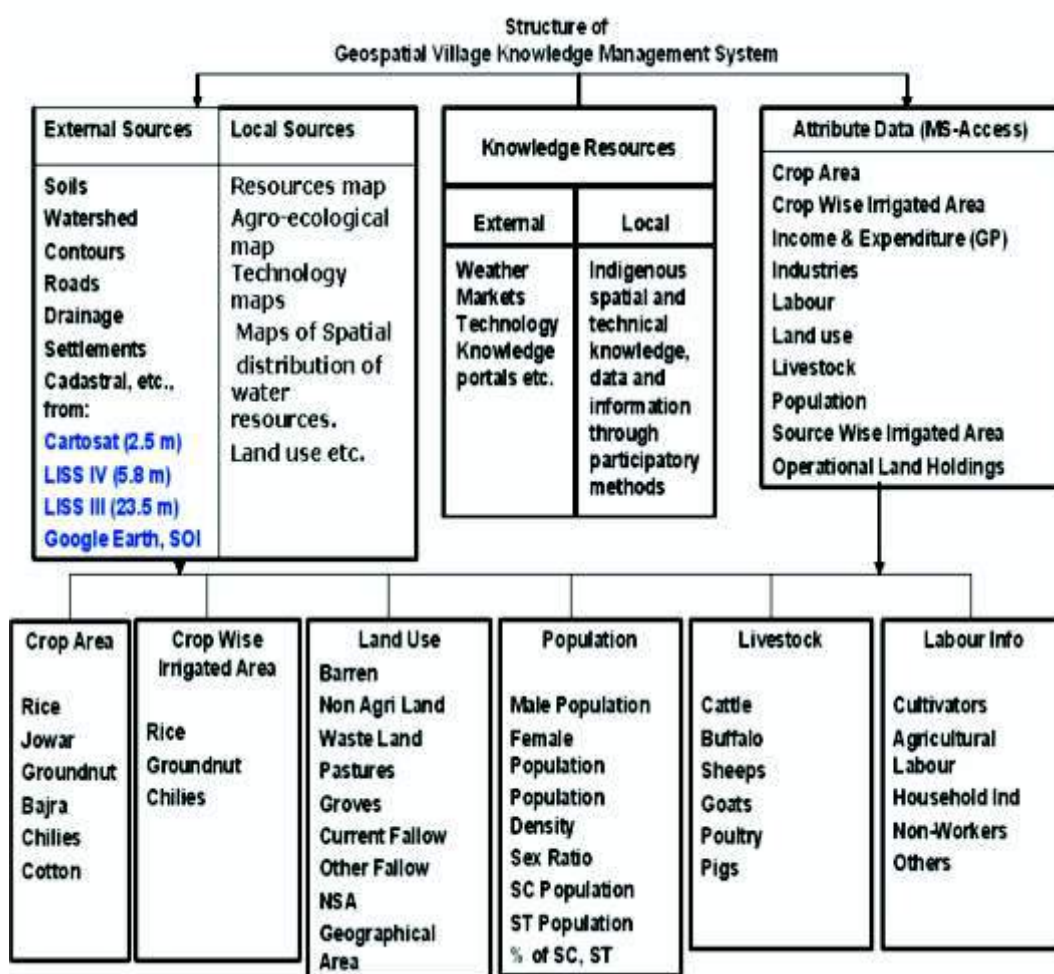


Fig 7: Structure of the Geospatial Village Knowledge Management System (GVKMS)

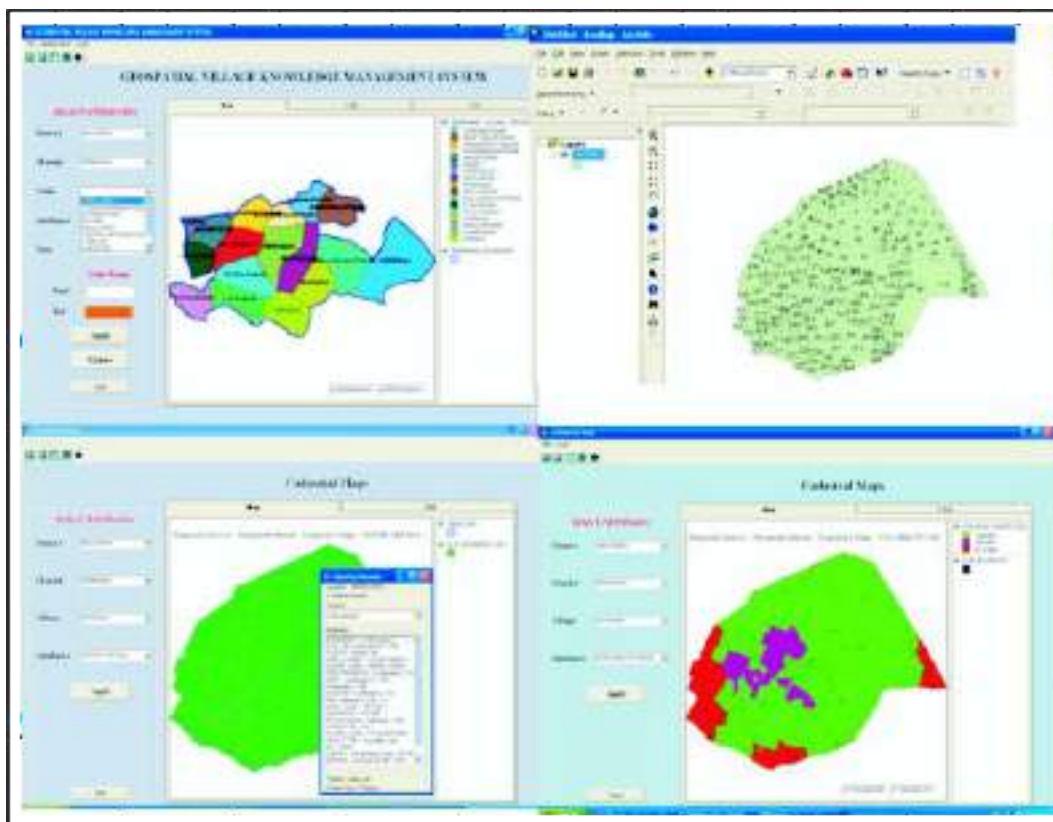


Fig 8: Village and cadastral information in GVKMS

In this study, the primary data from the farmers through structured questionnaire was collected in the study village. The sample was drawn based on spatially balanced sampling, when the sampling frame was the cadastral maps, rather than the list of farmers. The efficiency for a sample of size was 30 using this spatially balanced sampling compared to simple random sampling.

The prototype has been validated and tested by installing it and demonstrating it in the village knowledge centre set up by an NGO, Sri Aurobindo Institute of Rural Development (SAIRD).

Training Needs and Impact Assessment of L& CB under NAIP

B.S. Sontakki, T. Balaguru, K.H. Rao, G.P. Reddy and J. Challa

Objectives:

- ❖ To assess training needs for L&CB under NAIP
- ❖ To evolve a comprehensive L&CB plan and framework based on the identified training needs
- ❖ To identify indicators for impact assessment of L&CB under NAIP
- ❖ To assess impact of L&CB programme
- ❖ To evolve a comprehensive impact assessment strategy for L&CB

Progress:

This study has the overall objective of developing methodology for training needs and impact assessment. The study employed a combination of survey, brainstorming workshops, focused group discussions, observation and desk research to generate training



needs assessment protocols. Based on these core competencies, their priorities and corresponding knowledge, skill and attitude classes were identified for three levels of professionals in NARS – Research Managers, Scientists and Technical and administrative staff as well as for other stakeholder group (private and NGO sectors). Training impact indicators identified were content, relevance, design, resources, delivery, process, outcome, and logistics related to learning programs.

Strategies to Promote Use of ICTs in Agriculture Value Chains

D. Rama Rao and B.K. Sikka

Objective:

- ❖ To promote use of ICTs in agriculture value chains

Progress:

ICTs are in use for agriculture & rural development, literacy promotion, flow of information, business transactions, governance, and various other societal needs from jobs to marriages. The study focused on models and best practices for assessing, planning, designing and managing rural ICTs for use in small scale agribusiness. The current use of ICTs, their potential and strategies to promote their use in agriculture value chains were documented.

Numerous case studies dealt with ICT infrastructure for providing health, education and other services in rural areas. However, very few of these studies treated ICT infrastructure for providing traceability, and those that focused on the services provided rather than the infrastructure. Traceability is in place in export oriented agri-commodity trade. Export promotion of grapes has small farm experiences recording traceability information to describe crop protection chemicals, interventions, analysis (soil, residues, etc) in electronic form. This is promoted and supported by government through its export promotion agency namely APEDA.

The rapid growth in domestic food business through retail and super market chains has need and potential for use of ICTs in their supply chains. Certain niche product chains (eg: poultry, grapes and many horticultural products, organic food products) have well established market chain. Most of these chains at the farmers end function on the basis of trust and not through formal agreements. The business houses use modern communication technologies extensively in the higher levels of the supply chain. Some business houses are using ICTs for backend integration too, mobile being most preferred means now.

Traceability systems available in certain cases today maintain manual records as decided by the business requirement. In such cases information can be transmitted along the chain, for example to guarantee its quality (organic farming, fish farming, export of basmati rice, export trade in select fruits, etc). In these cases, the cost of accessing information exchange is low by using internet. Various rural institutions like cooperatives, Micro finance self-help groups, primary agriculture markets, commodity board market centres are in the process of computerization and networking. A good number of these institutions use computers for specific purposes. These are potential centres for linking farmers to market in the supply chain. However, none of these institutions are ready, as of now, aiming at such function. As they are in the service of farmers, networking markets through these centres will benefit the farmers.

A model Community Multi-link Point would be the first link in the value chain at block (mandal) level providing a host of services to farmers and also linking them to markets and supply chains. The system would lay foundation for knowledge sharing, market access and other value added services to rural areas. It would capture block level data from public



institutions and put them on national net work. It can provide last mile link to the proposed village knowledge centres (VKCs) or Common Service Centres (CSCs) in the villages and also functions as a window or channel for public and business houses to communicate with farmers.

Wireless technologies such as Wi-Max and 3G have the potential to solve the rural access issue due to their high scalability, low operational costs, open standards and their adaptability to voice and data transfer. The technology is now simple to use and can be layered for sharing by various actors using simple devices like mobiles and a variety of low cost hand held devices like PDAs and netbooks. Rural communities can easily install and operate wireless systems to their needs.

Job Training Interventions through Distance Digital Mode for Enhancing the Effectiveness of Research and Educational Institutes

M.N. Reddy, A. Dhandapani and J. Challa

Objective:

- ❖ To develop on the job training interventions through distance digital mode for enhancing the effectiveness of research and educational institutes

Progress:

Distance learning module on spreadsheet applications was modified by incorporating more interactivity and audio support. Modules on data analysis and data management with basic functionality were completed. This module was developed using the open source content management software (CMS) Drupal. Some of the features of this content management system are as follows:

- ❖ Capacity to identify all the key users
- ❖ Ability to assign roles and responsibilities to different content categories or types
- ❖ Ability to track and manage multiple versions of a single instance of content
- ❖ Ability to capture content (e.g. scanning) and the ability to publish the content to a repository content
- ❖ Separation of content's semantic layer from its layout (For example, the CMS may automatically set the color, fonts, or emphasis of text)
- ❖ Conformance to global e-learning standards
- ❖ Enhanced scalability and usability of content
- ❖ Ability to serve diverse learners needs

This module was made available on NAARM Intranet under distance education. This provided learning skills in using MS Excel which were needed in the courses related to quantitative methods. This module, further, will be strengthened by adding more applications in business as well as in agriculture.

IT-based Decision Support System for Effective Knowledge Technology Transfer

G.R.K. Murthy and K.M. Reddy

Objectives:

- ❖ To identify and assess the relevance of different content management strategies (CMS)
- ❖ To short list suitable management strategies for decision making purpose in agriculture
- ❖ To integrate decision support tools through content management models for effective knowledge and technology transfer



Progress:

- Various open source Learning Management Systems (LMS) and Content Management Systems (CMS) were studied and a prototype website using Joomla was developed hosted on intranet. Based on the experience, a similar platform of LMS was identified and developed a e-learning website using Moodle which is hosted on internet with <http://elearning.naarm.ernet.in> for the PGD program of the academy, where 59 e-resources were made available on the web for accessing the content with 24 / 7 Content Access, provision for uploading assignments of Students, Online quizzes, contributing to Glossary / Discussion fora and Instant evaluation of Results.
- An online examination system in intranet was deployed and successfully used for the evaluation of FOCARS probationers. Economic analysis indicated that the online system results in a saving of 70 per cent as compared to traditional mode of examination.

Empowering Rural Women through Information and Communication Technologies (ICTs)

Sandhya Shenoy and V.K.J. Rao

Objectives:

- ❖ To develop database on women agricultural professionals in India
- ❖ To develop training module for capacity building

Progress:

Updating Database on Women Agricultural Professionals in India (WAPI):

The database was strengthened by 720 entries of professional women in agriculture representing from SAUs, ICAR institutes, Government Organizations and NGOs. The 'online entry form with the link to database' was created to enable the online entry from the professional women in agriculture nationwide. This was tested and linked to NAARM website (<http://202.141.137.4/AgriProfWomen/Default.htm>). Apart from the online registration for New Users, updation facilities for the Registered Users were provided in the website.

Training modules for capacity building:

Training was provided to the project associates (KVK, Gaddipalli and DHAN Foundation) under two programmes, namely "Enhancing Skills in ICT Based DSS for Market and Agri Business Orientation of Research, and Sustaining Rural Livelihoods" and "Information and Communication Technologies in Agriculture". The contents in the former training programme included Introduction to various Office Management tools, MS-Project Software, applications of remote sensing and GIS in weather forecasting, agro-advisory service, soil health and degradation assessment, and functions to enhance their skills in making most appropriate use of these tools, exposure to multimedia and web design software for preparation of information modules. In the latter programme, exposure to ICTs application in agriculture, and hands-on practice of internet usage for e-mailing and group SMS in vernacular language were provided for the DHAN project associates along with facilitators of the village knowledge centre (VKC) situated in the Aepoor village.

Implementation of Integrated ICT model:

The activities of Integrated ICT model such as Participatory Needs Assessment and Problem Prioritization, Village resource mapping using Participatory GIS and GPS, and media research in the 'Process' component were completed. Based on the Participatory Needs Assessment and problem prioritization in agriculture, the relevant content organizations and experts were identified for problem solving and for content modules.



The media research helped in selection of ICTs for problem solving based on access, use and preference of farmwomen. In the 'content management component', the farm women database was linked to the cadastral map of the selected village, Aepoor, using GIS for thematic mapping. The outputs include the water, crop, animal resources mapping, as well as social issues mapping on livelihoods, distribution on leadership, ICTs access, usage and preferences, age, education, land holding, farming experience etc. The knowledge tests were designed for 4 main crops of the selected villages (paddy, castor, cotton and red gram) in consultation with the crop experts to determine the knowledge gaps and to design need based content modules using ICTs for farm women. The knowledge test for paddy was conducted. The test showed that 93.75 percent of the land preparation practices were answered by farm women indicating high level of knowledge. Regarding the seed rate, sowing and spacing queries, they were able to answer only up to 77.5 percent, while, the queries on fertilization management were answered up to 65.5 percent and irrigation management up to 50 percent. The knowledge level was the least in case of plant protection practices where only 39.30 percent of the queries were answered. Therefore, the content modules should be designed to be able to bridge these knowledge gaps of farm women.

Based on the media research, SMS based messaging interfacing through web portals was introduced for interfacing farmwomen with the content experts. The relevant content modules were placed in Village Knowledge Centre (VKC) at the selected villages. The master trainers and facilitators were trained to manage VKC. The ICT user group creation and capacity building for maintaining the web site are the activities to be taken up under the 'Sustainability initiatives' component.

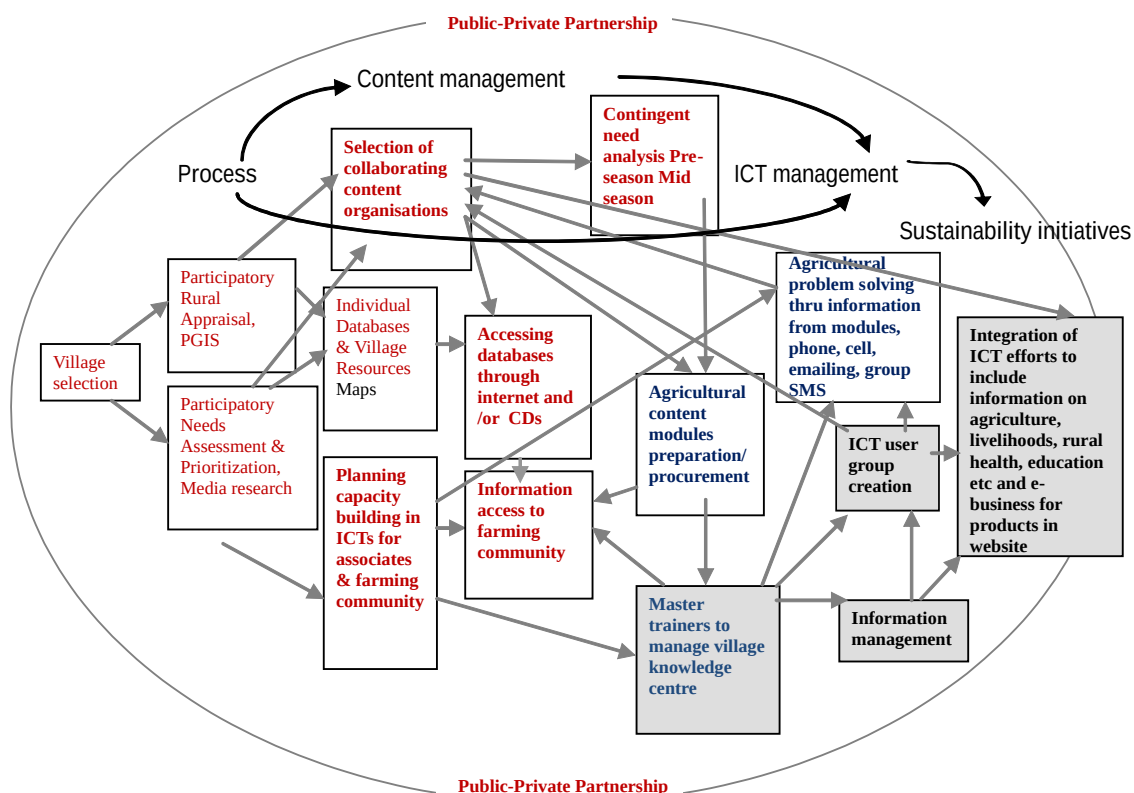
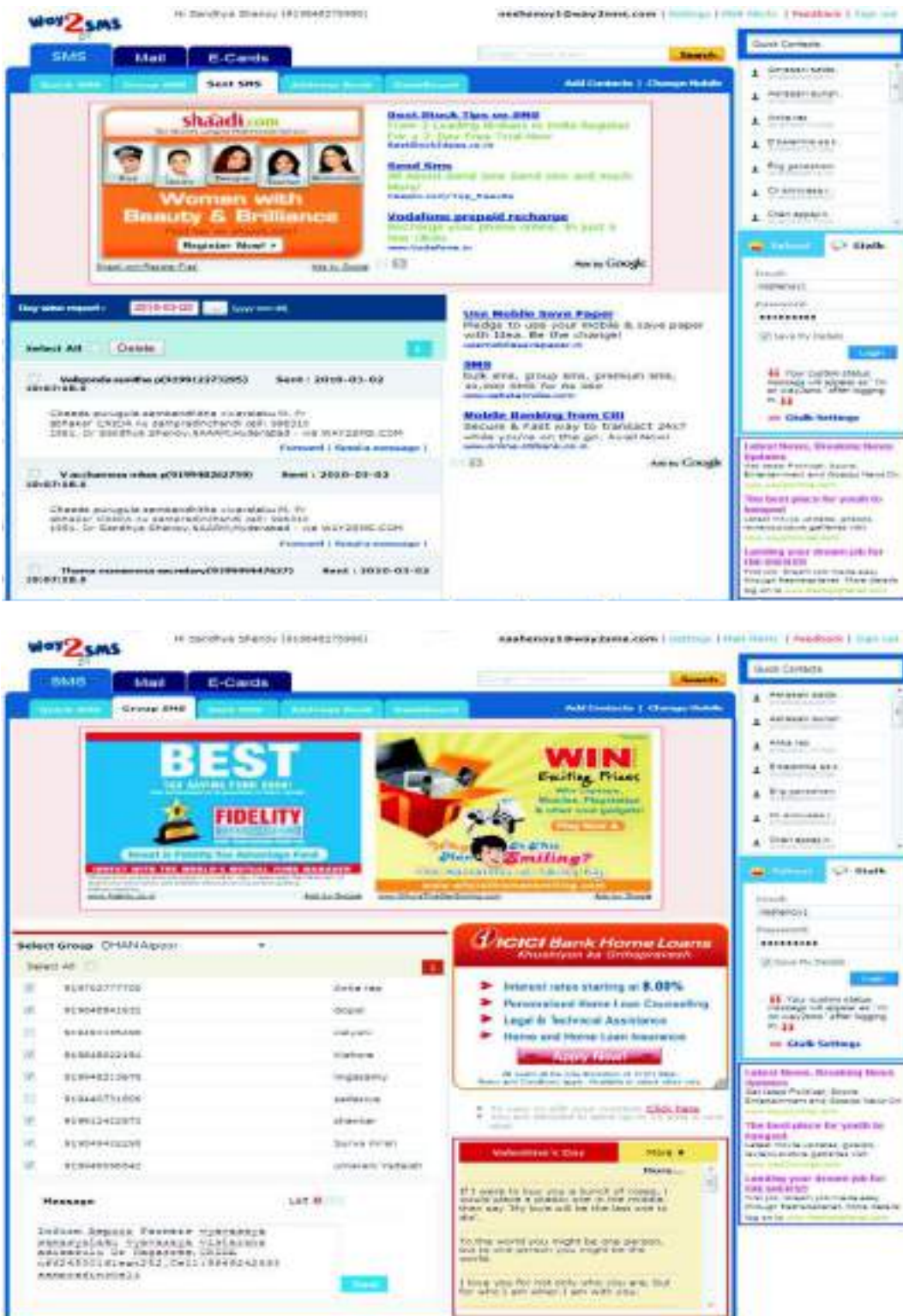


Fig 9: Screen shots of Group SMS for agri problem solving



The discussions were held with the women SHG group members on their information needs, livelihoods activities, interest in product generation for improvement and diversification in livelihoods. Based on this information, the website template was prepared for agriculture information and e-business under the 'Sustainability initiatives' component which was shown below.



Agriculture Information and e-Business Gateway
Women Self-Help Groups (SHG) in Apper and Dugulad, Rajguda District, Andhra Pradesh

Crops	Date	Available Quantity	Accepted Rate(Rs.)
Paddy	15.3.2010	20 tons	1000/- ton
Sorghum			
Red gram			
Green Gram			
Chick pea			
Groundnut			
Caster			
Tomato			
Cluster bean			
Mango			
Sweet Lime			
Stain			

Animals

Milk
Meat
Eggs

Processed foods

Fruits
Pickles
Ghee
Butter

Alled

Herbal products

Veterinary products

Handicrafts

Other Images

Collaborating partners

Development of Humane Action (DHAN) Foundation
Sri Aurobindo Integrated Rural Development (SARID)

Agriculture Information

Crops

Paddy
Sorghum
Cotton
Raddam
Green gram
Chick pea
Groundnut
Caster
Vegetables: Tomato, Shrub, Chilli, Green leafy vegetables, Beans
Mango
Citrus (Sweet lime)
Animal Husbandry
Dairy (Buffaloes, Cows)
Poultry
Sheep, Goats
Associate Content organizations
Directorate of Rice Research (DRI)
Directorate of Oilseeds Research (DOI)
Central Research Institute for Dryland Agriculture (CRIDA)
Directorate of Sorghum Research (DSR)
Acharya NG Ranga Agriculture University (ANGRAU)
Sri Vasukateswara Veterinary University (SVVU)
Sri Aurobindo Integrated Rural Development (SARID) Kirti
Vijayan Kendra (VKV)
Traditional Knowledge
Agro Biodiversity
Content Experts
Dr R. Mahender Kumar, DRS for Paddy
Dr M. Padmavathi, DRS for Oilseeds
Dr Nagendra, CRIDA for dryland crops
Dr Y.G. Prasad, CRIDA for Coffee
Others
Health
Education

Village Requirements

1. Want Vermi-compost 2 tons quantity by 5th April, Contact

Digital Multimedia for Agri-innovation Transfer

K.M. Reddy, V.K. J. Rao and G. R. K. Murthy

Objectives:

- ❖ To assess the extent of availability, use and need of Digital Resources in NARS
- ❖ To develop comprehensive digital resource management strategy

Progress:

- ❖ Complete documentation installation procedure of PLONE CMS for I-DMRC
- ❖ Replication procedure of PLONE CMS in other machines
- ❖ Back up procedure of PLONE CMS portals
- ❖ Port numbers setting for PLONE CMS

Digitally Enabled Customization of Information for Decision and Empowerment

V.K. J. Rao, N. Sandhya Shenoy and K.M. Reddy

Objectives:

- ❖ To assess the information requirement for effective decision making among farmers
- ❖ To suggest model for developing real time information system through customizing



Progress:

- ❖ Procedure and documentation notes of creating interactive learning objects like the Fish anatomy were developed
- ❖ SMS based messaging interfacing through web portals and CMS GUI was developed
- ❖ A PRA based framework to customise the content of DMAT for a village knowledge centre has been developed.
- ❖ The farmer information needs to make a workable Digital media forum has been initiated through PRA study at Zamistanpur village an adopted village of CRIDA under the Sustainable Rural Livelihoods project, where the Village Resource center of CRIDA which has a touch screen facility and a high end personal computer linked with the internet will be used to develop the Digital Media forum for which learning objects related to farmers needs will be developed and hosted and the interaction through SMS based services using Way2SMS is also planned and cell numbers of farmers to register in the digital media forum has been collected.

Technology Delivery Models for Less Favoured Areas

B.S. Sontakki

Objective:

- ❖ To design technology delivery models for promoting livelihoods security in less favoured areas

Progress:

This study attempted to analyze the existing technology delivery models in terms of their effectiveness to address the special challenges and needs of less favoured areas and suggest alternative models for the same. Extensive desk study was done to profile the characteristics of less favoured areas. Existing technology delivery models and mechanisms were analyzed.

Organization and Management for Strengthening Agricultural Research**Assessment of Future Human Capital Requirements in Agriculture**

D. Rama Rao, S.K. Nanda, N. Sandhya Shenoy, Bharat S. Sontakki and G.P. Reddy, NAARM
 Rashmi Agrawal, I.C.Awasthi, G.P. Joshi, Inder Kumar, Sanchita Bhattacharya and Sadanand Sahu, IAMR

Objectives:

- ❖ To assess the trend in supply-demand and employment of trained manpower in agriculture and allied sectors
- ❖ To develop a manpower information system for planning trained manpower in agriculture and allied sectors
- ❖ To evaluate institutional set up, interface between agricultural graduates with various stakeholders and the impact of diversification of agriculture on skill requirements



- Progress:

Simulation Model for Quality Assessment of Agricultural Colleges

Objectives:

- ❖ To conceptualize the quality dimensions in agricultural education
- ❖ To develop a simulation model for dynamic quality assessment
- ❖ To develop a tool for self-assessment of quality

Progress:

49 quality parameters were identified of educational grouped under seven broad dimensions (Faculty, Intake, Outturn, Infrastructure, Service, Employer and Management). The causal loop diagram (Figure 1) was constructed with these dimensions.

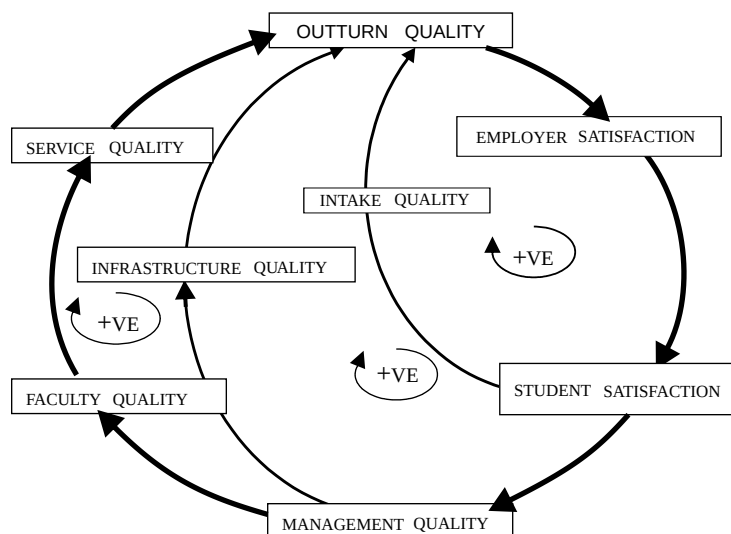


Figure 1: Causal loop diagram for equality of education

A system dynamics simulation model was developed on STELLA platform. The model was used for scenario analysis.

Performance Assessment of Agricultural Universities

J. Challa

Objectives:

- ❖ To develop an appropriate methodology for performance assessment of Agricultural Universities on benchmark indications such as education, research, extension, consultancy and governance
- ❖ To enhance global competitiveness of agricultural universities and to attract performance based funding
- ❖ To standardize and rationalize the mechanism of Sardar Patel Award for Outstanding Agricultural University

Progress:

Eight benchmark Indicators and 66 parameters were identified for the study, which are listed below:

- ❖ Education (22) - admission procedure, curriculum. Evaluation system, student advisory system/services, innovations in teaching. Faculty profile and evaluation, student placement/self employment, students' performance
- ❖ Research(18) - publications, patents/IPR, technologies developed, collaborative research, externally funded projects, applied vs basic research, need-based research, awards and recognition, research contingency per faculty, technology adoption
- ❖ Extension(15)- extension methods and methodologies, publications, advisory services, participatory research and extension activities, linkages, IT and mass media programmes, technology parks, impact assessment, private sector participation in extension, common property resource management
- ❖ Organization and management (10) - financial management, human resource management, monitoring and review, decentralization and devolution of powers, policy, planning and implementation, e-Governance,
- ❖ Social relevance (10) - social empowerment, food and nutritional security, environmental sustainability, livelihoods security, agri-business, social mobilization
- ❖ Resource generation (8) - consultancy services, commercialization of technologies, sale of university produce, sale of publications, externally funded projects, royalty from patents and other IPs, student fees and self financing programmes, endowments and voluntary assistance, sponsored programmes
- ❖ Infrastructure(10) - farm facilities, modern classrooms, modern library and digital resources, modern laboratories, hostels guest houses, co-curricular activities, transport, residential quarters, technical support facilities, welfare facilities
- ❖ Networking and collaboration (7) - national collaboration, international collaboration, collaboration with industries, exchange programmes, public private partnership, online networking



Change Management for Promoting Innovation

P. Manikandan and R.V.S. Rao

Objective:

- ❖ To synthesize the experiences of change management and identify issues for change management

Progress:

Data were collected through pre-tested questionnaires from 111 respondents. The questionnaires focused on the collection of information pertaining to (i) readiness to change; orientation to change; organizational innovation; organizational diagnosis; organizational culture; and (ii) assertiveness; power dimension; decision-making.

A case study of successful change in the value-chain project of sorghum was also undertaken. The analysis of the data collected indicated that readiness for change was moderate to low among scientists and teachers. Planners of change were found to be high, whereas initiators were very few. Organizational innovation and organizational culture were perceived to be moderate to low. Attitude towards change was not perceived to be encouraging; helpful mechanisms, leadership, and personal relationships were indicated as poor. Whereas assertiveness was good, expert power seems to have been eroded and personal power is very good. In general, the survey pointed out that institutional mechanisms and personal dimensions are not conducive for change.

A case study of the NAIP Project on "Creation of demand for millet foods through PCS value chain" under Directorate of Sorghum Research was taken up to study the success of change brought out through the project. Empirical models were used to describe the case study to understand the changes brought out through the project. The case study pointed out to the success of change.

Leadership Effectiveness for Promoting Innovation

R.V.S. Rao and P. Manikandan

Objectives:

- ❖ To identify the nature of leadership in research institutions of NARS
- ❖ To measure the leadership qualities and effectiveness of leaders
- ❖ To suggest measures for enhancing leadership
- ❖ To formulate recommendations for HRD in research organizations for leadership

Progress:

In order to understand the leadership effectiveness in the NARS, questionnaires were developed and data were collected from 101 scientists and teachers in the system. Decoding and analysis of the information collected is in progress.

Performance Enhancement for Promoting Innovation

R.V.S. Rao and P. Manikandan



Objectives:

- ❖ To identify key performance drivers that influence performance at the individual, team and organizational levels
- ❖ To identify different ways used by researchers and their superiors to encourage and motivate research scientists to develop and utilize their full potential
- ❖ To identify the key developmental and training needs for achieving excellence in researchers' performance
- ❖ To find out how does the existing employee performance management system support high performance
- ❖ To find out different formal and informal assessment methods and measures used to determine employee well being, satisfaction and motivation
- ❖ To find out how organizations maintain a safe and healthy work environment

Progress:

Three questionnaires were developed to study the performance management and enhancement in the system. The data from 101 scientists and teachers was collected. Further collection of data and decoding and analysis of data is in progress.



Assessment and Development of Organizational Citizenship Behavior for promoting efficiency and effectiveness in NARS

K.H. Rao, R.V.S. Rao, B.S. Sontakki and G.P. Reddy

Objectives:

- ❖ To identify the indicators of organizational citizenship behavior to work in teams cutting across the disciplines and institutions
- ❖ To measure and analyze organization citizenship behavior among personnel of NARS
- ❖ To determine the antecedents of organization citizenship behavior
- ❖ To develop strategic plan for promoting organization citizenship behavior of scientific personnel working in consortiums

Progress:

Survey schedule was used to collect data from scientists of ICAR institutes located in Hyderabad and Bhubaneswar and also from the faculty members of three agricultural universities namely ANGRAU, Hyderabad; OUAT, Bhubaneswar; and AAU, Guwahati. The questionnaire covered the perception of respondents on their role, management and organization. Organizational citizenship behavior indicators identified include individual innovativeness, superior-subordinate relationship, loyalty, obedience, social/advocacy/functional participation, identification with the organization, altruism towards colleagues, conscientiousness, interpersonal harmony and protecting organization resources. The respondents' perception on facilities and resources at work, work-load, job-autonomy, job-satisfaction and organizational climate was also collected.

Promotion of Inter-disciplinary Orientation and Inter-institutional Collaboration in Public Private Partnerships in NARS

T. Balaguru, P. Manikandan and S.K. Nanda

Objectives:

- ❖ What is the best way to conduct interdisciplinary research?
- ❖ Which are the most successful interdisciplinary research programs?
- ❖ How the appropriate collaborators are identified?
- ❖ How the research community overcome peer-review and publishing challenges?

Progress:

The study was based on the survey from scientists in SAUs, ICAR institutes and other R&D institutes. The maximum percentage of respondents stated that the inter-disciplinary orientation to research facilitates to address problems that lie beyond the disciplinary knowledge and skills of individual scientists. Large section of respondents identified that the farming systems / cropping systems development in the top ranked programme to be amenable for inter-disciplinary research. Most of the respondents reported that role clarity to each member of the team promotes inter-disciplinary orientation. Excessive disciplinary affinity of scientists was the major constraint hindering inter-disciplinary orientation. The senior-level functionaries can be sensitized by organizing discussions and meetings on importance of inter-disciplinary research. The skills of the scientists can be



developed by providing opportunities to experience working in inter-disciplinary projects. It is suggested that the research programmes are developed in inter-disciplinary mode, the process needs to be institutionalized. The majority of the respondents perceived that development of relevant HRD strategies would help in promoting inter-disciplinary research. is the policy intervention required.

Inter-Institutional Collaboration

The intended benefits of inter-institutional collaboration as per the views of majority of respondents of all organizational affiliations are the efficient utilization of facilities and expertise through sharing for mutual benefit. It is reported that, entering into proper MoU for sharing responsibility, benefit and risks is the prerequisite for entering into inter-institutional collaboration in public-private partnership. Purpose and commitment are the key factors for mapping the common interest space in PPP. Agri-business and value chain, biotechnology, seeds and crop varieties/ livestock (species) development are the potential areas for PPP in agricultural research. Relevant policy interventions to promote PPP by institutionalizing the process are the mechanism to be followed to build effective PPP. Differing work environment, perception of competition-inherent weakness, procedural weakness, incompatibility-inherent weakness are the major threats for building inter-institutional collaboration. Maximum percentage of respondents from all organizational affiliations suggested procedural simplification, both administrative and financial as the most important policy intervention.

Procedures for Total Quality Management in Multi-stakeholder and Multi-institutional Consortia-based Research Project

S.K. Nanda

Objective:

- ❖ To develop a sufficiently general analytical and knowledge based innovation management framework and innovation quality metrics that can be applied across different scales and sectors, for assessing the innovations and its quality in various agricultural R&D organizations

Progress:

There is a shift of competition from product or services based concept to innovations. In present scenario, organizations cannot rely alone on product and service quality, but require the combination of quality and innovations for better competitive advantage and sustainable growth. Innovation is a very complex and dynamic process and to gain maximum from its approach there is a need to understand the basic elements of innovations and its quality management. This study presents a general framework aiming to understand the basic elements required to implement innovation quality management process in the organization (Fig. 11). The approach is based on two major dimensions – Innovations and Quality Management. After analyzing the basic concepts of innovation and quality management the interpretation on the concept of “Innovation Quality Management” in contrast of quality management is given.



Principle of Innovation Quality Management

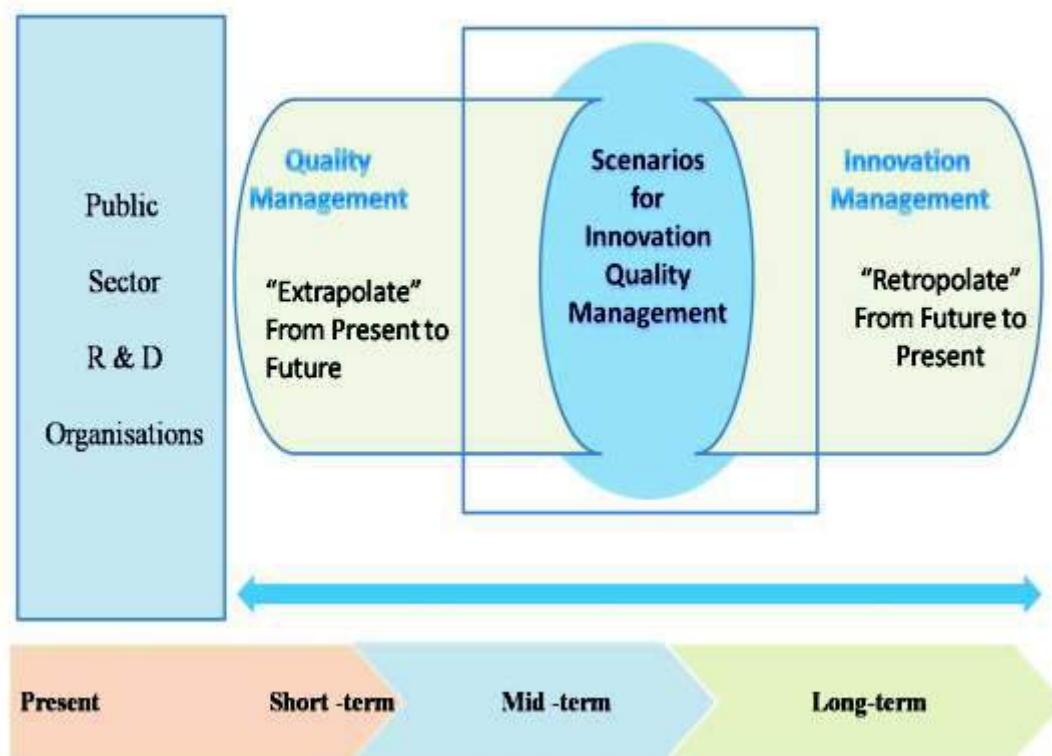


Fig-11: Innovation Quality Management

The innovation quality framework can be effectively used to for assessing, quantify, characterize and analyze the innovative performance and quality of R&D organizations (Fig. 12). In the innovation quality framework, the innovation quality depends on the antecedents related to the following elements: innovation actors, innovation inputs, innovation activities, innovation process, innovation output and innovation impact. The characterizing these elements are critical for assessing value of innovation quality in the R&D organizations and innovation system.



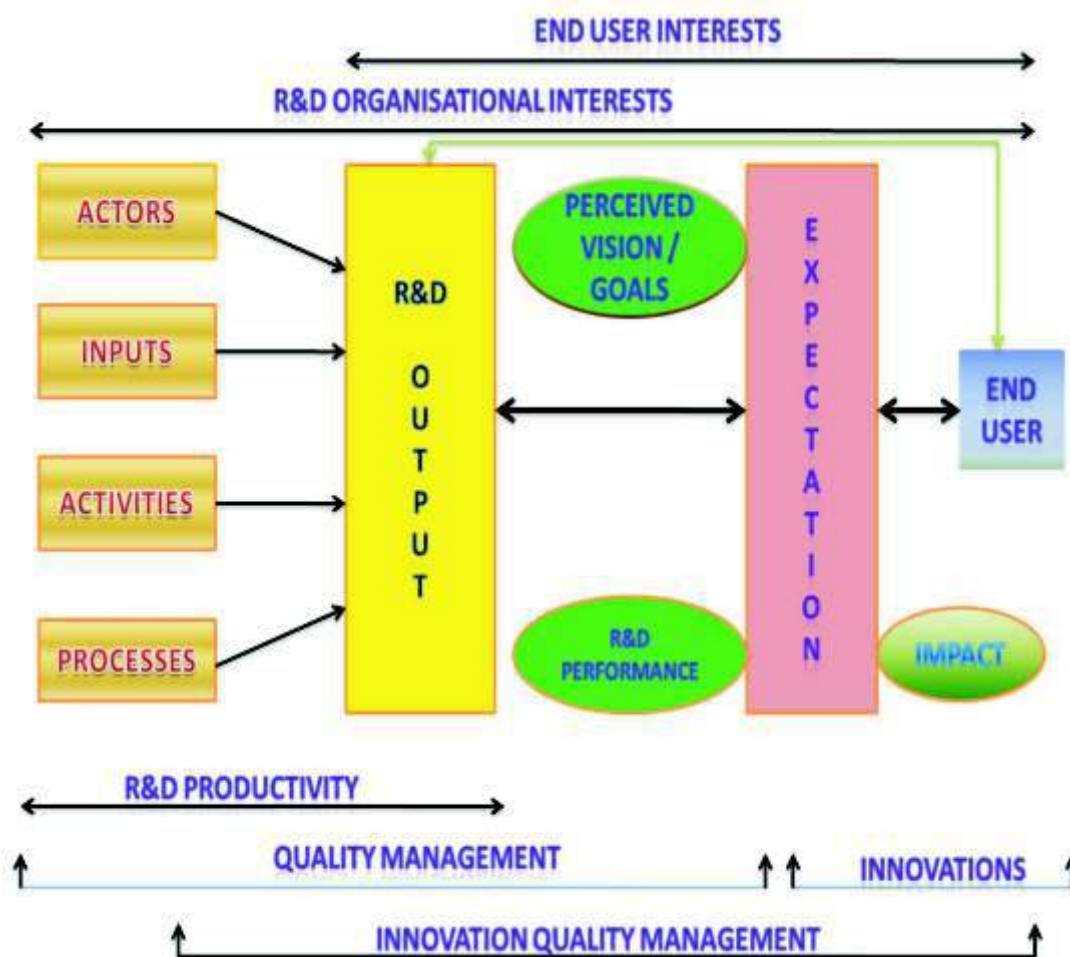


Fig. 12: The Conceptual Model for Innovation Quality Management in Public Sector R&D organizations

From the review of available literature, 95 variables related to the innovation quality in R&D organizations were identified. The elements identified have certain level of inter-relationship and dependency on each other to produce impact on the innovation quality. They collectively contribute for the innovation quality in R&D organizations. These elements provide strategic insights and practical thinking that will help in explaining the involvement of organization structure, work and integrated changes which can be implemented for improving innovation quality in these organizations. A pilot survey was conducted and data was collected from 50 principal and senior scientists working in agricultural R&D organizations in India for scope of improvement and modification if required. After the relevancy test, the data is collected from 613 scientists working in agricultural R&D organizations. From factor analysis, these 95 variables are reduced to 10 factors which represent the metrics for innovation Quality. The work is underway for assigning weights to these elements and validating these elements for developing the innovation quality metric for measuring innovation quality in public sector R&D organizations.



Agri-marketing and Value Chain Management

Policy Support for Strengthening the Value Chain to Address the Challenges of Globalisation

G.P. Reddy, K.H. Rao, M.N. Reddy and N.H. Rao

Objectives:

- ❖ To analyze rural agro-industrial systems and post-farm aspects of the supply chain
- ❖ To identify and analyze the market structure, marketing, processing enterprises and institutional mechanisms needed to raise the access of rural households to markets' information
- ❖ To identify and analyze alternatives for enhancing the contribution to income, food security and employment of the rural poor from value chain management
- ❖ To identify the constraints to the development of value added activities and rural agro based industries

Progress:

In India, there has been considerable growth in organized retailing business in recent years and it is poised for much faster growth in the future. Growing demand for fruits and vegetables due to rising income and change in the consumption pattern coupled with declining farm income due to rising costs and stagnation food grain productivity in recent times, necessitate the shift in production pattern towards high value commodities. Major industrial houses have entered in agribusiness and have announced very ambitious future expansion plans. Transnational corporations are also seeking to invest in India and set up retail chains in collaboration with big Indian companies. However, opinions are divided on the impact of the growth of organized retail in the country. Concerns have been raised that the growth of organized retailing may have an adverse impact on retailer in the traditional retail sector. It has also been argued that growth of organized retailing will yield efficiencies in the supply chain, enabling better access to markets to producers (including farmers and small producers) and enabling higher prices, on the one hand and, lower prices to consumers, on the other. In the context of divergent views on the impact of organized retail, it is essential that an in-depth analytical study on the possible effects of organized retailing in India is conducted.

An attempt has been made to identify supply chains in five important vegetable crops namely tomato, cabbage, egg plant (brinjal), okra and local beans grown in Andhra Pradesh. These vegetables were purposively selected for analysis due to their importance in the production and consumption patterns. The estimates on costs and returns identified different supply-chain systems and associated margins and finally the price integration among different markets for these crops.

Impact on consumers:

1. Consumers have definitely gained from organized retail on multiple counts.
2. Overall consumer spending has increased with the entry of the organized retail.
3. Lower income consumers saved more than other income groups. Thus, organized retail is relatively more beneficial to the less well-off consumers.
4. Proximity is a major comparative advantage of traditional outlets.



5. Traditional retailers have significant competitive strengths that include consumer goodwill, credit sales, and amenability to bargaining, ability to sell loose items, convenient timings, and home delivery.

Impact on Intermediaries:

1. The study did not find any evidence of adverse impact of organized retail on intermediaries.
2. There is, some adverse impact on turnover and profit of intermediaries dealing in products such as fruit and vegetables.
3. Over two-thirds of the intermediaries plan to expand their business in response to increased business opportunities opened by the expansion of retail.

Impact on Farmers:

1. Farmers benefit significantly from the option of direct sales to organized retailers.

Governance and Institutional Arrangements in Areas of Topical Issues

Intellectual Property Management in Public Private Partnerships - Patents, PVP and Copyrights

R. Kalpana Sastry and S.K. Soam

Objectives:

- ❖ To review current experiences in relation to developing patents, PVP and copyrights in Indian agriculture
- ❖ To analyze the effectiveness of current IP policy measures as related to patents, copyrights and PVP operating in public system
- ❖ To develop suitable management options of IPs resulting from blend of resources of PP sectors

Progress:

A review of the relevant legal and policy documents was done to understand the background of the obligations at national and at the international level. This was followed by a brief review of the role and functions played by some statutory agencies in India that are actively involved in IP facilitation. Later the National Agricultural Research System was studied with an intention to understand the complexities and magnitude *vis-a-vis* the IP policy. Then the provisions and governance model of the new IP policy of the ICAR was analyzed. Two constituent institutes namely Project Directorate of Poultry (PDP) and Directorate of Rice Research (R), animal-based and crop-based institutes respectively, were studied in detail from the IP policy perspective. The study highlights on the implementation of guidelines, structural adjustments in decision making activities in IP management at institutes and at understanding the specific issues of IP management at these institutes.

IP Management in Public Private Partnerships - Agro-biodiversity, Geographical Indications and Traditional Knowledge

S.K. Soam and R. Kalpana Sastry



Objectives:

- ❖ To develop comprehensive understanding of GIs/agro-biodiversity/traditional knowledge with respect to their availability and suitability to enhance the socio-economic conditions and livelihoods security of the rural poor
- ❖ To provide policy support for development of the integrated mechanism that protect GI, agro-biodiversity and traditional knowledge, and also incorporate innovative interventions for the socio-economic upliftment of rural poor and farmers

The analysis was done through development of case studies; the data was collected through administration of questionnaire, personal interviews and meeting with key stakeholders. The key findings are given below:

- ◆ The registration of agricultural products as GIs has become possible as a result of personal interests of few professionals/individuals. There are no policy guidelines of most of concerned departments/organizations/corporations as GI matters.
- ◆ There is no administrative process, which establishes that the applicant is the 'true representative' of interest of GI producers. Interventions of state governments must be at some point of process of GI registration. There should also be mechanism to incorporate the reputation of the applicant's organization in GI registration process with respect to social responsibilities fulfilled by them in the recent past.
- ◆ Post-registration efforts related to marketing are very poor. The product niche e.g. medicinal property of navara rice and organic nature of pokkali rice are the marketing strength.
- ◆ IP enforcement is almost absent. The efforts to assert the rights have not been observed e.g. the website 'www.pokkali.com' must be objected by the registered owners of Pokkali rice.
- ◆ Lot of research initiatives are required to establish the uniqueness of the agricultural products. For example from the consumers' point of view, what is the scientific basis of characterizing products? And whether competent authority approves that particular justification? In case of 'Allahabad surkha', the GI registry document says peel color 'pink' but in the image provided in the document, the peel color is yellow. Further it says the color of flesh whitish sometimes pink, the statement would create total confusion to the consumers. In the document elsewhere the Allahabad surkha is equated with 'Apple color variety', while the scientific literature suggests that 'Apple color variety' is a specific germplasm having pink color peel looking like apple.
- ◆ Scientific rigour in establishing uniqueness and specific characteristics of the products is very poor in terms of scientific analysis and past research review.
- ◆ Defining and documenting production protocols and grading & standards protocol is the big scientific challenge because producers hardly do any type of grading or packaging, if do, it is mostly based on the physical parameters.
- ◆ The study revealed that in none of the cases, no producers have applied for membership or NOC to become the registered producer. It's a challenge for GI development but GI can also be used to create and strengthen such commodity based groups.





Apple color variety at Allahabad, the unripe fruits are green

 Government Of India Controller General of Patents Designs and Trade Marks GI Application Status Search		 Intellectual Property INDIA Patents Designs Trademarks and Geographical Indications
Friday, March 19, 2010 1:14 PM		
GI Status Report		
Application Number	50	
Geographical Indications	Allahabad Surkha	
Applicant Name & Address	Diversified Agriculture Support Project, 4th Floor, PICUP Bhawan, Vibhuti Khand, Gomtinagar, Lucknow - 226010	
Date of Filing	13.02.2006	
Class	31	
Goods	Agricultural	
Geographical Area	Uttar Pradesh	
Status	Registered	
Image		
Back		

Whether Allahabad surkha has yellow peel

Annexure -1.

List of Research Projects

Project Title	Project Team	Sponsor
Simulation Model for Quality Assessment of Agricultural Colleges	S.K. Nanda, D. Rama Rao and J. Challa	NAARM
Performance Assessment of Agricultural Universities	J. Challa	NAARM
Agroweb - Digital Dissemination System for Indian Agricultural Research	G.R.K. Murthy, D. Rama Rao, N. Sandhya Shenoy and P. Mohan Singh	NAIP
Assessment of Developments in Nanotechnology for Agricultural R&D	N.H. Rao and R. Kalpana Sastry	NAIP
Geospatial data and knowledge organization systems for agricultural research, education and technology transfer (Geospatial Library and PGIS)	N.H. Rao, M.N. Reddy, G.P. Reddy, K.H. Rao, S.K. Soam and K.V. Kumar	NAIP
Assessment of Future Human Capital Requirements in Agriculture	D. Rama Rao, S.K. Nanda, Sandhya Shenoy, Bharat S.Sontakki and G.P. Reddy	NAIP
Strengthening Statistical Computing for NARS NAARM - Consortium Partner of the project under component 1 of NAIP)	M.N. Reddy, A. Dhandapani and G.P. Reddy	NAIP
Learning and Capacity Building Project	N.H. Rao	NAIP

Sub-projects under Learning and Capacity Building (L&CD) of NAIP

Sl.No.	Sub-project Title	Project Team
1.	Organizational change for promoting innovation through research consortia	PI: P. Manikandan
	A. Change management	P. Manikandan and R.V.S. Rao
	B. Leadership effectiveness	R.V.S. Rao and P. Manikandan
	C. Performance Enhancement	R.V.S. Rao and P. Manikandan



	D. Assessment and Development of organizational citizenship behaviour	K.H. Rao, R.V.S. Rao, G.P. Reddy and B.S. Sontakki
2.	Promotion of inter-disciplinary orientation and inter-institutional collaboration in NARS	T. Balaguru (till June 2009), P. Manikandan and S.K. Nanda
3.	Assessment and Development of Organizational Citizenship Behavior for promoting efficiency and effectiveness in NARS	K.H. Rao, R.V.S. Rao, B.S. Sontakki, G.P. Reddy
4.	Training Needs and Impact Assessment of L& CB under NAIP	B.S. Sontakki; Co-PIs: T. Balaguru (up to June 2009), K.H. Rao, G.P. Reddy and J. Challa
5.	Information and Communication Technologies (ICTs) for promoting agricultural innovation	PI: M. N. Reddy
	A. Strategies to promote use of ICTs in agriculture value chains	D. Rama Rao and B.K. Sikka
	B. Job training interventions through distance digital mode for enhancing the effectiveness of research and educational institutes	M. N. Reddy, J. Challa and A. Dhandapani
	C. IT-based decision support system for effective knowledge technology transfer	G.R.K. Murthy, K.M. Reddy
	D. Empowering rural women through Information and Communication Technologies (ICTs)	N. Sandhya Shenoy, V.K.J. Rao
	E. Digital multimedia for agri-innovation transfer	K.M. Reddy, V.K.J. Rao and G. R. K. Murthy
	F. Digitally enabled customization of information for decision and empowerment	V.K. J. Rao, N. Sandhya Shenoy and K.M. Reddy
6.	Policy studies for promoting research and innovation in agricultural value chains	PI: G.P. Reddy
	A. Promotion of interdisciplinary orientation and inter-institutional collaboration in public private partnerships in NAIS	P. Manikandan and S.K. Nanda
	B. Policy support for strengthening the value chain to address the challenges of globalisation	G.P. Reddy, K.H. Rao, M.N. Reddy and N.H. Rao



	C. Intellectual property management in public private partnerships - Patents, PVP and Copyrights	R. Kalpana Sastry and S.K. Soam
	D. IP management in public private partnerships - Agro-biodiversity, Geographical indications and Traditional Knowledge	S.K. Soam and R. Kalpana Sastry
7.	Knowledge systems for research policy and management for sustainable rural livelihoods security	PI: N.H. Rao
	A. Knowledge discovery and knowledge management tools to characterize livelihoods systems	M.N. Reddy, N.H. Rao, G.P. Reddy, K.H. Rao, K.V. Kumar
	B. Spatial framework for assessing vulnerability and adaptive capacity of livelihoods systems to climate and market stressors	N.H. Rao, M.N. Reddy, G.P. Reddy, K.H. Rao and K.V. Kumar
	C. Technology delivery models for less favoured areas	B.S. Sontakki
	D. Develop procedures for total quality management in multi-stakeholder and multi-institutional consortia-based research project aimed at improving agricultural innovations for livelihoods security	S.K. Nanda



Education



Education

In line with its new vision to be a global player in agricultural management education, and the approved strategy for the XI Plan, NAARM has launched the following post graduate education programmes:

Sl.No.	Title	Duration Participants	Number of	Programme Director(s)
1	One-year PG Diploma in IT Management in Agriculture (PGD-ITMA)	November 11, 2008 to November 10, 2009	24	D. Rama Rao
2	One-year PG Diploma in Intellectual Property and Technology Management in Agriculture (PGD-IPTMA)	January 28, 2009 January 27, 2010	16	R. Kalpana Sastry
3	Two-Year PG Diploma in Management (Agriculture) – PGDMA (2009-11 Batch)	July 25, 2009 – July 2011	24	G.P. Reddy K.H. Rao

All the three programmes have benefited greatly with vital inputs from and strong involvement of key stakeholders in the agribusiness, information technology, technology management, IP law, industry, retail, agri-finance and other sectors. The PG Diploma in Management (Agriculture) was launched after obtaining approval from AICTE. The first batch of 24 students was admitted in July 2009. The all-India competitive examination for admission to the second batch in July 2010 (NAARMAT-2010) was held in March 2010. Highlights of the programmes are given below:

Post Graduate Diploma in Information Technology Management in Agriculture (PGD-ITMA)

In translating its vision into an action plan in the XI plan, NAARM has initiated one year PG-Diploma in Information Technology Management in Agriculture (PGD-ITMA) on November 11, 2008. The rapid developments in information technology and its application in enhancing the productivity of agriculture and allied sectors, and the need to apply these technologies in agriculture have necessitated this course. Academy designed and developed the curriculum through deliberations with senior executives from agri-business industry, financial institutions, NGOs, consultants, academicians and policy-makers in government. Professional support from private sector was sought to translate the needs into an effective curriculum with a blend of skill, attitude and knowledge.

The programme aimed to build bridge professionals who are information and systems specialists to successfully hold first level managerial positions in the agri-business and related sectors. To fulfill this need, the faculty members were drawn from public and



corporate sectors related to agriculture and Information Technology (IT). Besides conceptual inputs on IT Management in Agriculture, the academic requirements included practical training, work visits, seminar, group assignments, project, internship, etc. As an integral part of curricula, pre-internship and internship placements for in-company and on-field trainings were provided. Placement assistance was also provided to students.

This one-year programme comprised four trimesters, each of three months. First three trimesters were conducted at the Academy. The fourth trimester was devoted to internship pursued by the students at various identified organizations.

Courses Offered

- Issues in Agriculture and Rural Development
- IT Overview, Computer Networking & Security
- OOPS Based SDLC and Project Management
- Architectures, Data Management & Quality Systems in IT
- Web Based Development with J2EE
- Quantitative Methods in Agriculture Management
- Principles of Agriculture Economics & Marketing
- Geographic Information Systems in Agriculture
- Digital Media Resources & Web Applications
- Principles of Organization & Management
- IT based Information / Decision Support systems in Agriculture & Rural Development
- Designing and documenting a Web Based Agriculture Information System with Regional Language Support
- Entrepreneurship Techniques
- ICT Case Studies – Electives
- Project Work
- Internship

Faculty

The faculty members were drawn from public and corporate sectors related to agriculture and IT. The Academy's faculty members, who are highly qualified with rich expertise in information technology, education management, strategic management and human resource planning and development, also provided adequate resource input in the course.



List of Guest Faculty

Name of the Expert

Dr P. Jyothi Prasad, School of Management, Hyderabad University, Hyderabad

P.S. Narayana, M/s R Subrahmanyam & Company Chartered Accountants, Hyderabad

Smart Steps Placement consultants

Lavanyakumar Nanabala, Managing Partner Human Capital Consulting
SMART STEPS, Hyderabad

Sreenath Raju, CEO, SMART STEPS, Hyderabad

Srinivas (Food technologies, CFTRI), Managing Partner, SMART STEPS, Hyderabad

Sanjay Borkar, CEO, Farm ERP, Shivraj Technologies, Pune

Santosh Shinde, CEO, Farm ERP, Sevanand Society, Pune

Vijay Jesudassan, A.G.M., National Operations & Development, Nagarjuna Fertilizers
and Chemicals Ltd / I-kisan Ltd, Hyderabad

sudhakar sathyanarayanan

K. Venkata Raju, Manager, ILFS, Hyderabad

P. Srinivas, Sr. Manager, ILFS, Hyderabad

Kavitha, Manager- Rural Health & Education Services, ITC, Secunderabad

Dr A. Amarendra Reddy, ASCI, Hyderabad

Nagarjun Malladi, Hyderabad

N. Vishram, Hyderabad

D.S.K. Rao, Hyderabad

K. Gangadhar, Hyderabad

Oruganti Srinivas, Hyderabad

U. Ramana (Satyam), Hyderabad

Dr I. Prince Devadason, Scientist, NRC MEAT, Hyderabad

K.V. Sudhakar Reddy, Hyderabad

Pridhvi, Hyderabad

K. Vizayakumar, VJIT, Hyderabad

Ms. Suma Murthy, Hyderabad

P. Bhanumurthy, Secunderabad

Kishore Mocherla, Hyderabad



Placement

The students were placed in various organizations, predominantly private sector agribusiness firms. These were:

- ❖ Microsoft
- ❖ Monsanto
- ❖ Jolary
- ❖ ICRISAT
- ❖ deVGen
- ❖ NCMSL
- ❖ Spandana
- ❖ Vista
- ❖ Du Pont
- ❖ Coramondel
- ❖ Drishti
- ❖ ILFS
- ❖ NIF
- ❖ NFCL

Board of Studies-PGITMA

The members of the Board of Studies for PGD-ITMA were as follows:

Dr D. Rama Rao	Chairman
Dr M.N. Reddy	Member
Dr K.M. Reddy	Member
Dr R.V.S. Rao	Member
Dr G.R.K. Murthy	Member-Secretary

The Board of Studies is responsible for overseeing the curriculum design, implementation and evaluation of the PGD-ITMA. The Diploma Award Ceremony was held on November 17, 2009. Dr G. Ravi Prasad, Senior Vice President (Sales and Marketing), Coromandal International, Hyderabad, was the Chief Guest on the occasion. Dr P.K. Joshi presided over the function. This programme has reinforced the public-private partnership initiative of the academy and also expanded its stakeholder base. The Academy's aim of placing the students in reputed organizations engaged in agri-business and rural development in the country was fulfilled through this course. Dr D. Rama Rao, Head, Information and Communication Management Division, was the Course Director for this programme.

Post Graduate Diploma in Intellectual Property and Technology Management in Agriculture (PGD-IPTMA)

NAARM initiated this one-year programme to train the agriculture and other science graduates in Intellectual Property and Technology Management in Agriculture (IPTMA). This course was first of its kind in the country and offered an excellent opportunity to systematically study the subject of Intellectual Property (IP) and Technology Management in Agriculture. The curriculum was designed with the help of leading experts from the academia of National Law Schools, research institutions in public and private sector, Law firms and KPOs with a focus on building strong linkages with the corporate sector, research institutions, and statutory and regulatory bodies. The aim of the course was to train selected students to enable them to successfully hold first- level executive position in



intellectual property and technology management assignments in the agriculture and related industries, and also to confidently chart their path in this area. Thus, it was structured to create 'bridge professional' who could develop IP assets for leveraging them for technology dissemination in the agri-food sector.

Programme Framework

This fully-residential programme was spread across four trimesters with a stress on various issues of intellectual property (IP) and technology management in agriculture along with internship in last trimester. Besides the conceptual inputs on IP and Technology Management in Agriculture, the academic inputs included practical training, work visits, seminars, group assignments, project, and internship.

Some Highlights

- ▶ Curriculum designed with the help of experts from cross-sectoral panorama including National Law Schools, research institutions, law firms, technology transfer consultancy firms, and others
- ▶ Core faculty drawn from the Academy
- ▶ Guest faculty was from research institute, industry, business groups, legal-organizations, National Law Schools, technology consulting firms and KPOs
- ▶ Students were exposed to real life challenges through interactions with experts and working professionals and internships
- ▶ Programme was delivered in the form of class room lectures, tutorials, group discussions, seminars, laboratory work, assignments, case studies, library research projects and other exercises including field visits
- ▶ Emphasis was given on achieving a fine balance between knowledge, skills and attitude
- ▶ Short course modules on specialized topics like agri-entrepreneurship, patent drafting, prosecution of patents, etc. conducted by internationally reputed faculty (Cornell University, US patent agents)
- ▶ Post third trimester, internship for 10 weeks under the guidance of functional area experts in the corporate sector

Courses offered

- Perspectives in Agriculture Systems and Technologies
- IT Applications and Agriculture Bioinformatics
- Project Management
- Jurisprudence of Intellectual Property Rights
- International IP Regime



- IP Regime in India
- Communication Skills
- Managerial Economics for Agri-business Management
- Laws Interfacing IP Regime
- Technology Forecasting
- Principles of Organization and Management
- Quality, Safety, Standards and Regulatory Mechanisms
- Quantitative Methods
- IP Information Management, Patent drafting, Filing and Registration
- Innovation and Technology Development
- IP Prosecution and Enforcement
- Venture Management, Technology Incubation and Entrepreneurship
- Technology Marketing
- Technology Landscaping and Risk Management
- Business Communication

Some of the major activities undertaken during the course included:

- a. Academic Blog: Students started an academic blog and continue to contribute brief articles. The external faculty, Prof V.C.Vivekanandan of NALSAR along with Course Director oversees this activity. The aim is to sensitize on issues of topical interest in the field of IP and develop the writing skills for communicating technical opinions of evolving cases and issues in IP on electronic media.
- b. Students participated in the National Seminar on WTO-TRIPs with special reference to transactional IP issues on May 3 and 4, 2009, which was sponsored by the Ministry of HRD and NALSAR University.
- c. Celebration of World IP Day at the Academy: The students of PGD-IPTMA celebrated this event by hosting a cultural-cum-literary event “*Sankalan*” on April 28, 2009. The event had a tremendous participation from students and trainees of other courses at the Academy.
- d. A special workshop was conducted on 'Agri-entrepreneurship' by Dr Richard Cahoon, Cornell University. The students were engaged in formulating business plans for start-up ventures on agri- based patented technologies.
- e. Two workshops on patent drafting were conducted with CII-APTDC where US patent agents were invited as guest faculty



List of Guest Faculty

KPO's, LPO's, Attorneys:

- ? Prof V.C. Vivekanandan Rajiv Gandhi School of Intellectual Property Law
- ? Prof VK Unni, IIM, Kolkata
- ? Prof S.Acharyalu, NALSAR University of Law
- ? Ms U. Parameswaran and
- ? Mr Manoj Pant, Scitech Patent Art Services Pvt. Ltd
- ? Mr Raju Bhupathiraju FoxMandal Little Solicitors and Advocates
- ? Dr Gopakumar Nair, GNS Associates
- ? Mr R.Muralidharan, Krishna & Saurastri Associates
- ? Dr Kaushalya Santhanam, Scivista IP and Communication
- ? Dr Anindya Sircar, Biocon Pvt Ltd

Academic Research Organizations:

- ? Dr S. Kochhar, IPR Unit, ICAR
- ? Dr Shobha Rani, Directorate of Rice Research
- ? Prof R.M.Prasad, National Institute of Rural Development
- ? Dr P. Jyothi Prasad and
- ? Dr Chetan Srivastava, University of Hyderabad
- ? Dr Prabhakar, Directorate of Oilseeds Research
- ? Dr Suhasini and Dr M. Ganesh, Archarya N.G. Ranga University (ANGRAU)
- ? Dr Subbaiah Arunachalam, MS Swaminathan Research Foundation

Industry- Seed/Pharmaceuticals/Biotechnology:

- ? Dr T.V. Ramanaiah, PHI Seeds Limited
- ? Dr N.P. Sarma, Kaveri Seed Company Limited
- ? Dr V. Shenoy, Monsanto India

Tech Transfer/Management Offices:

- ? Mr Subodh Kumar, CII- APTDC
- ? Mr K. Vijayaraghavan, Mr Prabhu Ram, Mr Akshat Medekkar, and Mr Prabir Mishra, Sathguru Management Consultants
- ? Brig Ganesham, AP Honey bee Network, (NIF)
- ? Mr N.Lavanya Kumar, Smart Steps
- ? Mr Pruthi, V3 Consultants
- ? Mr Bhanumurthy, PMI consultancy

International Faculty:

- ? Dr. Victoria Henson Apollonio, CAS-IP, Italy
- ? Mr. Joe Root, Qaulipat, USA
- ? Dr. Richard Cahoon, R&S Associates, USA
- ? Dr. S.M. Virmani, Formerly of ICRISAT



Internships	Placements
Advanta India Ltd.	John Deere Technology Center (India), Pune
FoxMandal Little Law Firm	Global Institute of Intellectual Property (GIIP), New Delhi
John Deere Technology Center (India)	Adhikari Intellectual Property Consultants Pvt. Ltd., Hyderabad
K&S Partners	Skyquest Technology Mgmt., Ahmedabad
KPIT Cummins Infosystems Ltd	Tech Bank, NIRD, Hyderabad
Monsanto India	Global IP solutions, Bangalore
Rural Technology Park, National Institute of Rural Development (NIRD)	SGS Company, Mumbai and Hyderabad
Sathguru Management Consultants	Nuziveedu Seeds (NSL), Hyderabad
Scitech Patent Art Services	Legalisis Co., Pune
CSIR Unit for Research and Development of Information Products (CSIR-URDIP)	National Innovation Foundation (NIF), Ahmedabad
CII-APTDC	BPD Units, ICAR

Board of Studies -PGIPTMA

The members of Board of Studies for PGD-IPTMA were as follows:

Dr T. Balaguru	Chairman
Dr R. Kalpana Sastry	Member
Dr M.N. Reddy	Member
Dr K.H. Rao	Member
Dr S.K. Soam	Member-Secretary

The Board of Studies was responsible for overseeing the curriculum design, implementation and evaluation of the PGD-IPTMA.

The course was successfully completed on February 11, 2010. Dr Biswajit Dhar, Director-General, Research and Information Systems for Developing Countries (RIS), New Delhi, was the Guest-of-Honour on the Graduation Award Ceremony. Mr R. Rajamani (IAS), Member, Board of Directors, ICICI Knowledge Park, Hyderabad, was the Chief Guest on the occasion. Dr P.K. Joshi, Director, NAARM, presided over the function and conferred Diplomas to all the students. Dr Dhar presented the gold medals to Mr Hari Mohan Naidu and Ms Ruchika Goel, and silver medal to Ms Poorva Rati.

Dr R. Kalpana Sastry, Head, Agricultural Research Systems Management and Policies Division, was the Course Director for PGD-IPTMA.



Post Graduate Diploma in Management (Agriculture)-PGDM (A)

The two-year Post Graduate Diploma in Management (Agriculture) was launched by Dr Mangala Rai, Secretary, DARE, and DG, ICAR at the Academy on July 25, 2009, Twenty four students were admitted in this course. Dr S.P. Tiwari, DDG (Education), ICAR, was the Guest-of-Honour on the inaugural ceremony. Dr S.M. Ilyas, Director, NAARM, presided over the function. The course has completed two trimesters during the period under report. The courses offered during the three trimesters are indicated below.

Courses Offered
Institutional and Policy Environment
Agricultural and Food systems
Business Statistics
Computers and Information Systems
Managerial Economics-I
Marketing Management
Business communication-I
Management of Information system
Managerial Economics
Legal Environment and Business Ethics
Quantitative Methods for Business Management
Business communication-II
Human Resource Management-I
Financial Accounting and Management
Supply Chain Management
Project Management
Management accounting
Organizational Behaviour
Business communication-III
Marketing Research
Technology Management

Students will attend summer internship in following organizations:

- ❖ Coromandal Fertilizers Ltd.
- ❖ Ganesh International (Spices exporter) Ltd.
- ❖ Hyderabad Foods Ltd
- ❖ INTAS Pharmaceuticals Ltd
- ❖ ITC Ltd.
- ❖ John-Deere (Tractor division)
- ❖ Mahyco



- ❖ MCX
- ❖ Nuziveedu Seeds Ltd.
- ❖ Reliance Retail Ltd.
- ❖ SGS Agri-food Services
- ❖ SIDBI
- ❖ Yes Bank

Several guest faculty from industry, academia and other stakeholders were part of the course.

- ◆ Dr Chetan Shrivatsava, Faculty of Management, Central University, Hyderabad
- ◆ Dr Laxmi Mantha, Osmania University, Hyderabad
- ◆ Prof. S.V.N. Sachendra, Administrative Staff College of India, Hyderabad
- ◆ Shri. Venkateswarlu, Former Director (Finance), National Mineral Development Corporation (NMDC), Hyderabad
- ◆ Shri. Kameshwar Rao, Former Manager (Finance)
- ◆ Shri. Bhanumurthy, faculty, PME
- ◆ Dr P.G. Chengappa, Vice-chancellor, University of Agricultural Sciences, Bangalore
- ◆ Dr Srinivasa Gowda, Former Head, Agricultural Economics Department, University of Agricultural Sciences, Bangalore
- ◆ Shri. Ashok C. Khosla, Agri-business consultant, Adjunct faculty, NIAM
- ◆ Shri. Ramakrishnan, Executive Director, SIDBI
- ◆ Shri. Rajendra Prasad, Nadella, Director, Indian Retail School
- ◆ Shri. Sundara Rajan, Former GM (Marketing), Parle Foods
- ◆ Dr Jagrook, Faculty, ICFAI Business School
- ◆ Dr Shanmugam, MCX (Agriculture Economics Specialist)
- ◆ Shri. P B Suryaprakash Rao, Adviser Priya Foods

International level guest-faculty:

- ◆ Prof. Vithala Rao, Cornell University, USA
- ◆ Dr Ashok Gulati, International Food Policy Research Institute

Board of Studies:

Following was the Board of studies for PGDM-A

Dr P. Manikandan	Chairman
Dr Jagannadham Challa	Member
Dr M.N. Reddy	Member
Dr S.K. Nanda	Member
Dr G.P. Reddy	Member-secretary

The Board of Studies was responsible for overseeing the curriculum design, implementation and evaluation of the PGDM (Agriculture).

Dr G.P. Reddy and Dr K.H. Rao, Principal Scientists, were the Course Directors for PGDM (Agriculture).



Consultancy



Consultancy

I. Agribusiness Knowledge Centre

Against a premise that establishment of a Agribusiness Knowledge Centre (AKC) can provide a conducive ecosystem that will nurture and facilitate value addition, business development and growth among the stakeholders across the agricultural supply chain, the Academy has established a centre which seeks to link stakeholders across the country and the world. The objective was to provide an enabling environment for establishing actual businesses through knowledge sharing. A state-of-art ICT based Agribusiness Knowledge Exchange (AKX) platform is being planned with the specific aim to facilitate registration into the Agribusiness Knowledge Center, Exchange of knowledge between different registered stakeholders, start discussion blogs, post knowledge sharing experiences, initiate/monitor commercial transactions between different stakeholders including web based monitoring on almost real time basis of the business transactions (Supply Chain, GIS based crop systems etc). The AKC and AKX will provide a platform for NAARM and agribusiness companies to establish backward and forward linkages to the research and education programme. This facility was also planned for providing a practical interface for students of ongoing educational programmes at the Academy.

II. Help Desk support for developing and managing consortia research projects of NAIP:

The objective of the Help Desk is to assist the prospective consortia partners of the NAIP sub-projects - public, private or civil society organizations in understanding the philosophy and requirements of NAIP, understand how best to compete for the sub-projects through presenting effective concept notes and full project proposals and, for sometime after the award of the projects, in overcoming the teething troubles in running the consortia. The help desk has provided handholding support for proposal development for all proposals approved under components 2,3, 4 of NAIP in 2009-10. In this year the help desk also initiated the second phase of its activities, namely, providing on-line support in project implementation. The sub-projects are now in execution phase and most queries pertain to monitoring and evaluation and submission of periodic reports. Once the query is received, it is analyzed and an appropriate solution is suggested through e-mail. The help desk also maintains a web portal for on line information dissemination and project support.

III. IP Management and Transfer/Commercialization of Agricultural Technology Scheme

April 2008-2012, Funding: ICAR

Objectives

- ❖ To study the literature and review the evolving trends of IP management and technology transfer in agriculture-based public sector institutions.
- ❖ To understand issues of automatic patent analysis (APA), and use it as benchmark as a R&D indicator for developing strategies to acquire and use technologies in agri-R&D.



- ❖ To develop resource materials on knowledge protection strategies for project management in emerging areas.
- ❖ To develop repository of prior art databases in emerging areas viz. agri-biotechnology.
- ❖ To study the literature and review the evolving trends of IP management and technology transfer in agriculture- based public sector institutions.
- ❖ To understand issues of automatic patent analysis (APA), and use it as benchmark as a R&D indicator for developing strategies to acquire and use technologies in agri-R&D.
- ❖ To develop resource materials on knowledge protection strategies for project management in emerging areas.

Achievements:

With rapid technological development and innovations occurring in Bt technology, the Bt stakeholders need to be aware of the usefulness and availability of patents associated with Bt technology. Therefore, project has initiated studies on IP landscaping of Bt cotton patents. Bibliometric, technology assessment analysis and citation analysis is being undertaken using relevant patent data sets. The database s used for this include commercial providers like QPAT, Delphion, Micropatent and derwent apart from databases of National Patent offices.

Initial results indicate that quantum of Bt cotton patents has tremendously increased from 1997, which is coinciding with the technological breakthrough in Bt technology, where all the major seed companies started up with their own R&D in this area. Maximum patents have been recorded in 2009.





Major Events / Achievements / Recognitions

Major Events / Achievements / Recognitions

Public-Private Partnership (PPP) Agribusiness Knowledge Centre

The Academy has entered into a Public-Private Partnership (PPP) mode to conceptualize and put into operation, an Agribusiness Knowledge Centre (AKC) to facilitate knowledge flows and network among stakeholders in the agri-value chain and provide need-based knowledge services.

The AKC is the outcome of the recognition of the need to create a platform for collaboration among the wide diversity of stakeholders across the agri-supply chain to consolidate the fragmented information chain and carry out business more effectively in a mixed policy environment. It consists of two components:

- (i) A physical component to house some of the stakeholders from the private sector, NGOs and farmer organizations at NAARM and to provide knowledge services like training, data products, and others.
- (ii) A virtual component, called the Agribusiness Knowledge Exchange (AKX) to orchestrate the collaboration interactively and on-line to provide value adding knowledge to the stakeholders.

Stakeholders from agribusiness would be expected to register both for the physical space in the AKC and the virtual space in the AKX after paying a fee. A significant share of the stakeholders would comprise emerging Small and Medium Enterprises (SMEs) that are expected to aggregate farmers and link them to larger processors and retailers, and those that provide a number of other local value added services. The resources generated from training and other activities of its physical component, and the additional value generated to the stakeholders as result of knowledge sharing on the AKX, would be the prime sources of income for the AKC.

The following are the broad activities of the AKC:

1. Organizing short term specialized training programmes for industry and other stakeholders
2. Undertaking contract research and consultancy for industry
3. Organizing national and international seminars and demand-driven events development activities and surveys
4. Facilitating and conducting primary surveys, and collating published information for informed decision support

More importantly, the AKC and AKX will provide a look-in opportunity to students and faculty of NAARM, and scientists of ICAR, to understand the processes of knowledge flows in the agri-food supply chain and lead to more demand-driven research within NARS. It will also be a window for access of industry to institutions of ICAR for research services and for the institutions of ICAR to industry for commercialization of research.



Recognitions

Dr P.K. Joshi, Director:

A. National Recognitions

- ▶ Chief Guest at National Institute of Nutrition, Hyderabad on the eve of World Food Day 2009, organized by the Association of Food Scientists and Technologies; delivered key note address on 'Achieving Food Security in Time of Crisis' (16 October 2009).
- ▶ Chief Guest at Dr Panjabrao Deshmukh Krishi Vidyapeeth (Akola) to inaugurate the Winter School on 'Emerging Issues in Agribusiness and Marketing' (2 December 2009)
- ▶ Chief Guest and key note speaker at Pondicherry University (Pondichery) to inaugurate workshop on 'Commodity Futures for Trainers' (8 January 2010)
- ▶ Chief Guest at Junagadh Agricultural University for 5th Annual Convocation of the University (29 January 2010)
- ▶ Chief Guest for a Training Program on 'Maize Singh Cross Hybrid Seed Production Technology' organized by the Directorate of Maize Research in Hyderabad (12 March 2010)
- ▶ Chief Guest at Project Directorate on Poultry (Hyderabad) to inaugurate and deliver key note address on 'Climate Change and Possible Impacts on Poultry Industry' (27 March 2010)
- ▶ Panelist on 'On-line Examination' organized by the Agricultural Scientists Recruitment Board, New Delhi (9 November 2009)
- ▶ Key note speaker in the National Conference on 'Operationalization of National Policy for Farmers' (ANGRAU, Hyderabad) on 'Bottom-up Prioritization of R&D Agenda in Agriculture', organized by Indian Agricultural University Association (2 March 2010)
- ▶ Key note speaker in the 33rd Session of Indian Social Science Congress at B R Ambedkar Open University (Hyderabad) on 'Indian Agriculture at the Crossroads' (12 March 2010)

B. International recognitions:

- ▶ Member, International Steering Committee on Climate Change, Agriculture and Food Security of CGIAR Challenge Program and Earth Sciences.
- ▶ Member, Independent Evaluation Group of the World Bank on 'Agriculture and Agribusiness', Washington, DC.

C. Recognitions by professional societies

- ▶ Re-elected Secretary, Agricultural Economics Research Association (India)
- ▶ Reviewer, Indian Journal of Agricultural Economics; Agricultural Economics Research Review; Food Policy; Land Economics, etc
- ▶ Member, Board of Management, Acharya N.G. Ranga Agricultural University, Hyderabad

Dr N.H. Rao, Joint Director:

- ▶ Invited reviewer of international journals :
 - (i) Environmental Technology (Springer)
 - (ii) Water Resources Management (Springer)
 - (iii) International Journal of Environmental Management (Elsevier)
 - (iv) Livestock Science (Elsevier)



- ▶ Member, Consortium Advisory Committee of the CRIDA subproject under Component 4, entitled 'Development of Decision Support Systems for Insect Pests of Major Rice and Cotton-based Cropping Systems'
- ▶ Member, Academic Council, Kerala Agricultural University
- ▶ Member, Committee on ICAR 2030
- ▶ Member, Committee on ICAR Rules and Guidelines for Professional Service Functions.

Dr R. Kalpana Sastry, Head, ARSMP division:

- ▶ Expert Member, AP Biodiversity Board from 2008 onwards.
- ▶ Referee for Food Policy Journal (Elsevier)
- ▶ Invited Panelist in Session on Agricultural Nanotechnology, Organization for Economic Co-operation and Development (OECD), Paris
- ▶ Invited in Nano applications panel session to speak on Role of nanotechnology in agriculture and food security planning in developing countries. Center for Nanotechnology in Society University of California, Santa Barbara and co-organizer of the "Emerging Technologies: Emerging Economies: Nanotechnology for Equitable Development conference"
- ▶ Vigilance Officer, ICAR
- ▶ Paper setting and Evaluation of papers- FBM 510 Course-MFSc Sem I for WTO and IP course, CIFE, Mumbai
- ▶ Invited as expert member in ITMU of DOR, CRIDA, PDP, NRC-Meat, IIHR from 2008 to till date
- ▶ TV Talks: Invited as panelist in panel discussion on World IP Day by Doordarshan, Hyderabad. Programme aired on April 26, 2009

Sports Achievements

ICAR Inter-Zonal Sports Tournament

NAARM contingent took part in the ICAR Inter-Zonal Sports Tournament held at NDRI, Karnal from December 12 to 15, 2009. Mrs Rukmini Ammal, Personal Assistant, stood first in discuss, Shot put throws and third in javelin throw. She also bagged the title in shuttle badminton. Mrs G. Aneeja, Technical Officer, won three second prizes in discuss, shot put and javelin throws. Mr M.K. Samson, UDC, won Carrom title along with two second prizes in Discuss and Shot put throws. Dr A. Debnath, Medical Officer, won third prize in javelin throw.



ICAR Inter-Institutional Sports Meet

NAARM contingent, consisting of 49 members, took part in the ICAR inter-institutional (South Zone) tournament organized by Central Institute of Fisheries Technology (CIFT) at Cochin from February 23 to 27, 2010. Mrs G. Aneeja won first prizes in Javelin and Discuss, second prize in Shotput and third prize in Long Jump; Mrs K.K. Rukmini Ammal won first



prizes in Shuttle Badminton (singles) and Shot Put, second prize in Discuss; Mrs N. Vijaya Lakshmi stood first in Chess; Mr M.K. Samson won second prizes in Shotput, Discuss; and Dr A. Debnath won second prize in Javelin throw. NAARM volleyball (Smashing) team stood runners-up. The team comprises of Mr M.K. Samson (Captain), Dr A. Debnath, Mr Sham Bahadur, Mr Chandramouli, Mr Laxman Ahire, Mr M. Srinivasa Rao, Mr P. Swamy, Mr P. Venkatesh and Mr K. Shivaiah.

NAARM bags Rose show

NAARM won Queen of the Show rolling trophy in the XXIV Annual Rose Show conducted by Secunderabad Horticultural Society on December 19 and 20, 2009 at Secunderabad Cantonment garden. The Academy also bagged cup in Hybrid Tea (HT) Specimen Bloom of Red group and five second prizes in different categories along with rolling trophies for the best Institutional garden, best collection of Hybrid Tea (HT) Roses, and Floriband roses.



Other Events



Other Events



Dr Mangala Rai laid Foundation Stones for the PG Complex Building and Sport Complex in the campus.

Foundation Day Celebrations

Marking the 34th year of its foundation, the Academy organized its Foundation Day on September 1, 2009. Dr H.S. Gupta, Director, Indian Agricultural Research Institute was the Chief Guest to deliver the Foundation Day Lecture on “Improving the Efficiency of a Research Organization”. Speaking on the occasion, Dr Gupta said along with developing technologies the steps should be taken to improve the overall effectiveness and efficiency of research organization. NAARM has set a standard in agricultural research management which is a role model for all ICAR institutes, he added. Dr N.H. Rao, Joint Director presided over the function. The Chief Guest distributed Best Worker awards under different categories announced on the occasion. Earlier, Dr P. Manikandan, Head, HRD division, welcomed the gathering and introduced the chief guest. Dr Jagannadham Challa, Principal Scientist proposed a vote of thanks. The scientists and employees of NAARM and ICAR sister institutions participated in the celebrations.

Hindi Cell Activities

Official Language (OL) policy was implemented in the Academy as per the guidelines of Government of India, Ministry of Home Affairs. Apart from its regular activities like compliance of section 3(3) documents in bilingual format, preparing bilingual stamps, noting and drafting, conducting quarterly meetings, and ensuring usage of official language policy implementation among the employees, Hindi Cell also implemented innovative concepts in various official activities as per the mandate of the Academy which are as follows:

- ◆ Established a self-learning language lab.
- ◆ FOCARS Scientists were provided awareness in OL policy of Government of India and enlightened with the primary responsibilities related to OL. Non-Hindi speaking FOCARS Scientists were also given training in verbal usage of Hindi in situational contexts. They were also involved in the competitions of Hindi Fortnight celebrations.
- ◆ Published a compilation on “*Krishi Prabandhan ke Naye Aayam Ank 4*”.



Rajbhasha Fortnight Celebrations:

The Rajbhasha Fortnight Celebrations were organized at the Academy from September 2 to 14, 2009. To mark the celebrations, various competitions were held which included noting and drafting, elocution, essay writing, dictation, translation, words making, just a minute, general knowledge, memory testing, etc. All categories of employees of the Academy participated in these competitions with much enthusiasm and interest. Dr Narahar Dev, Senior Manager (OL) (Rtd), Bharat Dynamics Limited, Hyderabad was the chief guest on the closing ceremony. Prizes were distributed to all the winners of various competitions. Dr S.K. Soam, OIC Rajbhasha, presented a brief report about the activities of the Hindi Cell in the Academy.

Workshops on Official Language:

Four internal Workshops on Official Language were organized to bring awareness and to promote Hindi in day-to-day official work among the employees of the Academy. Hands-on training on noting and drafting in Hindi was also given along with group discussion among the employees on the topic entitled, "My contribution to official language implementation in my office".

ICAR-SRF (PGS) Examination

The ICAR-SRF (PGS) Examination was conducted for the year 2009-10 by the SRF-Committee comprising of Dr P. Manikandan as Chairman, and Dr R.V.S. Rao, Dr G.R.K. Murthy, Mr P.P. Brahmaji, as Members. The examination was planned and organized successfully in 12 centres on December 20, 2009. Through this all-India competitive examination, 174 candidates were selected for fellowship for pursuing their Ph. D. degree programme and 374 candidates were qualified for admission to Ph. D. programme without fellowship. Dr P. Manikandan was the Controller of Examination.

Republic Day Celebration

Dr P.K. Joshi, Director, hoisted the National Flag on the eve of Republic Day celebration on January 26, 2010 at the Director's Office and addressed the gathering. All faculty members, officers, staff, trainees of various programmes, and temporary status workers attended the celebration.

Women's Day Celebration

To mark the International Women's Day on March 8, 2010, the women employees of the Academy generously helped the poor people by distributing clothes, fruits, other needy consumer products and groceries to the inmates of Orphanage Centre and the Old Age Home at Shivrampally. For this noble cause, contributions were made by the members of the NAARM family, FOCARS probationers and other trainees of the Academy. The Women's Cell President Dr N. Sandhya Shenoy coordinated this activity.

Skill Up-gradation Training for Group D Staff

A training programme on Skill Up-gradation for Group D (Non-metric and Non-ITI) staff of the Academy was conducted from August 22 to 28, 2009 as per the guidelines of VI Central Pay Commission. Thirty-three participants, including one employee from National Research Centre on Pomegranate, were sensitized in areas like values, ethics and behavioural issues, office procedures, basic knowledge of computers, typing, printing, vigilance and RTI, library and reception, etc. Mr Ashish Roy was the programme coordinator.





88th and 89th FOCARS scientists donated blood in the medical camp conducted by Kamineni hospitals, Hyderabad in the Academy.

Visitors

A six-member delegation from National Agricultural Research Organization (NARO), Uganda, comprising of Dr Cyprian Ebong, Director (Quality Assurance); Dr James Ogwant, Director (Crops Resources Research Institute); Mr John Ociti, (Chief Accountant); Dr George Maiteki, Director (Ngetta Zonal Agricultural Research and Development Centre); Dr Japheth Magyembe, Coordinator (Competitive Grants Scheme) and Ms Regina Musaaazi, Monitoring and Evaluation Officer, visited the Academy on June 05, 2009. They interacted with the Director and the Heads of Divisions, and expressed their interest in future collaboration with NAARM.

Committee Meetings

RAC meeting

Tenth Research Advisory Committee meeting, under the chairmanship of Prof. Pritam Singh, was organized on May 23, 2009 to review and discuss the activities of the Academy including teaching and training programmes, research, consultancy and policy support.

RAC members:



- Dr Pritam Singh, Ex-Director, IIM, Lucknow
- Prof. R.P. Singh, Director (ASRAM), Amity School of Rural & Agril Management, Amity University, Lucknow
- Dr V. Balaji, Head - ARSMP, Knowledge Management & Sharing, ICRISAT
- Dr Giri Rao, Director of Extension, ANGRAU, Hyderabad
- Prof. K. Pratap Reddy, Director, Sagar Food and Agribusiness School, Hyderabad
- Dr K. Vijaya Raghavan, Sathguru Management Consultants Pvt. Ltd., Hyderabad
- Dr P.K. Joshi, Director, NAARM
- Dr Arvind Kumar, DDG (Edn), ICAR
- Shri A. Mohan Dutta Deshmukh, Computer Point, Ahmednagar, Maharashtra
- Shri G. Narsimharaju Yadav, Progressive Farmer, Andhra Pradesh
- Dr R. Kalpana Sastry, Head, ARSMP Division, NAARM



IMC Meeting



The 44th Institute Management Committee meeting was held under the chairmanship of Dr S.M. Ilyas, Director, NAARM on June 27, 2009.

IMC members:

- Dr D.M. Hegde, Project Director, Directorate of Oilseeds Research, Hyderabad
- Dr P.K. Joshi, Director, National Centre for Agricultural Economics and Policy Research, New Delhi
- Dr P.K. Agarwal, National Professor, IARI, New Delhi
- Sh. Mohan Datta Deshmukh
- Sh. G. Narasimharaju Yadav
- Mr B.D. Sati, Finance and Accounts Officer, CRIDA, Hyderabad
- Dr N.H. Rao, Joint Director
- Drs D. Rama Rao, T. Balaguru and P. Manikandan, Heads of Divisions; NAARM
- Mr C. Bagaiah, FAO I/c, NAARM

Institute Research Council



The Institute Research Council meeting was held on March 29 and 30, 2010. During the meeting, on-going projects were reviewed and new projects were presented.

Review Meetings:

Agroweb Project: A review meeting of Agricultural Digital Dissemination System for Indian Agriculture Project under NAIP

was held on February 4, 2010 under the chairmanship of Dr S.L. Mehta to take stocks the progress of the Agroweb project. Twenty five participants from nine consortia partner institutions including NAARM attended the meeting.

Learning and Capacity Building: A review meeting of Learning and Capacity Building (L&CB) component of NAIP was held on March 20, 2010 under the chairmanship of Dr S.L. Mehta to take stock the progress of the project and discuss the issues pertaining to the implementation of the project. Thirty-four participants from four consortia partners including NAARM attended the meeting.





Publications

Publications

I. Papers Published in Journals

Ahire, L. M., Sontakki, B. S. and Punna Rao, P. 2009. Performance of agri-clinics and agri-business centres in Maharashtra. *Agricultural Extension Review* (July-September):3-13.

Amerender Reddy, A. and Reddy, G.P. 2010. Supply side constraints in production of pulses in India: A case study of lentil. *Agricultural Economics Research Review*. Vol. 23 Jan. – June 2010 pp: 129-136.

Aneeja, G. and Gutam, S. 2009. Prospects of open access to Indian agricultural research: A case study of ICAR. *First Monday.Org*. Vol. 14, Number 7 - 6 July 2009.

Karunasree, E. and Sandhya Shenoy, N. 2009. Information and Communication Technology (ICT) need assessment of agricultural professionals working in Krishi Vigyan Kendras in Andhra Pradesh. *Journal of Communication*, Vol.28, pp 24-35 2009.

Parthasarthy Rao, P. and Joshi, P K. 2009. Does unurbanization influence agricultural activities? A case study of Andhra Pradesh. *Indian Journal of Agricultural Economics* 64 (3): 401-408

Praveen, D. and Sandhya Shenoy, N. 2009. Knowledge of cotton farmers on health hazards of pesticides in Kurnool district of Andhra Pradesh – An analysis. *The Andhra Agricultural Journal*. Vol. 56, No 1, pp 1-14. 2009.

Raju, S S, Shinoj, P. and Joshi, P K. 2009. Sustainable development of biofuels: prospects and challenges. *Economic & Political Weekly XLIV* (52); 65-72

Rao, N.H., Reddy, M.N. and Kumar, K.V. 2010. Craving for spatial data. *Geospatial Today*, Feb. 2010, pp 36-39.

Sailaja, B., Nathawat, M.S. and Rao, N.H. 2009. A combination method for regional rice area estimation using remote sensing. *Asian Journal of Geoinformatics*, Sept. 2009, Vol. 9, No.3.

Sailaja, B., Voleti, S.R., Nathawat, M.S., and Rao, N.H. 2009. Validation of Oryza 2000 model with irrigated lowland field experiment of rice. *International Journal of Tropical Agriculture*, Vol. 27 (1-2), pp229-233.

Sastry, R.K., Rashmi, H.B, Rao, N.H. and Ilyas, S.M. 2009. Integrating nanotechnology into agri-food systems research in India: A conceptual framework. *Technological Forecasting and Social Change*, 77, 639-648 2010 (available on line from Dec 2009, <http://dx.doi.org/10.1016/j.techfore.2009.11.008> published in May 2010 issue).

Sastry, R.K., Rashmi, H.B. and Rao, N.H. 2010. Nanotechnology patents as R&D indicators for disease management strategies in agriculture. *Journal of Intellectual Property Rights*, Vol.15(3) pp197-205. 2010.

II. Book and Book Chapter(s)

Kalpana Sastry, R. *Intellectual Property Management Regime in the Indian National*



Agricultural Research Systems. In: CAS-IP NPI. 2009. Institutionalization of Intellectual Property Management: Case Studies from four Agricultural Research Institutions in Developing Countries. CAS-IP, Rome, Italy.
Online Publication April 2009. At: http://www.cas-ip.org/public/uploads/2009/04/compilation_of_4_working_papers_npi_2008.pdf

Rao, N.H. Water Resources (Chapter – 2). Natural Resources in State of Indian Agriculture, National Academy of Agricultural Sciences, New Delhi.

Reddy, G.P. Performance and Capital Formation in Fishes in India Inclusive Growth of Agriculture. Ed: Umesh et al., Department of Economics, Osmania University, Hyderabad.

Soam, S.K., Hanumanth Rao, B. and Victoria Henson-Apollonio. A Snapshot Guide to Intellectual Property Systems: A Compendium of Information from Developing Countries of Asia, Africa and Latin America. NAARM, Hyderabad and CGIAR CAS-IP, Rome, 290 pages (Eds) 2009.

III. Working papers

Hora M, Paranjape V and Sastry RK (2009). The evolving facets of PVP Laws in major jurisdictions of Asia-Pacific Region". Working paper in collaboration with Monsanto Research Center, Bangalore

Nanda, S.K., Sontakki, B.S., Rana, A.S. (2010). Innovation Quality Management in Agricultural Research and Development Organizations, Working paper No.6, NAIP L&CB, NAARM

Rao, K.H. (2010.) Working Paper on Quality, Safety and Traceability in Perishable Food Supply Chain: Indian Perspectives and Experiences, L & CB Project, NAIP

Rao, N.H., Reddy, M.N., Kumar, K.V., and Murthy, M.R.K. (2010). Vulnerability and adaptive capacity of livelihood systems: a GIS based framework, L&CB Working paper No. 2.

Reddy, M.N., Rao, N.H., Kumar, K.V., and Reddy, K.V. (2010). Use of data mining tools for assessment of vulnerability of livelihood systems, L&CB Working paper No. 4.

Sontakki, B. S., T. Balaguru, K. H. Rao, G. P. Reddy and J. Challa (2010). Training Needs and Impact Assessment of Learning and Capacity Building (L&CB) under National Agricultural Innovation Project (NAIP). Working Paper – 9. NAARM, Hyderabad.

IV. Research Reports

Rao, N.H., Reddy, M.N., Kumar, K.V., Latha, P. and Praveena Reddy. (2010) Knowledge Management through GeoSpatial Library and Rural GIS I. Geospatial Library, Research Report submitted to PMAC of NAIP Jan 2010, 54pp

Rao, N.H., Reddy, M.N., Kumar, K.V. and Murthy, M.R.K. (2009-10) Vulnerability and Adaptive Capacity of Rural Livelihood Systems: A GIS based Framework, NAARM, Hyderabad.

Rao, N.H., Reddy, M.N., Latha, P., Praveena, M.L., Reddy, K.V. and Madhu Sudhana Rao, S.



(2009-2010) Designing and Implementing Geo-spatial Library , NAARM, Hyderabad.
 Reddy, G.P. (2009). Food Supply Chain: Food Retail Management, NAARM, Hyderabad.

Reddy, M.N., Rao, N.H., Kumar, K.V., Madhusudan, P. and Reddy, K.V. (2010) Knowledge Management through GeoSpatial Library and Rural GIS II. Rural GIS, Research Report submitted to PMAC of NAIP, Jan, 2010, pp 164.

Reddy, M.N., Rao, N.H., Kumar, K.V. (2009-10) Characterization of Vulnerability of Livelihoods using Data Mining Tools, NAARM, Hyderabad.

Reddy, M.N., Rao, N.H., Kumar, K.V. (2009-10) Design and implementation of Rural GIS, NAARM, Hyderabad.

Sastry, R.K., Rao, N.H., Ilyas, S.M. and Rashmi, H.B. (2009) Assessing Interrelationships between Developments in Nanotechnology and Agriculture, Research Report submitted to PMAC of NAIP, Dec. 2009.

V. Base Papers

Rao, K.H. (2010) Working Paper on Quality, Safety and Traceability in Perishable Food Supply Chain: Indian Perspectives and Experiences, L & CB Project, NAIP

Jagannadham Challa., Tamboli, P. and Joshi, P. K. (2010) Revitalizing Higher Agricultural Education in India-Issues and Challenges, Base paper presented at the National Workshop on revitalizing Higher Agricultural Education in India held at NAARM, Hyderabad, February 24-25, 2010

Jagannadham Challa, D. Rama Rao and S. M. Virmani (2010) Developing Winning Research Proposals in Agricultural Research, 3rd Revised Edition, NAARM Publication (in advanced stage of completion)

VI. Others

Raju, P.N., Pal, D and Rao, K.H (2010). Traditional Indian Dairy Products with Enhanced Health Attributes, Paper presented at National Convention of Food Security and Preservation during January 27-28, 2010 held at Department of Food Technology and Management, Loyola Academy, Secunderabad, Proceedings, pp 6-13.

Kalpana Sastry, R., Rashmi, H.B., Rao, N.H. and Ilyas, S.M. (2009). Nanotechnology in Rice Production Systems - A Perspective. Paper presented in Workshop on "Forecasting Future Technological Needs for Rice in India". In: Proceedings of the Workshop on " Forecasting Future Technological Needs for Rice in India, CRRRI Cuttack; July 28-29, 2008. Editors: V. Ramasubramanian, Amerender Kumar, V.K. Bhatia and Satya Pal. pp 195-207.

Reddy, G.P., Reddy, M.N., Sontakki, B.S. and Kumar, K.V. (2009). Participatory GIS for Geospatial Planning and Environmental Impact Assessment of Aquaculture. Training Manual and Case Study. NAARM, Hyderabad.

Rama Rao, D. and Murthy, G.R.K. (2009). Technical Bulletin as Resource material for "Web-based E-learning and Content Management", NAARM, Hyderabad, 100p
 Murthy, G.R.K. (2009). Workshop Report on Sensitization of Networking and Web Hosting", NAARM, Hyderabad.



Murthy, G.R.K. (2009). User manual on three online modules developed (Online database, Online Examination, Online PDF search), NAARM, Hyderabad.

Sandhya Shenoy, N., and Reddy, M.N. (2009). Training Manual on Web Design Methodologies and Protocols under NAIP (AGROWEB). P- 131 NAARM, Hyderabad.

N.Sandhya Shenoy FET guidelines. 2009. P- 16 FET Committee NAARM, Hyderabad.

VII. Popular Articles

Joshi, P K. 2009. Need reforms in agriculture to contain rising food prices. Financial Express 30 Dec 2009.

Rao, N.H. and Joshi, P.K. (2009) Agriculture can be part of climate change mitigation strategy, Financial Express, Dec., 10, 2009.

VIII. Products Developed

E-learning

Academy developed customized open source software Moodle for e-learning. This is extensively used in the academic activities at NAARM. Academy's process is being used in six e-learning projects of NAIP being implemented in nine State Agricultural Universities. Besides, three more universities have implemented it in their systems. The e-learning frame work promoted by NAARM is now *de facto* standard followed in all e-learning initiatives in Indian NARS.

Web tool for Delphi Forecasting

An online Delphi process forecasting tool was developed. The software enables to solicit expert's opinion online and also in a single sitting thereby avoiding the delay in reaching consensus, cutting down the logistic costs like travel, paper and mailing work etc. The software is useful to develop forecasts and future strategies.





Faculty News

Faculty News

I. Papers presented in Conference/Workshop/Seminar

Avatar Rana, S. K. Nanda and B. S. Sontakki. (2010). Innovation Quality Management in Public Sector Research and Development Organizations. Paper presented by first author in National Conference on Business Innovation held at Apeejay Institute of Management, Jalandhar, Punjab held on February 27, 2010.

Kalpana Sastry, R. (2010). Ramifications of Nanotechnology in Agri-food Systems. CNS seminar series. Presented in Centre for Nanotechnology in Society. April 12, 2010. CNS, UCSB, California.

Kalpana Sastry, R., H. B. Rashmi, and N. H. Rao. (2009). Analysis of Nanotechnology Patents in Disease Diagnosis and Treatment in Agriculture. Presented in Conference on Recent trends in Patinformatics, 26th National Convention, December 9-12 2009, Pune.

Kalpana Sastry, R., H. B. Rashmi, and N. H. Rao. (2009). Nanotechnology as Tool in Enhancing Food Security in India. Presented in CNS-UCSB Conference on Emerging Technologies/Emerging Economies: [Nano]technology for Responsible, Equitable Development, November 4-6, 2009, Washington, DC.

Kalpana Sastry, R., H. B. Rashmi, N. H. Rao and S.M. Ilyas. (2009). Nanotechnology and Agriculture in India: The Second Green Revolution, Presented in OECD Conference on Potential Environmental Benefits of Nanotechnology: Fostering Safe Innovation-Led Growth July 15-17, 2009, OECD Conference Centre, Paris, France.

Kalpana Sastry, R., H.B. Rashmi, N.H. Rao and S.M. Ilyas.(2009). Nanotechnology and Agriculture. Panelist at Nanotechnology Business: "The Giant in Dwarfs" April 4-15, 2009: Shahjehan Hall: Hotel Taj Palace: New Delhi.

Raju, P.N., Paul, D., and Rao, K.H. (2010). Value Addition in Traditional Indian Dairy Products with Enhanced Health Attributes. Paper presented at the National Convention of Food Security and Preservation, held at Loyola Academy, Secunderabad on January 28-29, 2010. Proceedings, pp 6-13.

Rao, K.H. (2010). Supply Chain Management in Meat and Meat Products. Paper presented at the National Convention of Food Security and Preservation, held at Loyola Academy, Secunderabad on January 28-29, 2010.

Rao, N.H. (2009). e-Enabling Geospatial Knowledge Management for Adapting to Climate Change, e-Agriculture India, e-India Conference, August 25-27 2009, Hyderabad (invited presentation)

Reddy, G.P., Vyshnavi, P.V. and Rao, K.H. (2010). Management of Agri-food Supply Chains and Networks, National Seminar on Changing Ethos of Agribusiness Management and Education held at Mysore University, Mysore, March 18-20, 2010.

Reddy, G.P, Murthy, M.R.K., and Vyshnavi, P.V. (2010). Food Supply Chain: Food Retail Management, National Conference on Managing Agri-food Supply Chain, to be held at IIM, Lucknow, April 9-11, 2010.

Reddy, G.P., and Murthy, M.R.K. (2010). Agri-food Retail Supply Chain Management – 2010 China-India Consumer Insights Conference at Tsinghua University, Beijing, China, July 1-3, 2010.



Ruchica Goyal, and Rashmi, H.B. (2009). Trends in Nanobiotechnology in Agriculture using Patent as Indicators. Poster presentation in Conference on Recent trends in Patinformatics, 26th National Convention, Pune, December 9-12, 2009.

Sailaja, B., Voleti, S.R., Nathawat, M.S. and Rao, N.H. (2009). Rice Yield Estimation by Integration of Crop Models, GIS and Remote Sensing, Paper accepted for presentation in International Symposium on Plant Growth Modelling, Simulation, Visualization and their Applications, Beijing, China, November 9-13, 2009.

Sandhya Shenoy, N. (2010) "Information and communication Technologies in Agriculture – Indian experience" on 25.2.2010, at Extension and Outreach Department, Iowa State University, Iowa State, USA. <http://connect.extension.iastate.edu/p83672031/>

Sontakki, B. S. (2010). Strategies for Developing Need-based Content. Invited paper presented in the Content Development Workshop organized by the Directorate of Rice Research under the NAIP Component – I Consortium Project on Development and Maintenance of Rice Knowledge Management Portal at DRR, Hyderabad during February 5-6, 2010.

Thomas Abraham, Nagaraj Vaidya, Vimal Kumar, Sridhar Gutam, Aneja Guttikonda (2009). Open Access Journal Publishing in the Agricultural Sciences. Paper presented at Second International PKP Scholarly Publishing Conference July 8-10, 2009 at Vancouver, Canada.

II. Special Lectures Delivered by:

B.S. Sontakki:

"Frontiers in Rural Communication in Developing Nations" to the students of Master of Professional Studies in International Agricultural and Rural Development of CALS-IP, Cornell University, Ithaca, NY, USA on November 20, 2010.

"Strategies for Developing Need-based Content" - invited paper presented in the Content Development Workshop of RKMP Project at Directorate of Rice Research, Hyderabad held during February 5-6, 2010.

"Effective Communication, Classroom Management, Student-centred Teaching Learning and Instructional Evaluation and Testing" - Invited lectures delivered in the Training cum Workshop on Optimization of Teaching and Learning organized by College of Veterinary Science, Khanapara of the Assam Agricultural University, Guwahati during January 17-19, 2010.

G.R.K. Murthy:

"Content Management in Joomla" in the training-cum- workshop on "Distributed Ownership of Institutional website for content management and updates" on 16-3-2010 held at CIBA, Chennai.

Jagannadham Challa:

"Developing Winning Research Proposals" for the Indian Council of Forestry Research and Education, Dehradun for 60 Indian Forest Service Officers during January 13-14, 2010.

"Educational Technology" guest lecture to the University of Agricultural Sciences, Raichur, Karnataka on for January 18, 2010.

M.N. Reddy:

"MS Excel and Its Applications" in the Refresher training course for the faculty of ANGRAU teaching Computer Course STCA 301 at Acharya N.G. Ranga Agricultural University, Hyderabad on June 10, 2009.

"Introduction to agricultural statistics" to the participants of 10 month post-graduate diploma course in plant protection at National Institute of Plant Health Management, Hyderabad on July 21, 2009.



“Design of Experiments” to the participants of 10 month post-graduate diploma course in plant protection at National Institute of Plant Health Management, Hyderabad on July 21, 2009.

“Web design” to the participants of IT based DSS for market and agribusiness orientation at MANAGE on August 20, 2009.

III. Academic Guidance:

Dr B.S. Sontakki, Principal Scientist, guided Shri C. B. Meti, Associate Professor, College of Agriculture, University of Agricultural Sciences, Dharwad who has registered for Ph.D. in Agricultural Development at School of Agriculture, Yashwantrao Chavan Maharashtra Open University (YCMOU), Nashik. The title of his doctoral research was Impact Assessment of Micro Irrigation Scheme Implemented in Dharwad District of Northern Karnataka.

Dr N.H. Rao, Joint Director, supervised one Ph.D. student for thesis research on 'Development of GIS Based DSS for Management of Rice based Production Systems'- Birla Institute of Technology and Science, Mesra.

Dr N. Sandhya Shenoy, Principal Scientist, guided for M.Sc. and Ph.D (Agriculture Extension, Agricultural Communication and Agricultural Development) students at the Yashwant Rao Chauhan Maharashtra Open University (YCMOU), Nashik. Research guidance was provided to four Ph.D. students (Agriculture Extension) of YCMOU on the topics, “Scope for Public –Private Partnership in Agriculture Extension Delivery System in Andhra Pradesh”, “A Critical Study on Livelihoods of Tribal Farmers in Andhra Pradesh”, “Impact of Farmer Field Schools on Cotton farmers of Warangal District in Andhra Pradesh”, and “Paradigm shift in livelihood patterns of small and marginal farmers of AP”.

Dr N. Sandhya Shenoy, Principal Scientist, evaluated M.Sc. thesis of University of Agricultural Sciences, Dharwad, Karnataka, on 'Contribution of Livestock Production System to Farmers Livelihood in Western Region of Maharashtra, in the area of Agricultural Extension.

IV. Participation in Seminars/Workshops/Symposia/Conferences/ Meetings by Faculty and Staff:

Name of the Programme	Organizers and Period	Attended by
Training Programme on “Open Source Technologies”	AMR Andhra Pradesh Academy of Rural Development, Hyderabad in collaboration with Enterprise DB, Pune and National Informatics Centre, New Delhi July 20–Sept. 10, 2009	Mohan Singh, Technical Officer
Meeting on “STEM professional development programme for technical managers”	Society for Technology and Management Aug.19 to 21, 2009	R. Kalpana Sastry, Principal Scientist
Geospatial Knowledge Management for Adaptation to Climate Change in Session on e-Agriculture India	e-India Conference, Hyderabad Aug. 25-27, 2009	N.H. Rao, Joint Director
Consultation on “Enhancing Open Access in Indian Agriculture: Prospects, Opportunities, Advantages and Challenges”	ICRISAT organized in association with NAIP-ICAR and the FAO Sept. 6-7, 2009	D. Rama Rao, Principal Scientist, and G. Aneja, Technical Officer



Training Programme on "Cyber Laws, Information Security & Computers for Administrative Personnel"	Indian Institute of Public Administration, New Delhi, sponsored by Department of Science and Technology, Govt. of India, New Delhi. Sept. 21-27, 2009	Y. Anuradha, Personal Assistant
International Clinical Oncology Congress	Indo-American Cancer Institute and Research Centre at Hyderabad. Oct. 24-25, 2009	A. Debnath, Medical Officer
Workshop on "Communicating Science: Tools for Scientists and Engineers"	American Association for the Advancement of Science and the National Science Foundation of USA at Statler Hotel, Cornell University, Ithaca, NY, USA. Nov. 5, 2009	B.S. Sontakki, Principal Scientist
23 rd National Conference on "Agricultural Marketing"	CRIDA, Hyderabad. Nov.12-14, 2009	G.P. Reddy, Principal Scientist
Conference on "Institutions, behaviour and escape from persistent poverty"	Industrial and Labour Relations School, Cornell University, Ithaca, NY, USA Nov. 16-17, 2009	B.S. Sontakki, Principal Scientist
17 th Annual AERA conference on "Technology, Environment and livelihood security"	Tamil Nadu Agricultural University, Coimbatore Nov. 19-21, 2009	G.P. Reddy, Principal Scientist
17 th Annual AERA conference on "Technology, Environment and livelihood security"	Tamil Nadu Agricultural University, Coimbatore Nov. 19-21, 2009	P.K. Joshi, Director
Round Table discussion on 'Research and extension with immigrant farm workers'	CaRDI, CALS, Cornell University, Ithaca, NY, USA Nov. 23, 2009	B.S. Sontakki, Principal Scientist
Conference on "Recent Trends in Patinformatics"	National Chemical Laboratories, Pune Dec. 9-12, 2009	R. Kalpana Sastry, Principal Scientist
5 th DST-NIAS Workshop on "Nanotechnology: Dimensions of Nanoscience, Nanotechnology and Society for Women Scientists and Technologists"	National Institute of Advanced Studies (NIAS), Bangalore. Dec.14-18, 2009	R. Kalpana Sastry, Principal Scientist
69 th Annual conference of Indian Society of Agricultural Economics (ISAE)	Indian Society of Agricultural Economics (ISAE) at Amritsar Dec. 17-19, 2009	G.P. Reddy, Principal Scientist
Workshop of RKMP Project on "Strategies for Developing Need-based Content' at Content Development"	Directorate of Rice Research, Hyderabad Feb.5-6, 2010	B.S. Sontakki, Principal Scientist
Management Development Programme on "Food Supply Chain Management"	Indian Institute of Management (IIM), Ahmedabad Feb. 7-13, 2010.	G. P. Reddy, Principal Scientist



Training programme on "Establishment Rules"	Institute of Secretariat Training and Management, New Delhi Feb. 8-12, 2010	T. Srinivas, Assistant
Training programme on "Purchase Policy and Procedure in Government Departments and Government Sector"	Centre for Training and Social Research, New Delhi Feb. 24-26, 2010	L. Jhansi Lakshmi, Private Secretary
Workshop on "Pay Fixation"	Institute of Secretariat Training and Management, New Delhi Mar. 3-5, 2010	M. Dinesh, Assistant
workshop on "Use of Fly Ash in Agriculture"	Centre for Fly Ash in Agriculture, New Delhi Mar. 9, 2010	S.K. Nanda, Principal Scientist
International Conference on "Tackling CVD in Developing World"	Medwin hospitals at Hyderabad Mar.13, 2010	A. Debnath, Medical Officer
"National Update on Head and Neck Onco Surgery"	Kamineni hospitals, Hyderabad Mar. 14, 2010	A. Debnath, Medical Officer
National Workshop on "ICT Initiatives of the NAIP with Special Reference to the Uniformity Guidelines for ICAR Websites"	NBPGR, New Delhi Mar. 19, 2010	G.R.K. Murthy, Principal Scientist

V. Foreign Visits

Dr S.K. Nanda, Principal Scientist underwent training in Agricultural Research Management at International Food Policy Research Institute (IFPRI), Washington, DC, USA under L&CB Component-1 of NAIP from March 29 to April 28, 2009.

Dr R. Kalpana Sastry, Principal Scientist, and Dr S.K. Soam, Principal Scientist, attended Annual Meeting of National Partners Initiative at The Hague, The Netherlands from September 1 to 5, 2009 conducted by CGIAR Central Advisory Service on Intellectual Property (CAS-IP), Rome, Italy.

Dr P.K. Joshi, Director, visited University of Copenhagen, Denmark, and attended the Steering Committee meeting of the Climate Change, Agriculture and Food Security (CCAFS) as a Member of its Steering Committee at University of Copenhagen, Denmark from October 26 to 30, 2009.

Dr B.S. Sontakki, Principal Scientist, attended one-month International Training on "Marketing Management, Agricultural Marketing and Business Communication" at Cornell University, Ithaca, NY, USA from October 28 to November 27, 2009.

Dr R. Kalpana Sastry, Principal Scientist, attended a conference on "Emerging Technologies/ Emerging Economic, Nanotechnology for Responsible, Equitable Development" organized by Centre for Nanotechnology in Society, University of California, Washington DC, USA from November 4 to 6, 2009.

Dr P.K. Joshi, Director, attended the roundtable session on 'Policies and Institutions for Resilient Development in the Face of Climate Change' organized by Agriculture and Rural Development Day (ARDD) at University of Copenhagen, Denmark from December 11 to 13, 2009.



Dr S.K. Soam, Principal Scientist and Dr R. Kalpana Sastry, Head, Agricultural Research Systems and Management Policies Division, attended CGIAR Central Advisory Service on Intellectual Property (CAS-IP) Meeting, Indonesia on January 18, 2010.

Dr P. Manikandan, Head, Human Resource Development Division, was on study visit to the College of Agriculture and Life Sciences (CALS) at Cornell University, Ithaca, NY, USA, from February 1 to 14, 2010.

Dr N. Sandhya Shenoy, Principal Scientist, underwent international training, at University of Florida, USA from February 1 to 28, 2010 under the Learning and Capacity Building (L&CB) Project of NAIP, on "ICTs in Agriculture and Natural Resources focused on Development, Authoring and Management of Reusable Learning Objects (RLOs)".

Dr R.V.S. Rao, Principal Scientist, was deputed on a study visit to Cornell University, Ithaca, New York from February 1 to March 3, 2010 under L&CB Component of National Agricultural Innovation Project (NAIP).

Dr K.H. Rao, Principal Scientist, was deputed to Cornell University, Ithaca, New York, USA from February 1 to March 3, 2010 under L&CB Component of National Agricultural Innovation Project (NAIP) to study "Supply Chain Management and Human Resource Management in Agriculture".

Dr A. Dhandapani, Principal Scientist, was deputed on a study visit on "Management Information System" at Cornell University, Ithaca, New York from February 1 to March 3, 2010 under L&CB Component of National Agricultural Innovation Project (NAIP).

Dr P. K. Joshi, Director attended the External Advisory Panel Meeting for the Independent Evaluation Group (IEG) of the World Bank support for Agriculture and Agribusiness at The World Bank, Washington DC on March 18 and 19, 2010.

Dr M.N. Reddy, Principal Scientist, was deputed on a study visit to University of Georgia, Griffin, Georgia, USA from March 22 to April 4, 2010. He underwent training programme on "Crop simulation models and geospatial knowledge management" at Department of Biological and Agricultural Engineering of the university.
Dr Jagannadham Challa, Principal Scientist, was deputed on a study visit to University of Maryland, USA from March 22 to April 2, 2010 under L&CB component of National Agricultural Innovation Project (NAIP).

Dr S.K. Soam, Principal Scientist, was deputed on a study visit to Washington State University Research Foundation (WSURF), Office of Intellectual Property Administration, Washington State University at Pullman, USA from 30 March to 29 April, 2010 Under the NAIP (L&CB) project.

VI. Higher Qualification Acquired

Dr Ahire Laxman Maharu, Technical Officer (T-5), NAARM, acquired his Doctoral Degree from School of Agriculture Science, Yashwantrao Chavan Maharashtra Open University, Nashik, Maharashtra for his thesis titled "A study on management of training in agricultural training organizations".

Dr V. Murali, Farm Superintendent, NAARM, acquired his Doctoral Degree from Department of Agronomy, Acharya N.G. Ranga Agricultural University, Hyderabad for his thesis titled "Integrated Nutrient Management (INM) practices for sunflower - groundnut cropping sequence".

Dr M.A. Basith, Technical Officer, NAARM, acquired Ph. D. in Agronomy from Acharya N.G. Ranga Agricultural University (ANGRAU), Hyderabad for his thesis titled "Recent techniques to evaluate the comparative performance of castor based intercropping systems under different plant geometrics".





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Personalia

Faculty

Name and Designation

Dr S.M. Ilyas, Director (Till July 31, 2009)
 Dr P.K. Joshi, Director (w.e.f. September 19, 2009)
 Dr N. Hanumantha Rao, Joint Director*
 Dr P. Manikandan, Head, HRD
 Dr D. Rama Rao, Head, ICM
 Dr T. Balaguru, Head, ARSMP (till June 30, 2009)
 Dr (Mrs)R. Kalpana Sastry, Head, ARSMP (from Feb. 6, 2010)
 Dr R.K. Samanta, Principal Scientist
 Dr A. Gopalam, Principal Scientist (till July 31, 2009)
 Dr Jagannadham Challa, Principal Scientist
 Dr K.M. Reddy, Principal Scientist
 Dr M. Narayana Reddy, Principal Scientist
 Dr (Mrs.) N. Sandhya Shenoy, Principal Scientist
 Dr Santosh Kumar Nanda, Principal Scientist
 Dr G.P. Reddy, Principal Scientist
 Dr R.V.S. Rao, Principal Scientist
 Dr S.K. Soam, Principal Scientist
 Dr Bharat S. Sontakki, Principal Scientist
 Dr K.H. Rao, Principal Scientist
 Dr A. Dhandapani, Principal Scientist
 Dr V.K. Jayaraghavendra Rao, Principal Scientist
 Dr G.R. Ramakrishna Murthy, Senior Scientist
 Mr M. Suresh Kumar, Chief Admn. Officer (till June 30, 2009)
 Sh. Ashish Roy, Chief Admn. Officer (w.e.f. July 14, 2009)

*Acting Director from 1 August to 18 September, 2010

Technical Personnel

Grade T-9
 Dr V. Murali, Garden Superintendent
 Dr A. Debnath, Medical Officer
 Mr Zameer Ahmed, Manager (Hostel)

Grade T(7-8)
 Dr M.A. Basith, Tech. Officer

Grade T-6
 Mrs P.V. Nirmala (till May 1, 2009)
 Mr Ch. Janardhan Rao
 Mr K.V. Kumar
 Mr P. Vijender Reddy
 Mr P. Namdev
 Mr Sohail Ahmed Khan

Grade T-5 (Tech. Officers)
 Mr N.R. Nageswara Rao
 Mr Bansidhar Nayak



Mrs G. Aneja
Mr P. Mohan Singh
Mr L. Ramesh
Mr M. Shekher Reddy
Mr B. Veeraiah
Dr Ahire Laxman
Mr N. Naresh Kumar
Mr M. Ravi

Administration and Finance

Mr Y. Sankara Rao, AAO
Mr P.P. Brahmaji, AAO
Mr C. Bagaiah, Junior Accounts Officer
Mr J. Renuka, Asst. Director (Hindi)
Mr B.Ch. Satyanarayana, Security Officer

Private Secretaries
Mrs L. Jhansi Lakshmi
Mrs Sarada Samanta
Mr Raghunath

Welcome



Dr A. Dhandapani joined the Academy on June 15, 2009 as Principal Scientist. Prior to this he was scientist since 1996 at National Centre for Integrated Pest Management (NCIPM), New Delhi and promoted to Senior Scientist (Agricultural Statistics) in 2005. He is a Ph.D. in Agricultural Statistics from IARI, New Delhi.



Dr Pramod Kumar Joshi assumed charge as Director for National Academy of Agricultural Research Management (NAARM), on September 19, 2009. Before joining the Academy, he served at National Centre for Agricultural Economics and Policy Research (NCAP), New Delhi as Director. Dr Joshi was South Asia Coordinator with the International Food policy Research Institute (IFPRI), Washington D.C. and also served International Crops Research Institute for the Semi-Arid Tropics (ICRISAT). Dr Joshi started his career as a Scientist at Indian Agricultural Research Institute, Pusa, New Delhi and later served at Central Soil Salinity Research Institute, Karnal. He is a recipient of prestigious D.K. Desai Award (1991) and Doshi Award (1998 and 2006) in recognition of his excellent contribution to the agricultural economics and policy research. He is fellow of the National Academy of Agricultural Sciences (NAAS) and Treasurer of the Trust for Advancement of Agricultural Sciences (TAAS).



Mr Ashish Roy assumed charge as Chief Administrative Officer of the Academy on July 14, 2009. Mr Roy holds degrees in B. Pharm (Hons.), Human Resources Management, LLB and M.A (Sociology). He joined ICAR system as Administrative Officer at Central Institute for Cotton Research, Nagpur in 1988 and later worked as Senior Administrative Officer at National Bureau of Soil Survey & Land Use Planning, Nagpur. Mr



Roy served at Indian Veterinary Research Institute, Izatnagar, U.P; Central Research Institute for Dryland Agriculture, Hyderabad, and Directorate of Oilseeds Research, Hyderabad as Senior Administrative Officer from 2001 to 2009 before joining the Academy.



Dr R.K. Samanta, Principal Scientist, joined the Academy on January 12, 2010 after relinquishing the position of Vice-Chancellor, Bidhanchandra Krishi Viswavidyalay, Mohanpur, West Bengal.

Farewell



Mrs P.V. Nirmala, Technical Officer (T-6) voluntarily retired from service on May 1, 2009 after serving the Information Centre of the Academy for nearly 18 years. Mrs Nirmala looked after library management, procurement of digital resources, and provided library services.



Dr T. Balaguru, Head, ARSMP division, superannuated on June 30, 2009 after serving the Academy for three decades. Dr Balaguru joined the ICAR system as Scientist (S-1) at Regional Centre of NBSS&LUP, Nagpur. Later he moved to NAARM as scientist (S-1) on Aug. 13, 1979. He belonged to the first batch of ARS. His thrust areas were agricultural research systems and management, farming systems approach for sustainable agricultural development, research prioritization, monitoring and evaluation, etc.



Mr M. Suresh Kumar, Chief Administrative Officer, superannuated on June 30, 2009. He started his career as Administrative Officer at CTRI, Rajahmundry in 1977. Later, Mr Suresh Kumar joined Directorate of Oilseeds Research on promotion as Senior Administrative Officer in 1983. He was elevated to the position of Chief Administrative Officer to IVRI, Izatnagar in 1988. He served the ICAR system for more than three decades including fourteen years at NAARM. His areas of interest were human resource development, administration and financial management, rules and regulations, vigilance and RTI Act, etc.



Dr S.M. Ilyas, Director, superannuated on July 31, 2009. He joined the Academy on February 8, 2007. He served the ICAR system for more than 33 years in different capacities. Dr Ilyas joined the ARS service in September 1, 1975 at Central Rice Research Institute (CRRI), cuttack as Research Engineer. He worked at IARI as scientist (S-3), and Principal Scientist. He served at the ICAR headquarters as ADG (IPR) and ADG(TC) till 2000 before moving to Central Institute of Post-harvest Engineering and Technology, Luthiana as Director. Later he became Vice Chancellor at Narendra Dev University of Agriculture and Technology, Faizabad in 2005. Dr Ilyas has published more than 150 papers



in both national and international journals, and he has six patents to his credit of which three technologies were commercialized. He has been awarded fellowship and gold medals by Indian Society of Agricultural Engineers for his outstanding contributions in the field. He is a member of different Boards of Management in the country.



Dr A. Gopalam, Principal Scientist, superannuated on July 31, 2009. He joined the Central Tobacco Research Institute, Rajahmundry as Senior Research Assistant in 1974 and became Scientist (S-1) in 1979. Later he served at Central Plantation Crops Research Institute, Calicut and National Research Centre for Spices, Calicut as Scientist (S-2) before moving to NAARM in 1989. Since then Dr Gopalam served the Academy as Senior Scientist and Principal Scientist in the discipline of Educational Methodology and Instructional Technology, and authored 52 papers and published 12 books. He was also OIC Hindi Cell of the Academy.

Mr Eeswari, T-2, superannuated on July 31, 2009. He joined the Academy as SSG-1 in 1987 and later on appointed as Power Tiller Driver (Aux.) in 1989. He was promoted to T-2 in 2001. He served the Academy for more than 21 years.

Promotions

Dr A. Debnath (Medical Officer) was promoted from T-6 to T (7-8) with effect from June 29, 2008.

Mr Sohail Ahmed Khan (Junior Engineer) was promoted from T-5 to T-6 with effect from January 1, 2009.

Dr R. Kalpana Sastry, Principal Scientist, joined as Head, Agricultural Research Systems Management and Policies division at the Academy on February 6, 2010.

Obituary



Mrs Y. Dhanalakshmi, Library Assistant (T-4), passed away on January 29, 2010. She joined NAARM on August 29, 1985, and rendered service for more than 24 years at Documentation Centre of the Academy in different positions. NAARM lost a wonderful staff and friend. May her soul rest in peace!





राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी

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